

SenseTime

SenseStudio V2

API Document

www.sensetime.com

PRODUCT
MANUAL



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1 Introduction

1.1 Overview

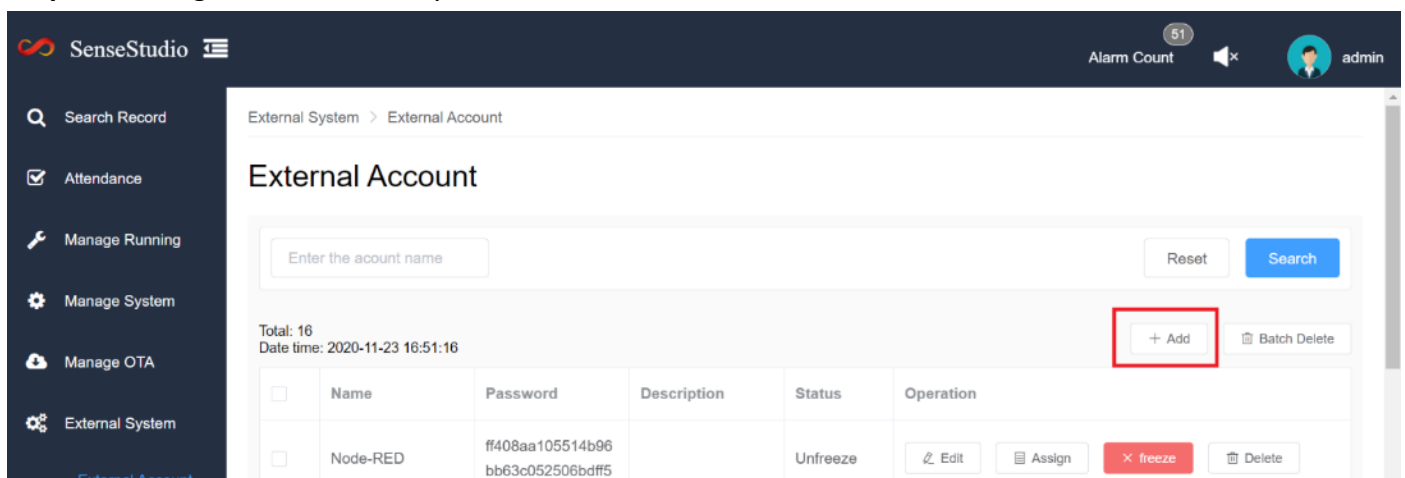
SenseStudio provides RESTful APIs to expose the capabilities. Applications can leverage the RESTful APIs to interact with the data from SenseStudio. Third-party systems can also integrate with SenseStudio using APIs. Besides, SenseStudio can push internal events to external systems through HTTP Push for further business processing.

1.2 Invoking the APIs

1.2.1 External Account Setup

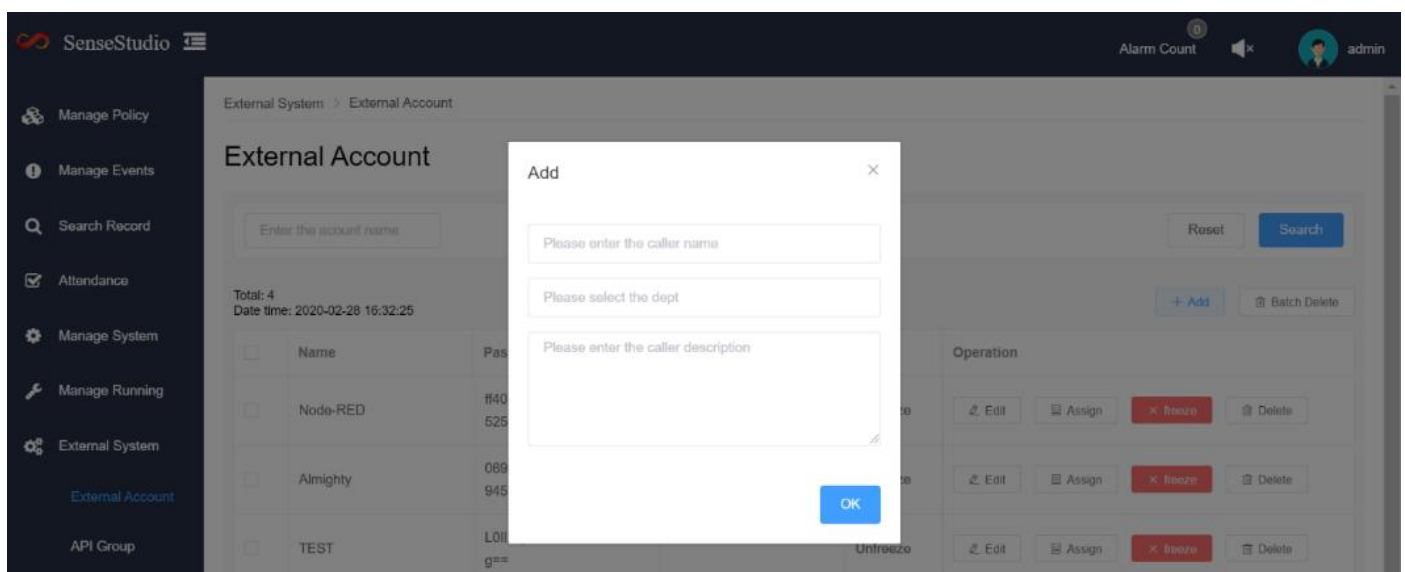
You have to obtain an external account for API authentication and authorization.

Step 1. Navigate to “External System” > “External Account”, and choose “Add”.

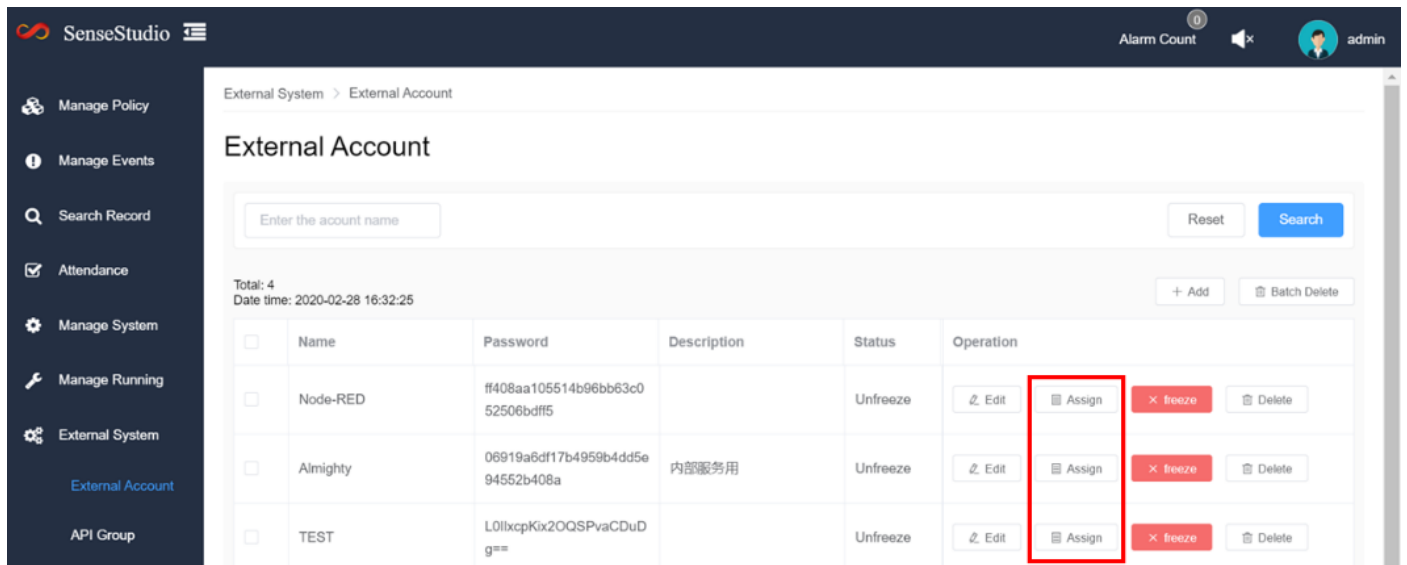


Step 2. Enter the external account information, and then choose “OK”.

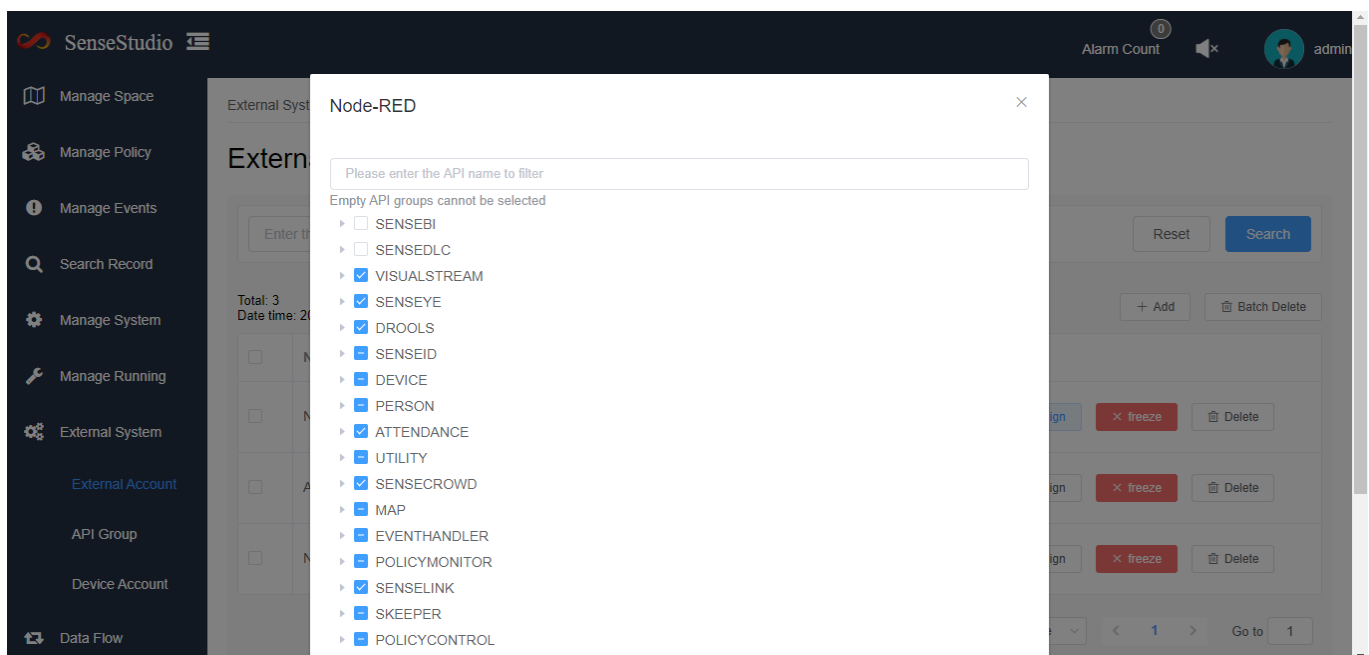
Caller Name	API caller account, mandatory.
Dept	The department of the API caller, mandatory.
Caller Description	Description of the API caller.



Step 3. Choose “Assign” on the row of the external account you just created.



Step 4. Select the APIs that you would like to assign to the external account, then choose “OK”.

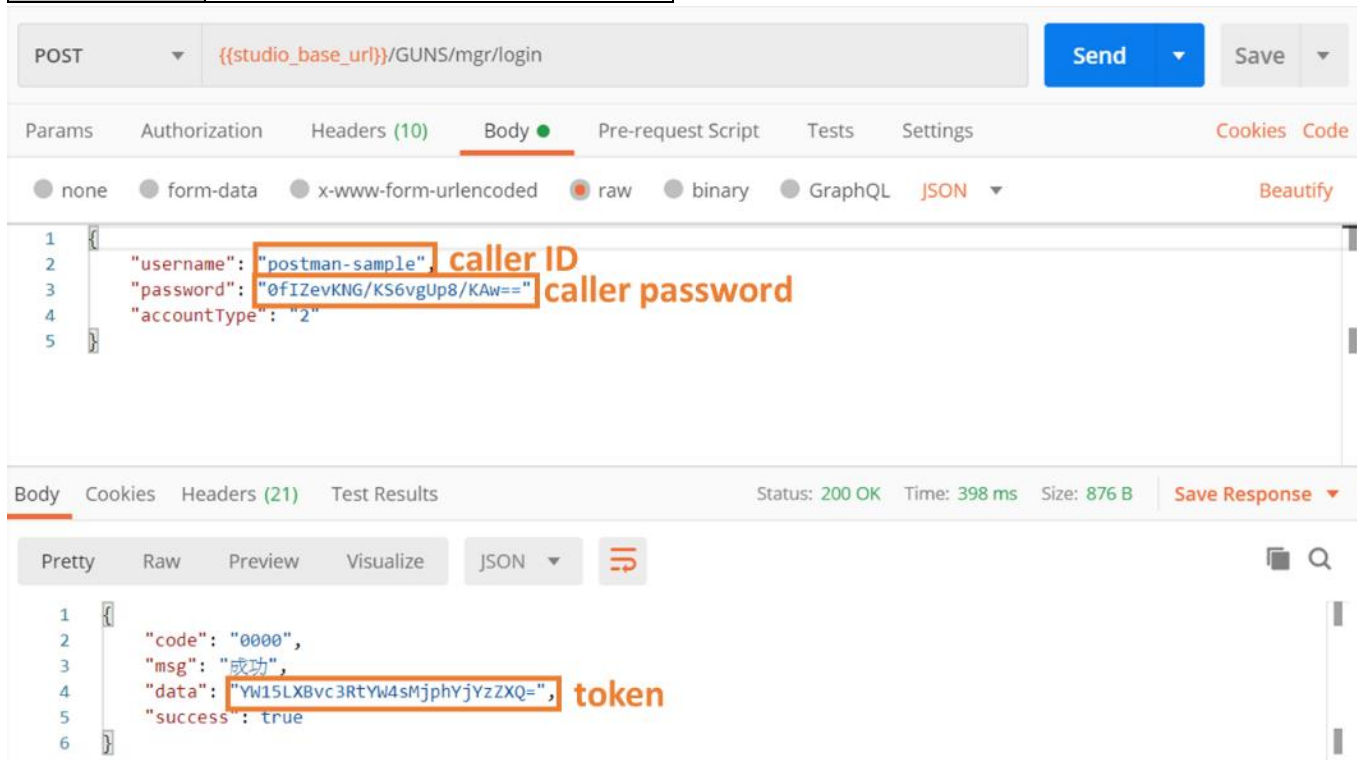


1.2.2 Get Access Token

Prior to invoking the APIs, you have to obtain an access token.

Step 1. Obtain an access token by invoking `/GUNS/mgr/login`.

username	Name of the external account.
password	Password of the external account.
accountType	2 for API call



The screenshot shows a Postman interface for a POST request to `{{studio_base_url}}/GUNS/mgr/login`. The request body is JSON, containing:

```
{  "username": "postman-sample",  "password": "0fIZevKNG/KS6vgUp8/KAw==",  "accountType": "2"}
```

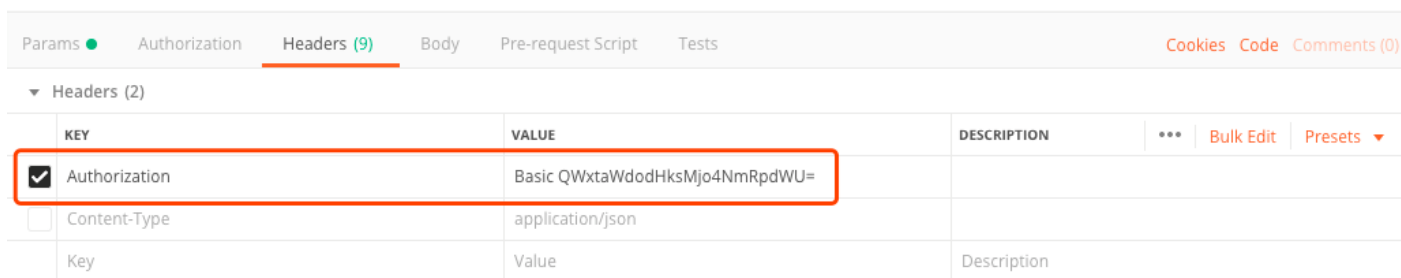
Annotations in the image identify `postman-sample` as the **caller ID** and `0fIZevKNG/KS6vgUp8/KAw==` as the **caller password**. The response status is 200 OK. The response body is JSON, containing:

```
{  "code": "0000",  "msg": "成功",  "data": "YW15LXBvc3RtYW4sMjphYjYzZXQ=",  "success": true}
```

An annotation identifies the `data` field as the **token**.

Step 2. Put the token into the header for basic auth.

KEY	Authorization
VALUE	Basic {{token}}



The screenshot shows the Postman Headers tab. The 'Authorization' header is checked and its value is set to `Basic QWxtaWdodHksMjo4NmRpdWU=`. Other headers like 'Content-Type' and 'Key' are also visible.

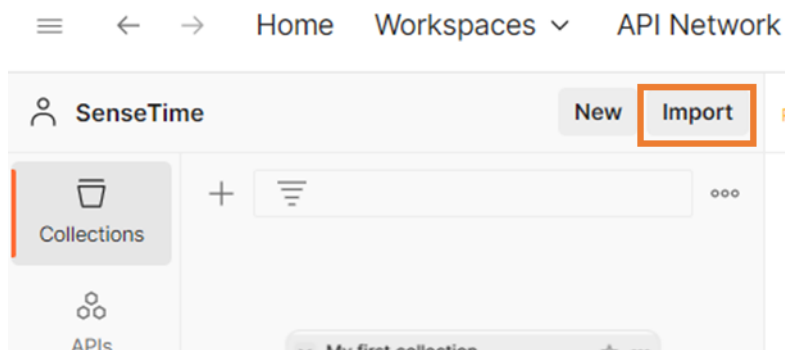
1.2.3 Postman Sample

SenseStudio provides API request samples for Postman. Please contact SenseTime team for the sample file "SenseStudio Postman Sample-en.json".

Step 1. Launch the Postman app.

Importing Files

Step 2. Click "Import" in the upper-left corner of the Postman app.



Import

File Folder Link Raw text Code repository **New** API Gateway **New**

Choose Files

Step 3. Upload the above json file and click "Import".

Import Elements

 Search files and folders

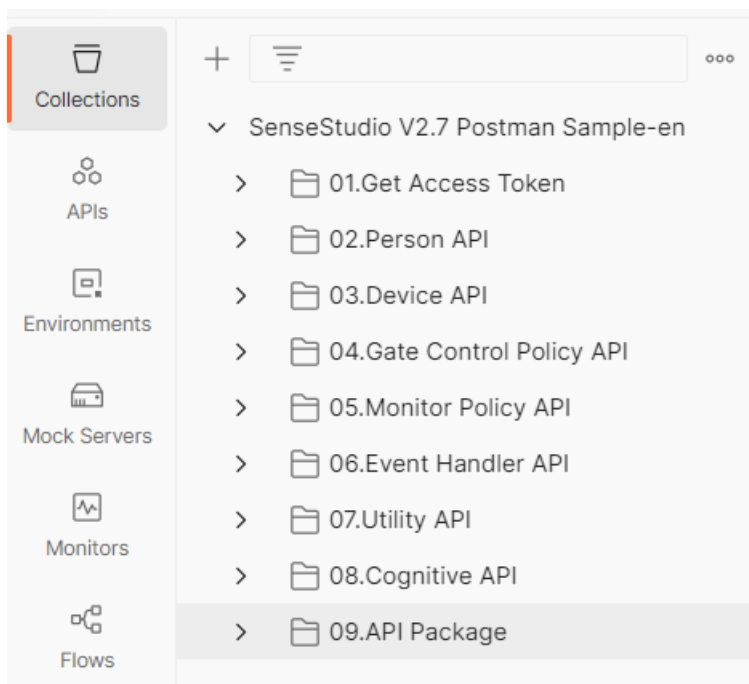
Collections

☒ Collection Name	Format
☒ SenseStudio V2.7 Postman Sample-en	Postman Collection v2.1

Import

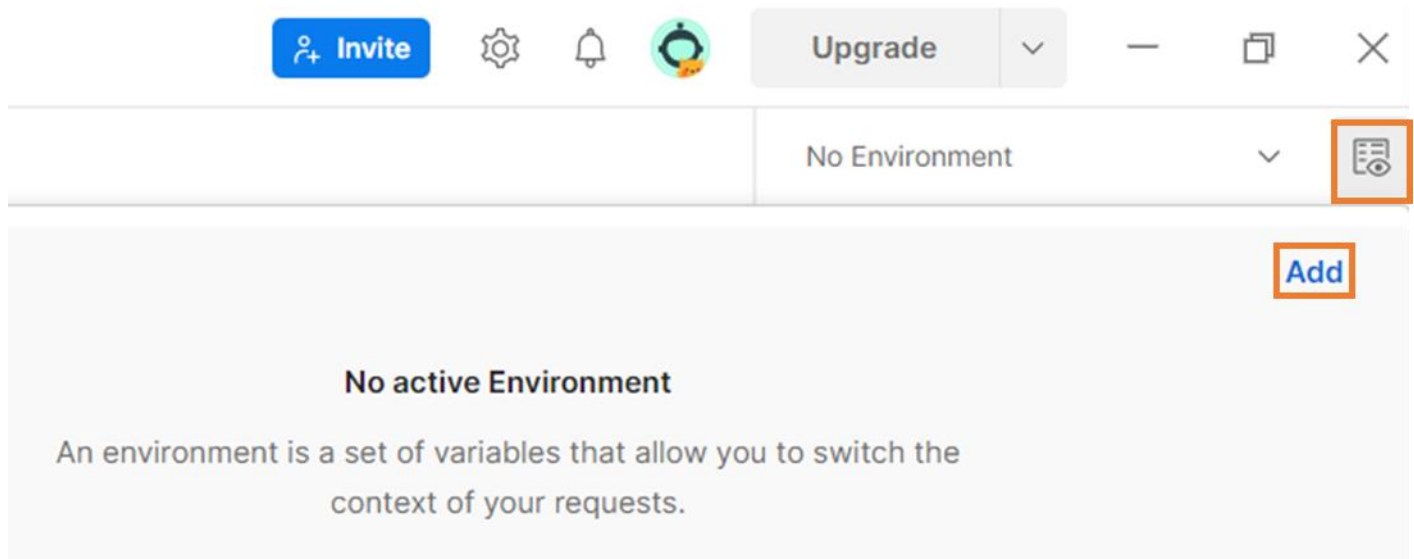
Cancel

The imported data will be presented on the left.



Creating Environments

Step 4. Create a new environment by clicking "Manage Environments" at the top right corner and click "Add" to create a new environment.



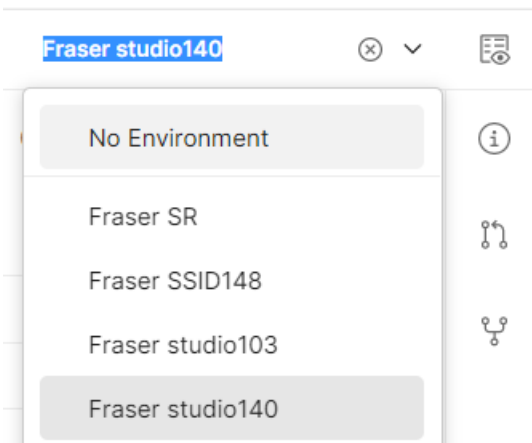
Step 5. Name the environment and add required variables for SenseStudio sample. Click "Add" to save the configurations.

VARIABLE	VALUE
studio_base_url	SenseStudio server base URL. E.g. https://hostserver
username	The API account username
password	The API account password
token	Access token retrieved from /GUNS/mgr/login

Fraser studio140

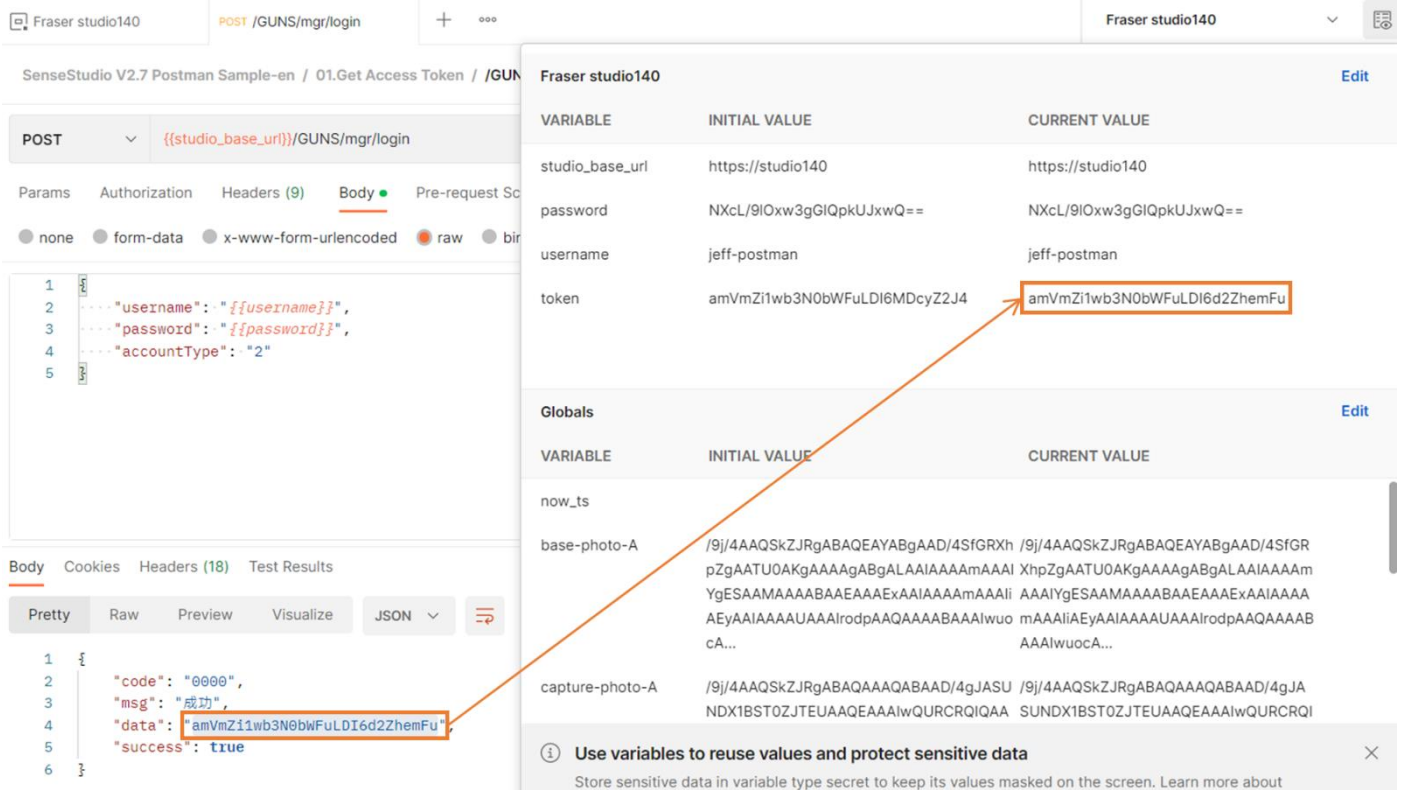
	VARIABLE	TYPE ⓘ	INITIAL VALUE ⓘ
☒	studio_base_url	default ▾	https://studio140
☒	password	default ▾	
☒	username	default ▾	
☒	token	default ▾	

Step 6. Select the new created environment for invoking SenseStudio APIs.



Obtain token and update environment variable

Step 7. Obtain the token by invoking `/GUNS/mgr/login`. Update the token to the environment setting.



The screenshot displays the SenseStudio interface for a POST request to `/GUNS/mgr/login`. The request body is a JSON object with fields `username`, `password`, and `accountType`. The response is a JSON object with fields `code`, `msg`, `data`, and `success`. The `data` field contains a token value: `"amVmZi1wb3N0bWVudDI6d2ZhemFu"`. This token value is highlighted with an orange box, and an orange arrow points from it to the `token` variable in the `Fraser studio140` environment settings table.

VARIABLE	INITIAL VALUE	CURRENT VALUE
studio_base_url	https://studio140	https://studio140
password	NXcL/9lOxw3gGIQpkUJxwQ==	NXcL/9lOxw3gGIQpkUJxwQ==
username	jeff-postman	jeff-postman
token	amVmZi1wb3N0bWVudDI6d2ZhemFu	amVmZi1wb3N0bWVudDI6d2ZhemFu

Globals

VARIABLE	INITIAL VALUE	CURRENT VALUE
now_ts		
base-photo-A	/9j/4AAQSkZJRgABAQEAYABgAAD/4SfGRXh /9j/4AAQSkZJRgABAQEAYABgAAD/4SfGRpZgAATU0AKgAAAABgABgALAAIAAAAmAAAI XhpZgAATU0AKgAAAABgABgALAAIAAAAmYgESAAIAAAABAAEAAEAAIAAAAmAAAI AAAYgESAAIAAAABAAEAAEAAIAAAAmAAIAAAAEyAAIAAAAUAAAIrodPAAQAAABAAAIwuo mAAAIiAEyAAIAAAAUAAAIrodPAAQAAABAAAIwuocA...	
capture-photo-A	/9j/4AAQSkZJRgABAQAAQABAAQ/4gJASU /9j/4AAQSkZJRgABAQAAQABAAQ/4gJANDX1BSt0ZJTEUAAQEAIAwQURCRQIAA SUNDX1BSt0ZJTEUAAQEAIAwQURCRQI	

Body Cookies Headers (18) Test Results

Pretty Raw Preview Visualize JSON

```
1 {
2   "code": "0000",
3   "msg": "成功",
4   "data": "amVmZi1wb3N0bWVudDI6d2ZhemFu",
5   "success": true
6 }
```

Use variables to reuse values and protect sensitive data

Store sensitive data in variable type secret to keep its values masked on the screen. Learn more about

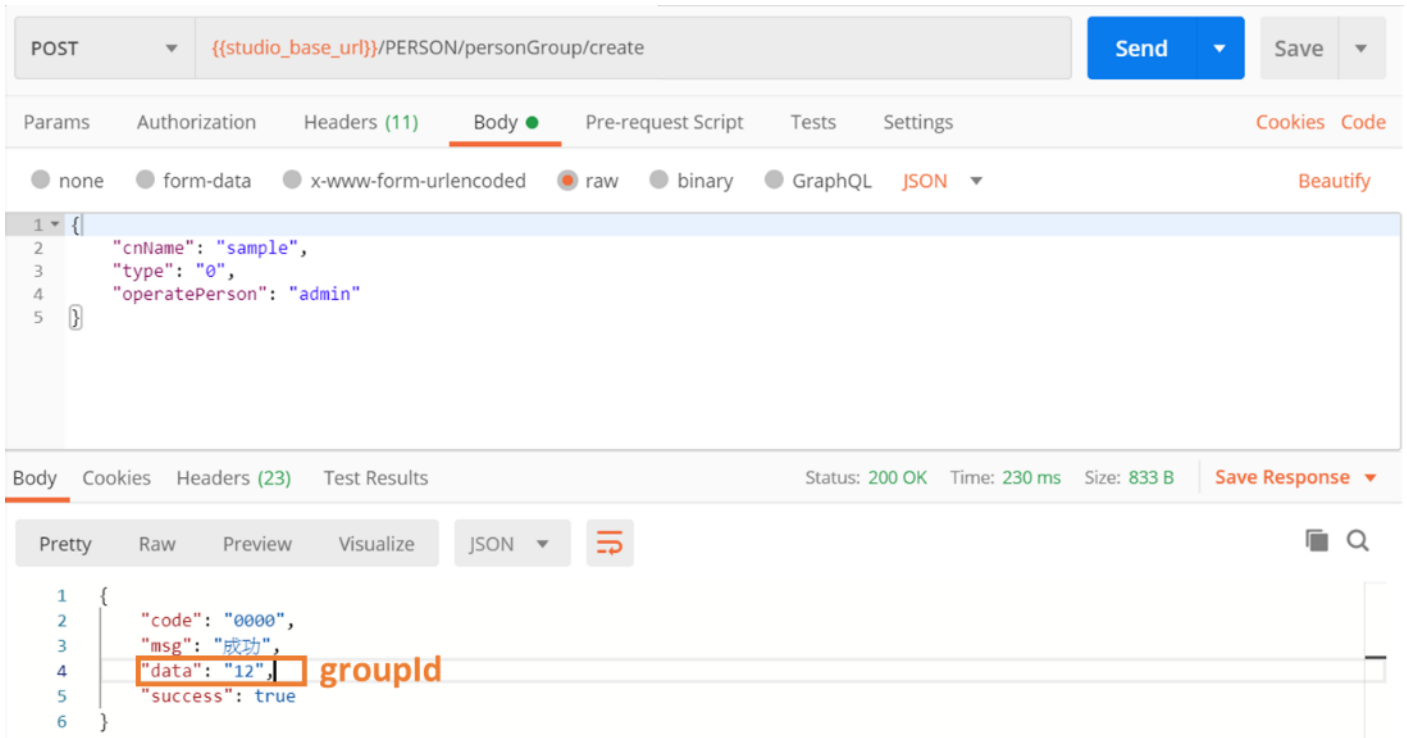
1.3.1 Face Enrollment

Step 1. Invoke `"/UTILITY/files/uploadImageBase64"` to upload face image to SenseStudio. Input `"serviceType"="person"`, SenseStudio will do required check for face images, such as quality check and face detection. You'll get an image URL in this API. Use this URL for creating a person.

Step 2. Invoke `"/PERSON/person/create"` to create a person profile.

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Step 3. Invoke `"/PERSON/personGroup/create"` to create a person group. Skip this step if the group already exists.

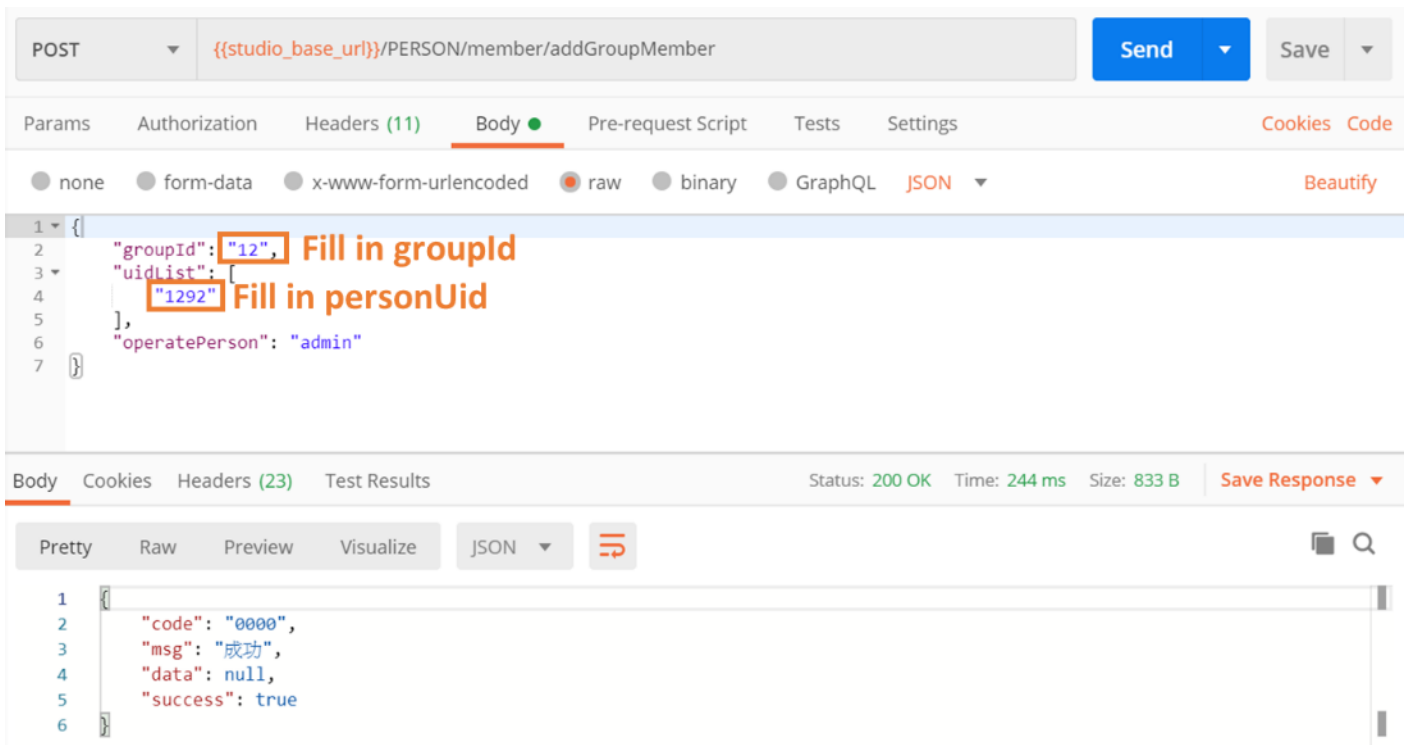


```
POST {{studio_base_url}}/PERSON/personGroup/create
```

```
{
  "cnName": "sample",
  "type": "0",
  "operatePerson": "admin"
}
```

```
{
  "code": "0000",
  "msg": "成功",
  "data": "12",
  "success": true
}
```

Step 4. Invoke `"/PERSON/member/addGroupMember"` to add the person into the group.



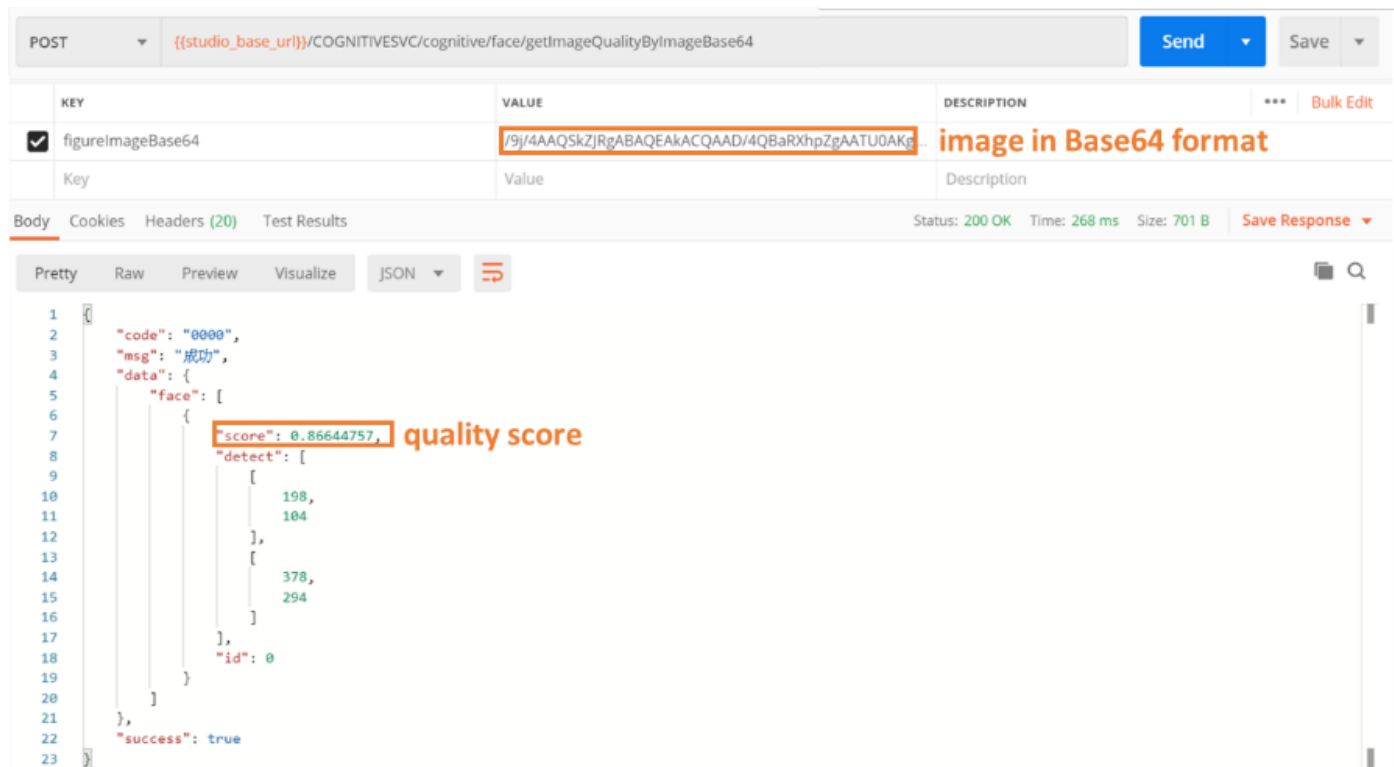
```
POST {{studio_base_url}}/PERSON/member/addGroupMember
```

```
{
  "groupId": "12",
  "uidList": [
    "1292"
  ],
  "operatePerson": "admin"
}
```

```
{
  "code": "0000",
  "msg": "成功",
  "data": null,
  "success": true
}
```

1.3.2 Photo Quality Check

Invoke `"/COGNITIVESVC/cognitive/face/getImageQualityByImageBase64"` to get image quality score.



The screenshot shows the SenseStudio API interface for the endpoint `POST {{studio_base_url}}/COGNITIVESVC/cognitive/face/getImageQualityByImageBase64`. The request body is a JSON object with the following structure:

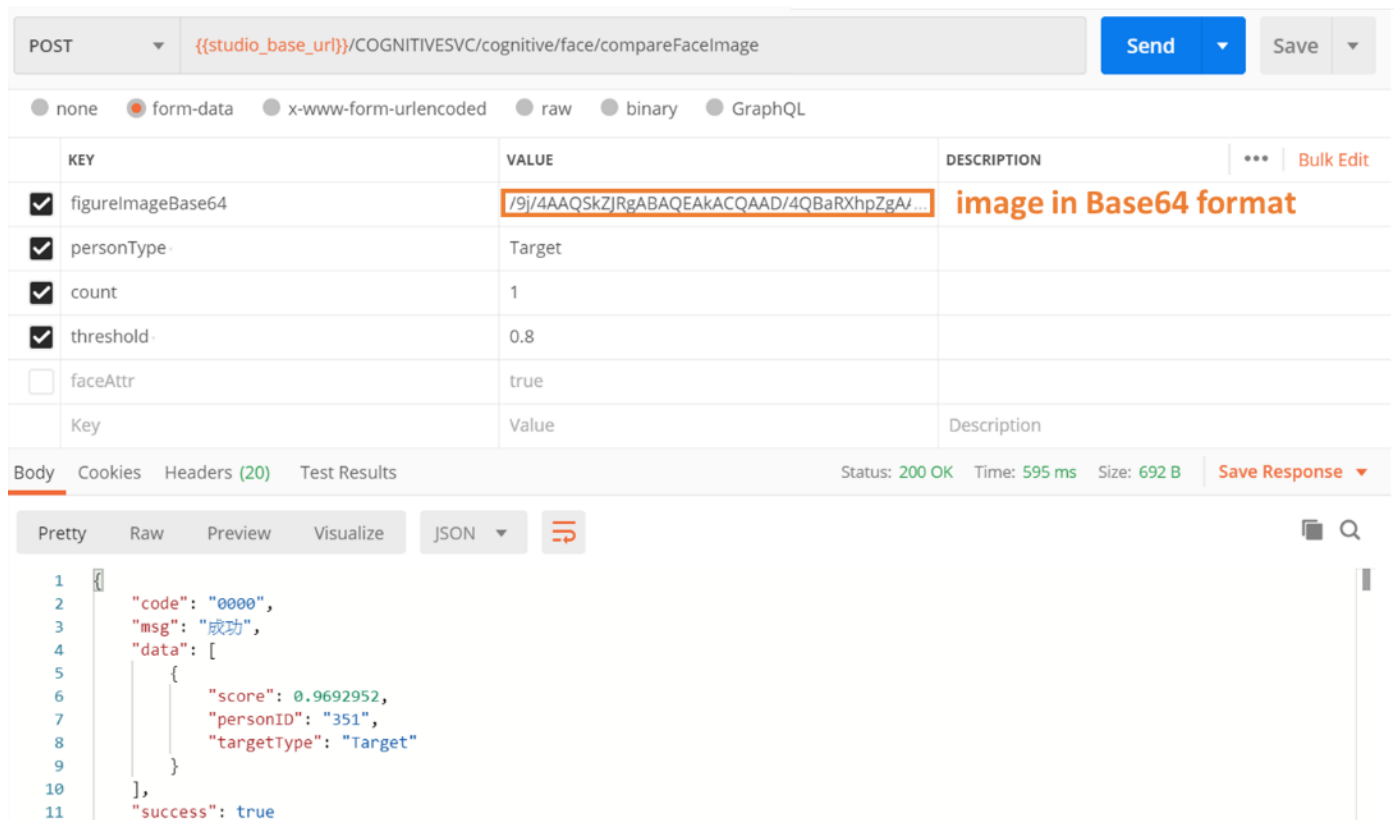
```
{  "code": "0000",  "msg": "成功",  "data": {    "face": [      {        "score": 0.86644757,        "detect": [          [            198,            104          ],          [            378,            294          ]        ],        "id": 0      }    ]  },  "success": true}
```

The response status is 200 OK, Time: 268 ms, Size: 701 B. The response body is a JSON object with the following structure:

```
{  "code": "0000",  "msg": "成功",  "data": {    "face": [      {        "score": 0.86644757,        "detect": [          [            198,            104          ],          [            378,            294          ]        ],        "id": 0      }    ]  },  "success": true}
```

1.3.3 1:N Search

Invoke `"/COGNITIVESVC/cognitive/face/compareFacelImage"` to do 1:N search.



The screenshot shows the SenseStudio API interface for the endpoint `POST {{studio_base_url}}/COGNITIVESVC/cognitive/face/compareFacelImage`. The request body is a JSON object with the following structure:

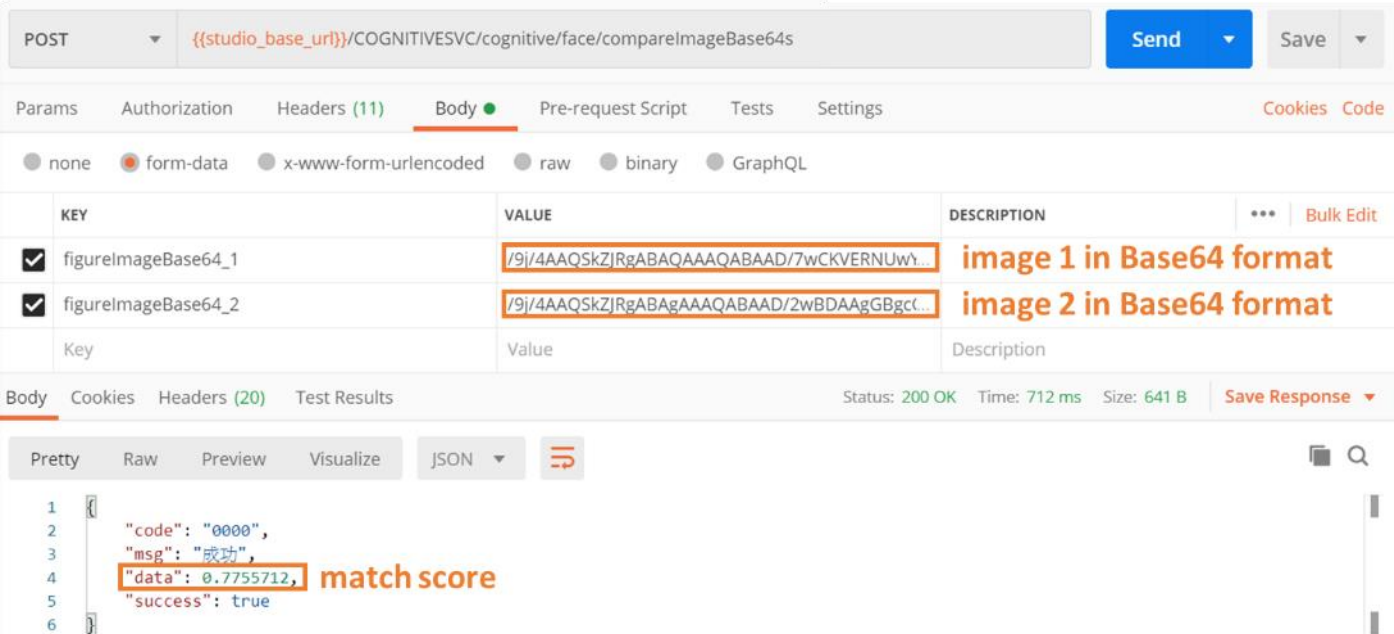
```
{  "code": "0000",  "msg": "成功",  "data": {    "face": [      {        "score": 0.9692952,        "personID": "351",        "targetType": "Target"      }    ]  },  "success": true}
```

The response status is 200 OK, Time: 595 ms, Size: 692 B. The response body is a JSON object with the following structure:

```
{  "code": "0000",  "msg": "成功",  "data": {    "face": [      {        "score": 0.9692952,        "personID": "351",        "targetType": "Target"      }    ]  },  "success": true}
```

1.3.4 1:1 Compare

Invoke "/COGNITIVESVC/cognitive/face/compareImageBase64s" to do 1:1 compare.



POST `{{studio_base_url}}/COGNITIVESVC/cognitive/face/compareImageBase64s` Send Save

Params Authorization Headers (11) **Body** Pre-request Script Tests Settings Cookies Code

none form-data x-www-form-urlencoded raw binary GraphQL

KEY	VALUE	DESCRIPTION
☒ figureImageBase64_1	/9j/4AAQSkZJRgABAQAAQABAAQ/7wCKVERNWY...	image 1 in Base64 format
☒ figureImageBase64_2	/9j/4AAQSkZJRgABAQAAQABAAQ/2wBDAAGBgC...	image 2 in Base64 format
Key	Value	Description

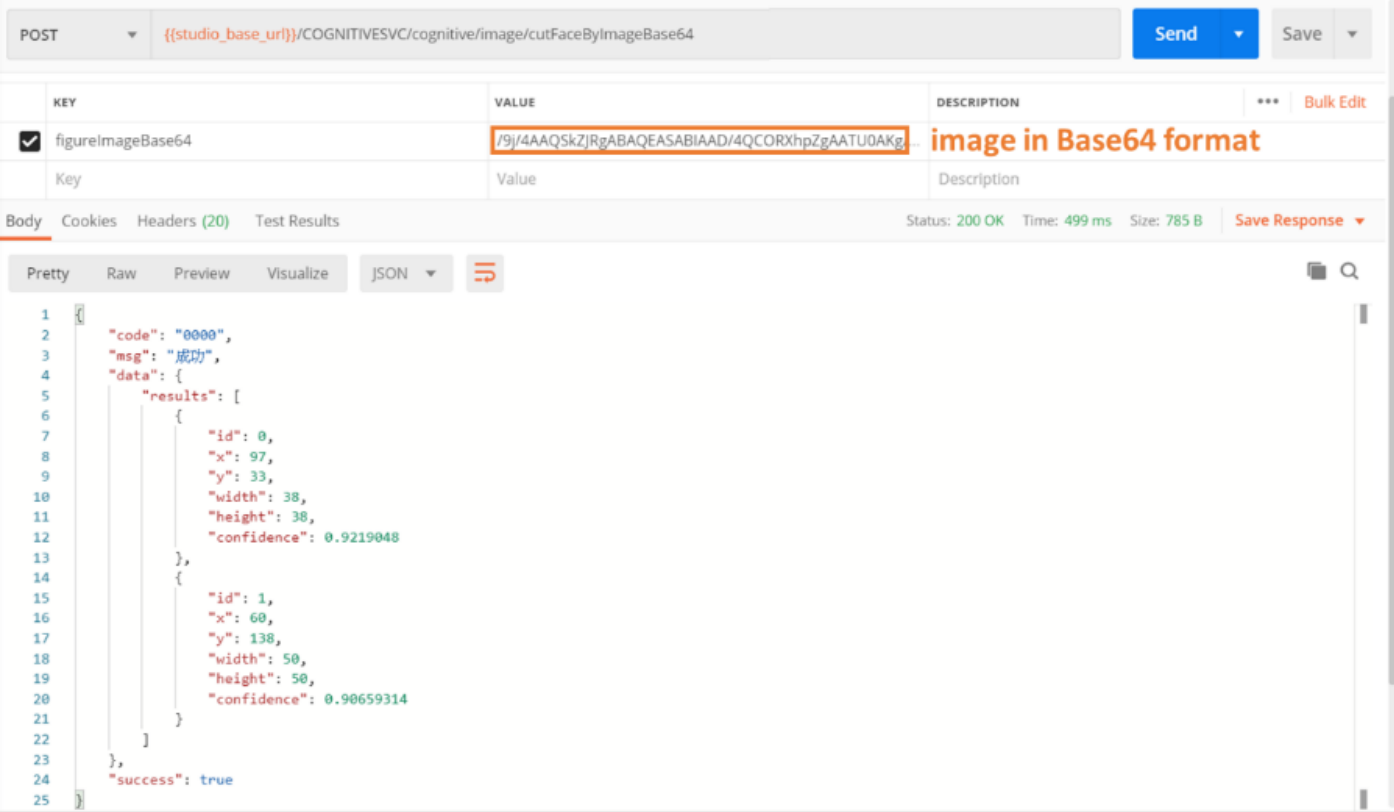
Body Cookies Headers (20) Test Results Status: 200 OK Time: 712 ms Size: 641 B Save Response

Pretty Raw Preview Visualize JSON

```
1 {
2   "code": "0000",
3   "msg": "成功",
4   "data": 0.7755712, match score
5   "success": true
6 }
```

1.3.5 Face Detection

Invoke "/COGNITIVESVC/cognitive/image/cutFaceByImageBase64" to detect faces in an image. This API will respond all the detected faces with coordinates.



POST `{{studio_base_url}}/COGNITIVESVC/cognitive/image/cutFaceByImageBase64` Send Save

KEY	VALUE	DESCRIPTION
☒ figureImageBase64	/9j/4AAQSkZJRgABAQEASABIAAD/4QCORXhpZgAATU0AKg...	image in Base64 format
Key	Value	Description

Body Cookies Headers (20) Test Results Status: 200 OK Time: 499 ms Size: 785 B Save Response

Pretty Raw Preview Visualize JSON

```
1 {
2   "code": "0000",
3   "msg": "成功",
4   "data": {
5     "results": [
6       {
7         "id": 0,
8         "x": 97,
9         "y": 33,
10        "width": 38,
11        "height": 38,
12        "confidence": 0.9219048
13      },
14      {
15        "id": 1,
16        "x": 60,
17        "y": 138,
18        "width": 50,
19        "height": 50,
20        "confidence": 0.90659314
21      }
22    ]
23  },
24  "success": true
25 }
```

1.4 Common API Return Format

```
{  
  "code": 0000,           // API return code, 0000 is success  
  "msg": "return msg",    // API return message  
  "data": JSON data       // Business data returned by the API. The API response mentioned in  
                           following chapters include only this part.  
}
```

2 Person API

2.1 Person API Return Code

Return Code	Return Code Description
0000	Success
0001	Failed
02002000	The data has already existed
02002001	Fail to update the checkout result
02012002	Operating data does not exist
02012003	[PersonListReq]bad request,uidList is required
02002004	uuid and avatarimageid con not all empty
02002008	Create a failure, more than the maximum number of valid users
02002009	Fetching image feature values failed
02002010	Extract image feature value data is empty
02002011	Characteristic value is empty
02002012	The model version is empty
02002013	Batch import user excel is empty, please re-upload
02002014	Batch import abnormality for users
02002015	Batch creation in this directory
02002016	Save batch importing personnel records failed
02002017	Update batch importing personnel records failed
02002018	Folder pictures are empty
02002019	File does not exist
02002020	Picture naming format does not meet the requirements. It should be name + blank + certificate number.
02002021	The naming format does not match the requirement. There is no space between the file name and the file extension
02002022	Failure to insert personnel when batch insertion
02002023	Failure to insert pictures when batch insertion
02002024	Failure to update personnel when batch insertion
02002025	Failure to update pictures when batch insertion
02002026	Uploading file format is incorrect
02002027	Call control to remove the failure of a human group
02002028	Call monitor to remove the failure of a human group
02010001	[PersonBaseReq] cnName is required
02010001	[PersonCreateReq] cnName is required
02010001	[PersonCreateReq] sex is required

02010001	[PersonCreateReq] birthday is required
02010001	[PersonCreateReq] idNumber is required
02010001	[PersonDeactivateReq] uidList is required
02010001	[PersonDeactivateReq] operatePerson is required
02010001	[PersonListByIdNumbers] idNumber is required
02010001	[PersonUpdateReq] uid is required
02010001	[PersonUpdateReq] operatePerson is required
02010001	[PersonUpdateReq] isUpdatelImage is required
02010001	[AssignMemberReq]groupId is required
02010001	[AssignMemberReq]uidList is required
02010001	[RemoveMemberReq] groupId is required
02010001	[RemoveMemberReq] uidList is required
02010001	[ListGroupInfoReq] groupIds is required
02010001	[PersonGroupCreateReq] cnName is required
02010001	[PersonGroupCreateReq] type is required
02010001	[PersonGroupDeactivateReq] groupId is required
022038	delete group fail
0202999	request deptId is required
0203999	you do not have the privilege

2.2 Person Management

2.2.1 Create Person

API Endpoint	{{studio_base_url}}/PERSON/person/create			
Method	POST			
Request Format	JSON			
Request Parameters	Name	Required	Type	Description
	imageURI	Yes	String	Person image path Get this URI from 7.2.1 or 7.2.2
	cnName	Yes	String	Person name
	idNumber	Yes	String	Identification number
	operatePerson	Yes	String	Operator
	personTag	No	String	User-defined person tag
	enName	No	String	Person nick name
	sex	No	String	Gender(male: 0, female: 1)
	birthday	No	String	Birthday, format: yyyyMMdd
	desc	No	String	Description

	personType	No	String	Person type
	personFeature	No	String	Person feature
	documentId	No	String	Document ID
	externalUid	No	String	External primary key
	idCountry	No	String	ID issuing country
	idExpiryDate	No	String	ID expiration date Format: yyyy-MM-dd HH:mm:ss
	idType	No	String	ID type 1: ID Card; 2: Passport; 3: Driving License; 4: FIN; 5: Others
	remark	No	String	Remark
	privilege	No	String	Privilege (Department ID)
	expireDate	No	String	Person expiration date Format: yyyy-MM-dd HH:mm:ss
	phone	No	String	Phone number
	externalInfo	No	Object	Additional information of the profile
	enOrDisableReason	No	String	Reason of activation/deactivation
Response Parameters	Name	Required	Type	Description
	data	Yes	String	Person unique ID, generated by the system, be used to uniquely identify the person
Sample Request	<pre>{ "imageURI": "/images/person/20200416/144/49273ec38a0e92e9fbbf22b37f4adbf0.png", "cnName": "John", "idNumber": "0001", "operatePerson": "1", "personTag": "", "enName": "", "sex": "", "birthday": "", "desc": "", "personType": "", "personFeature": "", "modelVersion": "", "documentId": "", "externalUid": "", "idCountry": "", "idExpiryDate": "",</pre>			

	<pre>"idType": "", "remark": "", "privilege": "", "expireDate": "", "phone": "" }</pre>
Sample Response	<pre>{ "code": "0000", "msg": "成功", "data": "1436", "success": true }</pre>

2.2.2 Deactivate Person (Soft Delete)

API Endpoint	{{studio_base_url}}/PERSON/person/deactivate			
Method	POST			
Request Format	JSON			
Request Parameters	Name	Required	Type	Description
	uidList	No	Array[String]	Set of person unique IDs
	operatePerson	Yes	String	Operator
	personTag	No	String	User-defined person tag
	timeField	No	String	Filter person by create time or last modified time 1: createTs, 2: lastModTs Default is 1 (createTs)
	startTime	No	String	Deactivate persons whose createTs/lastModTs(depends on timeField) is after this timing. Format: yyyy-MM-dd HH:mm:ss
	endTime	No	String	Deactivate persons whose createTs/lastModTs(depends on timeField) is before this timing. Format: yyyy-MM-dd HH:mm:ss
	beforeDays	No	String	Deactivate persons whose createTs/lastModTs(depends on timeField) is before "beforeDays"
	expireDate	No	String	Person expiration date

				Format: yyyy-MM-dd HH:mm:ss
Sample Request	{ "uidList": ["1","2"], "operatePerson": "1" }			
Sample Response	{ "code": "0000", "msg": "成功", "data": null, "success": true }			

2.2.3 Delete Person (Hard Delete)

API Endpoint	{{studio_base_url}}/PERSON/person/delete			
Method	POST			
Request Format	JSON			
Request Parameters	Name	Required	Type	Description
	uuids	Yes	Array[String]	Set of person unique IDs
	startTime	No	String	Delete persons whose create time is after this timing. Format: yyyy-MM-dd HH:mm:ss
	endTime	No	String	Delete persons whose create time is after this timing. Format: yyyy-MM-dd HH:mm:ss
	beforeDays	No	String	Delete persons whose create time is before "beforeDays"
	personTag	No	String	User-defined person tag
	expireDate	No	String	Person expiration date Format: yyyy-MM-dd HH:mm:ss
Sample Request	{ "beforeDays": "", "endTime": "2020-03-11 23:59:59", "personTag": "", "expireDate": "", "startTime": "2020-03-11 00:00:00", "uuids": ["4"] }			

	<pre>] }</pre>
Sample Response	<pre>{ "code": "0000", "msg": "成功", "data": null, "success": true }</pre>

2.2.4 Update Person

API Endpoint	{{studio_base_url}}/PERSON/person/update			
Method	POST			
Request Format	JSON			
Request Parameters	Name	Required	Type	Description
	uid	Yes	String	Person unique ID
	isUpdateImage	Yes	String	Whether to update person image(Yes:0, No:1)
	imageURI	Yes	String	Person image path
	idNumber	Yes	String	Identification Number
	operatePerson	Yes	String	Operator
	sts	No	String	Status(Valid: 0, Invalid: 1)
	cnName	Yes	String	Person name
	enName	No	String	Person nick name
	sex	No	String	Gender(male: 0, female: 1)
	birthday	No	String	Birthday, format: yyyyymmdd
	desc	No	String	Description
	personType	No	String	Person type
	personFeature	No	String	Person feature
	modelVersion	No	String	Model version
	documentId	No	String	Document ID
	personTag	No	String	User-defined person tag
	externalUid	No	String	External primary key
	idCountry	No	String	ID issuing country
	idExpiryDate	No	String	ID expiration date Format: yyyy-MM-dd HH:mm:ss
	remark	No	String	Remark

	idType	No	String	ID type 1: ID Card; 2: Passport; 3: Driving License; 4: FIN; 5: Others
	expireDate	No	String	Person expiration date Format: yyyy-MM-dd HH:mm:ss
	privilege	No	String	Phone number
	phone	No	String	Privilege (Department ID)
	externalInfo	No	Object	Additional information of the profile
	welcomeWords	No	String	Welcome message
	enOrDisableReason	No	String	Reason of activation/deactivation
Response Parameters	Name	Required	Type	Description
	data	Yes	String	Person unique ID
Sample Request	<pre>{ "imageURI": "/images/person/20200311/93/eae3d34d7071a6bc54dbc4e44a47c643.jpg", "isUpdateImage": "0", "cnName": "Peter", "idNumber": "0002", "birthday": "", "desc": "", "documentId": "", "enName": "", "externalUid": "", "idCountry": "", "idExpiryDate": "", "idType": "", "modelVersion": "", "operatePerson": "1", "personFeature": "", "personTag": "", "personType": "", "sex": "", "sts": "", "uid": "4" }</pre>			
Sample Response	<pre>{ "code": "0000",</pre>			

```

    "msg": "成功",
    "data": "11",
    "success": true
  }

```

2.2.5 List Person (Pagination)

API Endpoint	{studio_base_url}/PERSON/person/pageList			
Method	POST			
Request Format	JSON			
Request Parameters	Name	Required	Type	Description
	current	Yes	Integer	Query results on page #. Start from 1.
	size	Yes	Integer	Number of results per page
	cnName	No	String	Person name
	enName	No	String	Person nick name
	sex	No	String	Gender(male:0, female: 1)
	birthday	No	String	Birthday, format: yyyyMMdd
	idNumber	No	String	Identification Number
	desc	No	String	Description
	sts	No	String	Status(Valid: 0, Invalid: 1)
	personTag	No	String	User-defined person tag
	externalUid	No	String	External primary key
	createTsEnd	No	String	End time of the createTs (yyyy-MM-dd HH:mm:ss)
	createTsStart	No	String	Start time of the createTs (yyyy-MM-dd HH:mm:ss)
	lastModTsEnd	No	String	End time of the lastModTs (yyyy-MM-dd HH:mm:ss)
	lastModTsStart	No	String	Start time of the lastModTs (yyyy-MM-dd HH:mm:ss)
	documentId	No	String	Document ID
Response Parameters	Name	Required	Type	Description
	data	Yes	Array[Person]	Person list
Person	externalUid		String	External primary key
	idCountry		String	ID issuing country

	idExpiryDate		String	ID expiration date Format: yyyy-MM-dd HH:mm:ss
	idType		String	ID type 1: ID Card 2: Passport 3: Driving License 4: FIN 5: Others
	remark		String	Remark
	uid		String	Person unique ID
	cnName		String	Person name
	enName		String	Person nick name
	sts		String	Status(Valid: 0, Invalid: 1)
	type		String	User type
	personTag		String	User-defined person tag
	feature		String	User characteristic
	modelVersion		String	Model version
	idNumber		String	Identification Number
	imageURI		String	Person image path
	sex		String	Gender(male: 0, female: 1)
	documentId		String	Document ID
	desc		String	Description
	birthday		String	Birthday, format: yyyyMMdd
	groupIdNameList		Array[GroupIdName]	
	lastModTs		String	Last modified time Format: yyyy-MM-dd HH:mm:ss
	createTs		String	Created time Format: yyyy-MM-dd HH:mm:ss
	imageLastModTs		String	Image last modified time Format: yyyy-MM-dd HH:mm:ss
	privilege		String	Privilege (Department ID)
	expireDate		String	Person expiration date Format: yyyy-MM-dd HH:mm:ss
	externalInfo		Object	Additional information of the profile
GroupIdName	groupId		String	Person Group ID

	groupCnName		String	Chinese name of person group
	groupEnName		String	English name of person group
Sample Request	<pre>{ "birthday": "", "cnName": "", "current": 1, "desc": "", "enName": "", "externalUid": "", "idNumber": "", "personTag": "", "sex": "", "size": 10, "sts": "", "uid": "", "remark": "" }</pre>			
Sample Response	<pre>{ "data":[{ "externalUid": null, "idCountry": null, "idExpiryDate": null, "idType": null, "remark": "", "uid":"657166", "cnName":"Real_217972", "enName": "", "sts":"0", "type":"0", "personTag": "", "feature": "", "modelVersion": "kestrel", "idNumber":"1545014640342-889046", "imageURI":"/images/person/20181217/83/2f54db34bda0c6a8d1576812b82ad1d3.jpg", "sex":"0", "documentId": "" }] }</pre>			

```

"desc": "description",
"birthday": "20181127",
"groupIdNameList": [

],
"lastModTs": "2018-12-17 10:44:01",
"createTs": "2020-04-01 14:22:13",
"imageLastModTs": "2020-04-01 14:22:13",
"privilege": "0",
"expireDate": null
}
],
"size": 1,
"total": 302025,
"code": "0000",
"msg": "成功"
}

```

2.2.6 List Person By UIDs

API Endpoint	{{studio_base_url}}/PERSON/person/list			
Method	POST			
Request Format	JSON			
Request Parameters	Name	Required	Type	Description
	uidList	Yes	Array[String]	List of person unique IDs
Response Parameters	Name	Required	Type	Description
	data		Array[Person]	Person list
Person	uid		String	Person unique ID
	cnName		String	Person name
	enName		String	Person nick name
	sts		String	Status(Valid: 0, Invalid: 1)
	type		String	Person type
	feature		String	Person feature
	modelVersion		String	Model version
	documentId		String	Document ID
	personTag		String	User-defined person tag
	idNumber		String	Identification Number

	imageURI		String	Image URI
	sex		String	Gender (0: male, 1: female)
	birthday		String	Birthday Format: yyyyMMdd
	createTs		String	Created time Format: yyyy-MM-ddTHH:mm:ss.sssZ
	lastModTs		String	Last modified time Format: yyyy-MM-dd HH:mm:ss
	desc		String	Description
	avatarId		String	Person image ID
	externalUid		String	External primary key
	idCountry		String	ID issuing country
	idExpiryDate		String	ID expiration date Format: yyyy-MM-dd HH:mm:ss
	idType		String	ID type 1: ID Card 2: Passport 3: Driving License 4: FIN 5: Others
	remark		String	Remark
	privilege		String	Privilege (Department ID)
	expireDate		String	Person expiration date Format: yyyy-MM-dd HH:mm:ss
	externalInfo		Object	Additional information of the profile
Sample Request	<pre>{ "uidList": ["53"] }</pre>			
Sample Response	<pre>{ "code": "0000", "msg": "成功", "data": [{ "uid": "1895", "cnName": "Maria", </pre>			

```
"enName": "",
"sts": "0",
"type": "0",
"feature": null,
"modelVersion": null,
"documentId": "",
"personTag": "",
"idNumber": "A123456789",
"imageURI":
"/images/person/20200628/69/763b1bcc79e658beb572d86f28544caa.jpg",
"sex": "1",
"birthday": "19900316",
"createTs": "2020-06-28T09:46:10.000+0000",
"lastModTs": "2020-06-28 17:46:10",
"desc": "",
"avatarId": "1277176456497074178",
"externalUid": null,
"idCountry": "202",
"idExpiryDate": "2026-06-30 00:00:00",
"idType": "1",
"remark": null,
"privilege": null,
"expireDate": null
    }
  ],
  "success": true
}
```

2.2.7 Query Person Information by Person ID

API Endpoint	{{studio_base_url}}/PERSON/person/info			
Method	GET			
Request Format	form-data OR parameters in URL			
Request Parameters	Name	Required	Type	Description
	uid	Yes	String	Person unique ID
Response Parameters	Name	Required	Type	Description
	data		PersonInfo	

PersonInfo	idNumber		String	ID number
	documentId		String	Document ID
	sex		String	Gender(male: 0, female: 1)
	birthday		String	Birthday, format: yyyyMMdd
	imageURI		String	Person image path
	desc		String	Description
	uid		String	Person unique ID
	cnName		String	Person name
	enName		String	Person nick name
	type		String	Person type
	personTag		String	Person tag
	feature		String	Person feature
	modelVersion		String	Model version
	sts		String	Status(Valid: 0, Invalid: 1)
	lastModTs		String	Last modified time
	externalUid		String	External primary key
	idCountry		String	ID issuing country
	idExpiryDate		String	ID expiration date Format: yyyy-MM-dd HH:mm:ss
	idType		String	ID type 1: ID Card; 2: Passport; 3: Driving License; 4: FIN; 5: Others
	remark		String	Remark
	privilege		String	Privilege (Department ID)
	expireDate		String	Person expiration date Format: yyyy-MM-dd HH:mm:ss
	externalInfo		Object	Additional information of the profile
Sample Request	{{API Endpoint}}?uid=83			
Sample Response	{ "code": "0000", "msg": "成功", "data": { "idNumber": "A123456789", "documentId": "",			

```
"sex": "1",
"birthday": "19900316",
"imageURI":
"/images/person/20200628/69/763b1bcc79e658beb572d86f28544caa.jpg",
"desc": "",
"uid": "1895",
"cnName": "Maria",
"enName": "",
"type": "0",
"personTag": "",
"feature": null,
"modelVersion": null,
"sts": "0",
"lastModTs": "2020-06-28 17:50:33",
"externalUid": null,
"idCountry": "202",
"idExpiryDate": "2026-06-30 00:00:00",
"idType": "1",
"remark": null,
"privilege": "0",
"expireDate": "2020-06-30 00:00:00"
},
"success": true
}
```

2.2.8 Query Group of Person

API Endpoint	{{studio_base_url}}/PERSON/person/queryPersonGroupIds			
Method	GET			
Request Parameters	Name	Required	Type	Description
	uid	Yes	Integer	Person unique ID
Response Parameters (data)	Name	Required	Type	Description
	personGroupIds		Array[String]	Group IDs where the person belongs to
Sample Request	{{API Endpoint}}?uid=11			
Sample Response	{ "code": "0000", "msg": "成功",			

```
"data": {  
  "personGroupIds": [  
    "1"  
  ]  
},  
"success": true  
}
```

2.3 Group Management

2.3.1 Create Person Group

API Endpoint	{{studio_base_url}}/PERSON/personGroup/create			
Method	POST			
Request Format	JSON			
Request Parameters	Name	Required	Type	Description
	cnName	Yes	String	Group name
	type	Yes	String	Group type, fixed to "0"
	operatePerson	Yes	String	Operator
	enName	No	String	Group nick name
	desc	No	String	Description
Response Parameters	Name	Required	Type	Description
	data		String	Group ID, generated by the system, be used to uniquely identify the group
Sample Request	{ "cnName": "Sample Group 2", "desc": "test group", "enName": "", "operatePerson": "developer A", "type": "0" }			
Sample Response	{ "code": "0000", "msg": "成功", "data": "5", "success": true }			

2.3.2 Deactivate Person Group

API Endpoint	{{studio_base_url}}/PERSON/personGroup/deactivate			
Method	POST			
Request Format	JSON			
Request Parameters	Name	Required	Type	Description
	groupId	Yes	String	Group ID
	operatePerson	Yes	String	Operator
Response Parameters	Name	Required	Type	Description
	data		String	Group ID
Sample Request	{ "groupId": "1", "operatePerson": "1" }			
Sample Response	{ "code": "0000", "msg": "成功", "data": "1", "success": true }			

2.3.3 Update Person Group

API Endpoint	{{studio_base_url}}/PERSON/personGroup/update			
Method	POST			
Request Format	JSON			
Request Parameters	Name	Required	Type	Description
	groupId	Yes	String	Group ID
	operatePerson	No	String	Operator
	cnName	No	String	Group name
	type	No	String	Group type, fixed to "0"
	enName	No	String	Group nick name
	desc	No	String	Description
Response Parameters	Name	Required	Type	Description
	data		String	Group ID
Sample Request	{ "groupId": "5", "cnName": "Group with new name", }			

	<pre>"desc": "update the name through API", "enName": "", "operatePerson": "1", "type": "0" }</pre>
Sample Response	<pre>{ "code": "0000", "msg": "成功", "data": "5", "success": true }</pre>

2.3.4 List Person Group (Pagination)

API Endpoint	{{studio_base_url}}/PERSON/personGroup/list			
Method	POST			
Request Format	JSON			
Request Parameters	Name	Required	Type	Description
	current	Yes	Integer	Query results on page #. Start from 1.
	size	Yes	Integer	Number of results per page
	cnName	No	String	Group name
	enName	No	String	Group nick name
	type	No	String	Group type, fixed to "0"
	sts	No	String	Status(Valid: 0, Invalid: 1)
	desc	No	String	Description
Request Parameters	Name	Required	Type	Description
	data		Array[Group]	Person group information List
Group	groupId		String	Person group ID
	cnName		String	Group name
	enName		String	(Not in used for now)
	type		String	Person group type
	createTs		String	Created time
	lastModTs		String	Last modified time
	privilege		String	Privilege (Department ID)
	sts		String	Status(Valid: 0, Invalid: 1)
	memberCount		String	Number of members in the group

Sample Request	<pre>{ "current": 1, "size": 10, "cnName": "Team", "desc": "", "enName": "", "type": "" }</pre>
Sample Response	<pre>{ "data": [{ "privilege": "0", "groupId": "3", "cnName": "Test Team", "enName": null, "type": "0", "sts": "0", "memberCount": "3", "createTs": "2021-06-14 14:59:19", "lastModTs": "2021-06-14 14:59:58", "inGroupValidHour": "" }], "size": 10, "total": 1, "code": "0000", "msg": "成功" }</pre>

2.4 Member Management

2.4.1 Add Group Members

API Endpoint	{{studio_base_url}}/PERSON/member/addGroupMember			
Method	POST			
Request Format	JSON			
Request Parameters	Name	Required	Type	Description
	groupId	Yes	String	Group ID
	operatePerson	Yes	String	Operator

	uidList	No	Array[String]	Set of person unique IDs
Response Parameters	Name	Required	Type	Description
	data		String	null
Sample Request	{ "groupId": "5", "uidList": ["9", "10"], "operatePerson": "admin" }			
Sample Response	{ "code": "0000", "msg": "成功", "data": null, "success": true }			

2.4.2 Add Person to Multiple Groups

API Endpoint	{{studio_base_url}}/PERSON/member/addMemberToGroups			
Method	POST			
Request Format	JSON			
Request Parameters	Name	Required	Type	Description
	uid	Yes	String	Person unique ID
	groupIdList	Yes	Array[String]	Group ID
	operatePerson	Yes	String	Operator
	privilege	Yes	String	Privilege (Department ID)
Response Parameters	Name	Required	Type	Description
	data		String	null
Sample Request	{ "uid": "12", "groupIdList": ["5", "6"], }			

	<pre>"operatePerson": "1" }</pre>
Sample Response	<pre>{ "code": "0000", "msg": "成功", "data": null, "success": true }</pre>

2.4.3 Remove Group Members

API Endpoint	{{studio_base_url}}/PERSON/member/remove			
Method	POST			
Request Format	JSON			
Request Parameters	Name	Required	Type	Description
	groupId	Yes	String	Group ID
	uidList	Yes	Array[String]	Person unique ID
	operatePerson	Yes	String	Operator
Response Parameters	Name	Required	Type	Description
	data		String	null
Sample Request	<pre>{ "groupId": "2", "uidList": ["15", "10"], "operatePerson": "1" }</pre>			
Sample Response	<pre>{ "code": "0000", "msg": "成功", "data": null, "success": true }</pre>			

2.4.4 Remove Person from Multiple Groups

API Endpoint	{{studio_base_url}}/PERSON/member/removeMemberFromGroups			
Method	POST			
Request Format	JSON			
Request Parameters	Name	Required	Type	Description
	uid	Yes	String	Person unique ID

	groupIdList	Yes	Array[String]	Group ID
	operatePerson	Yes	String	Operator
Response Parameters	Name	Required	Type	Description
	data		String	null
Sample Request	{ "uid": "2", "groupIdList": ["15", "10"], "operatePerson": "1" }			
Sample Response	{ "code": "0000", "msg": "成功", "data": null, "success": true }			

2.4.5 List Member (Pagination)

API Endpoint	{{studio_base_url}}/PERSON/member/pagelist			
Method	POST			
Request Format	JSON			
Request Parameters	Name	Required	Type	Description
	current	Yes	Integer	Query results on page #. Start from 1.
	size	Yes	Integer	Number of results per page
	groupIdList	Yes	Array[String]	List of group IDs
	cnName	No	String	Person name
	enName	No	String	Person nick name
	sex	No	String	Gender(male: 0, female: 1)
	birthday	No	String	Birthday, format: yyyyymmdd
	idNumber	No	String	Identification Number
	documentId	No	String	Document ID
Response Parameters	Name	Required	Type	Description
	data		Array[Member]	
Member	idNumber		String	Identification Number
	imageURI		String	Person image path
	lastModTs		String	Last modified time

	sex		String	Gender(male: 0, female: 1)
	birthday		String	Birthday, format: yyyyymmdd
	sts		String	Status(Valid: 0, Invalid: 1)
	idCountry		String	ID issuing country
	idExpiryDate		String	ID expiration date Format: yyyy-MM-dd HH:mm:ss
	type		String	Person type
	personTag		String	User-defined person tag
	idType		String	ID type 1: ID Card; 2: Passport; 3: Driving License; 4: FIN; 5: Others
	externalUid		String	External primary key
	documentId		String	Document ID
	uid		String	Person unique ID
	cnName		String	Person name
	enName		String	Person nick name
	groupIdNameList		Array[GroupInfo]	
GroupInfo	groupId		String	Person group ID
	groupCnName		String	Group name
	groupEnName		String	(Not in used for now)
Sample Request	<pre>{ "groupIdList": ["5"], "current": "1", "size": "1" }</pre>			
Sample Response	<pre>{ "data": [{ "birthday": "19960115", "idType": "", "sex": "1", "lastModTs": "2021-06-14 14:50:56", "idNumber": "A008", "type": "0", "idCountry": "106", "uid": 9, </pre>			

```
"groupIdNameList": [  
  {  
    "groupId": 1,  
    "groupCnName": "staff"  
  },  
  {  
    "groupId": 2,  
    "groupCnName": "VIP"  
  },  
  {  
    "groupId": 5,  
    "groupCnName": "DEV Team",  
    "groupEnName": ""  
  }  
],  
"imageURI":  
"/images/person/batchImport/20210614144904_import_sample/Gina A008.jpg",  
"sts": 0,  
"cnName": "Gina",  
"phone": "99990007",  
"enName": "Nick7",  
"documentId": "D007",  
"personTag": "test_tag"  
}  
],  
"size": 1,  
"total": 2,  
"code": "0000",  
"msg": "success"  
}
```

2.4.6 List Member by Group ID

API Endpoint	{{studio_base_url}}/PERSON/member/groupsMemberList			
Method	POST			
Request Format	JSON			
Request Parameters	Name	Required	Type	Description
	groupIds	Yes	Array[String]	List of group IDs
	Name	Required	Type	Description

Response Parameters	data		Array[GroupMember]	Person Group Information List
GroupMember	uid		String	Person unique ID
	groupId		String	Person group ID
	groupCnName		String	Chinese name of person group
	groupEnName		String	English name of person group
	cnName		String	Person name
	enName		String	Person nick name
	birthday		String	Birthday, format: yyyyymmdd
	idNumber		String	Identification Number
	imageURI		String	Person image path
	sex		String	Gender(male: 0, female: 1)
	sts		String	Status(Valid: 0, Invalid: 1)
	lastModTs		String	Last modified time Format: yyyy-MM-dd HH:mm:ss
	externalUid		String	External primary key
	idCountry		String	ID issuing country
	idExpiryDate		String	ID expiration date Format: yyyy-MM-dd HH:mm:ss
	idType		String	ID type 1: ID Card; 2: Passport; 3: Driving License; 4: FIN; 5: Others
	personTag		String	User-defined person tag
	type		String	Person type
Sample Request	{ "groupIds": ["2"] }			
Sample Response	{ "code": "0000", "msg": "成功", "data": [{ "uid": "1806", "groupId": "1", "groupCnName": "Staff",			

```
{
  "groupEnName": null,
  "cnName": "Amelia",
  "enName": "",
  "birthday": "",
  "idNumber": "6666",
  "imageURI":
"/images/person/20200624/166/2ff9c4a88048b68321efce7b041deaed.jpeg",
  "sex": "",
  "sts": "0",
  "lastModTs": "2020-06-24 19:14:14",
  "externalUid": null,
  "idCountry": "",
  "idExpiryDate": "",
  "idType": "",
  "personTag": "",
  "type": "0"
},
"success": true
}
```

2.4.7 Query Member IDs from Group ID

API Endpoint	{{studio_base_url}}/PERSON/member/getMemberIdsFromGroup			
Method	POST			
Request Format	JSON			
Request Parameters	Name	Required	Type	Description
	groupId	Yes	String	Group ID
Response Parameters	Name	Required	Type	Description
	data		Array[String]	All person unique IDs under the group
Sample Request	{ "groupId": "1" }			
Sample Response	{ "code": "0000", "msg": "成功", "data": ["367",] }			

```
"313",  
"578",  
"1806",  
"10",  
"76",  
"791"  
],  
"success": true  
}
```

2.5 Stranger Management

2.5.1 Delete Stranger

API Endpoint	{{studio_base_url}}/PERSON/stranger/delete			
Method	POST			
Request Format	JSON			
Request Parameters	Name	Required	Type	Description
	beforeDays	No	String	Delete strangers whose last modified time is before "beforeDays"
	endTime	No	String	Delete strangers whose last modified time is before this timing. Format: yyyy-MM-dd HH:mm:ss
	startTime	No	String	Delete strangers whose last modified time is after this timing. Format: yyyy-MM-dd HH:mm:ss
	uuids	No	Array[String]	Set of stranger unique IDs
Response Parameters	Name	Required	Type	Description
	data		String	null
Sample Request	{ "beforeDays": "", "endTime": "", "startTime": "", "uuids": ["793"] }			

Sample Response	<pre>{ "code": "0000", "msg": "成功", "data": null, "success": true }</pre>
------------------------	---

2.5.2 List Stranger

API Endpoint	{{studio_base_url}}/PERSON/stranger/pageList			
Method	POST			
Request Format	JSON			
Request Parameters	Name	Required	Type	Description
	current	Yes	Integer	Query results on page #. Start from 1.
	size	Yes	Integer	Number of results per page
	deviceSource	No	String	Device unique ID
	endAge	No	String	Age upper bound
	endTime	No	String	Whose last detected time is before this timing Format: yyyy-MM-dd HH:mm:ss
	personTag	No	String	User-defined person tag
	privilege	No	String	Privilege (Department ID)
	sex	No	String	Gender
	startAge	No	String	Age lower bound
	startTime	No	String	Whose last detected time is after this timing Format: yyyy-MM-dd HH:mm:ss
	strangerId	No	String	Stranger unique ID
Response Parameters	Name	Required	Type	Description
	data		Array[Stranger]	List of stranger data
Stranger	uuid		String	Stranger unique ID
	imageURI		String	Stranger image path
	description		String	Stranger description
	personType		String	Fix to "passer" for stranger

	personSex		String	Detected gender (male: 0, female: 1)
	personAge		String	Detected age
	personTag		String	User-defined person tag
	deviceSource		String	Device unique ID
	privilege		String	Privilege (Department ID)
	createTs		String	Stranger create time (yyyy-MM-dd HH:mm:ss)
	remark		String	Remark
Sample Request	<pre>{ "current": 1, "size": 2 }</pre>			
Sample Response	<pre>{ "data": [{ "uuid": "1452", "imageURI": "/images/cognitivesvc/20200612/f7bbe036-09fd-4b88-bc09-e188f7cbf5b3.jpg", "description": "[{"image":"/images/cognitivesvc/20200612/f7bbe036-09fd-4b88-bc09-e188f7cbf5b3.jpg","deviceId":"1242699162579898369","captureTime":"2020-06-12 15:19:18"}]", "personType": "passer", "personSex": "0", "personAge": "31", "personTag": "", "deviceSource": "1242699162579898369", "privilege": "0", "createTs": "2020-06-12 15:19:18", "remark": null }, { "uuid": "1451", "imageURI": "/images/cognitivesvc/20200612/d002af10-71d4-4416-9c9c-6a40f4544a1d.jpg", </pre>			

```
"description": "[{"image": "/images/cognitivesvc/20200612/d002af10-71d4-4416-9c9c-6a40f4544a1d.jpg", "deviceId": "1242646289435332610", "captureTime": "2020-06-12 15:02:51"}],  
  "personType": "passer",  
  "personSex": "0",  
  "personAge": "24",  
  "personTag": "",  
  "deviceSource": "1242646289435332610",  
  "privilege": "0",  
  "createTs": "2020-06-12 15:02:52",  
  "remark": null  
}  
],  
"size": 2,  
"total": 253,  
"code": "0000",  
"msg": "success"  
}
```

2.5.3 Update Stranger

API Endpoint	{{studio_base_url}}/PERSON/stranger/update			
Method	POST			
Request Format	JSON			
Request Parameters	Name	Required	Type	Description
	strangerId	Yes	String	Stranger unique ID
	deviceSource	No	String	Device unique ID
	personType	No	String	Fix to "passer" for stranger
	personTag	No	String	User-defined person tag
	personSex	No	String	Gender (male: 0, female: 1)
	desc	No	String	Stranger description
Response Parameters	Name	Required	Type	Description
	data		String	Stranger unique ID
Sample Request	{ "strangerId": "793",			

	<pre>"personSex": "0" }</pre>
Sample Response	<pre>{ "code": "0000", "msg": "成功", "data": "793", "success": true }</pre>

3 Device API

3.1 Device API Return Code

Return Code	Return Code Description	Possible Causes	Solution Suggestion
0000	Success		
01001000	device serialNo has already existed	Duplicate device serial	Modify device serial
01001001	exceed the upper limit of the number of devices	Exceed the limit of device creation	Remove unused device
01001002	create device failed	Invalid input parameter / Database exception / Application exception	Investigate the cause by checking the backend logs
01001003	update device failed	Invalid input parameter / Database exception / Application exception	Investigate the cause by checking the backend logs
01001004	deactivate device failed	Invalid input parameter / Database exception / Application exception	Investigate the cause by checking the backend logs
01011005	date format error	Invalid input date format	Modify date to correct format
01001006	deactivate device and controlPolicy relation failed	Invalid input parameter / Database exception / Application exception	Investigate the cause by checking the backend logs
01001007	deactivate device and monitorPolicy relation failed	Invalid input parameter / Database exception / Application exception	Investigate the cause by checking the backend logs
01001008	deactivate device and map relation failed	Invalid input parameter / Database exception / Application exception	Investigate the cause by checking the backend logs
01001009	this edge has already existed	Duplicate track	Do not create duplicate tracks between devices
01001010	create edge failed	Invalid input parameter / Database exception / Application exception	Investigate the cause by checking the backend logs
01001011	update edge failed	Invalid input parameter / Database exception / Application exception	Investigate the cause by checking the backend logs

01001012	deactivate edge failed	Invalid input parameter / Database exception / Application exception	Investigate the cause by checking the backend logs
01001013	fill device node into edge failed	Invalid input parameter / Database exception / Application exception	Investigate the cause by checking the backend logs
01001014	ingressAngleThreshold's format is incorrect	Invalid input angle detection threshold format	The parameter format is "xx,yy,zz". And the angle needs to be between 0 and 180.
01001015	faceAttribute need to be 0 or 1	Invalid input attribute extraction switch format	Check if the parameter is 0 or 1
01001016	livenessCheck need to be 0 or 1	Invalid input liveness check threshold format	Check if the parameter is between 0 and 1
01001017	ingressQCThreshold need to be 0~1	Invalid input quality check threshold format	Check if the parameter is between 0 and 1
01001018	frameRate need to be 0~9999	Invalid input device frame rate format	Check if the parameter is between 0 and 9999
0100999	deptId can not be empty	Missing input parameter deptId	Input deptId
01001020	device does not exist	Device not exists	Check if the device exists in DB
01001021	you don't have the privilege	No permission to query the information	Check the privilege of the caller account. And check if the device exists.
01001022	set BI device config deviceId is required	Missing input parameter deviceId	Input deviceId
010010123	remark value is an invalid json string	Remarks is an invalid JSON string	Check if the input remark is a valid JSON string
010010124	matchingThreshold need to be 0~1	Invalid input matching threshold format	Check if the parameter is between 0 and 1

3.2 Parameter Definition

3.2.1 Device Type Definition

Device Type ID	Device Type
0	Terminal
1	Keeper

2	Camera
3	GrabMachine
6	GB2818
7	Pad
9	Server
10	VirtualGate
11	VirtualId
12	VirtualMatch
14	compareCamera
15	sensePass
16	CaptureCamera-T
17	sensePass Pro
18	Face Detecting camera
19	Terminal-Visitor
20	CaptureCamera-11
21	Senseye-Region
22	Senseye-X
23	Huawei snapshot camera
24	dlc-D
25	haikang snapshot camera
26	sensegate-H
27	dahua snapshot camera
28	senseyeApp

3.2.2 Product Type Definition

Product Type ID	Product Type
0	senseID
1	senseKeeper
2	senseFace
3	senseCity
5	senseCrowd
6	DLC
7	senseGo
9	senseHello
10	ViperImage
12	SensEye
13	sensePass

14	SenseCass
15	ViperFace

3.3 Device Management

3.3.1 Create Device

API Endpoint	{{studio_base_url}}/DEVICE/device/create			
Method	POST			
Request Format	JSON			
Request Parameters	Name	Required	Type	Description
	deviceCnName	Yes	String	Device name
	deviceSerial	Yes	String	Device serial
	deviceType	Yes	String	Device type ID Please refer to 3.2.1 Device Type Definition
	deviceVersion	Yes	String	Device version
	operatePerson	Yes	String	Operator
	productType	Yes	String	Product type ID Please refer to 3.2.2 Product Type Definition
	desc	No	String	Device description
	deviceConnType	No	String	Device connection type 0: Direct connection 1: Transcoding
	deviceEnName	No	String	Device nick name
	deviceIP	No	String	Device IP
	devicePort	No	String	Device port
	deviceProtocol	No	String	Device protocol 0: TCP 1: UDP
	deviceSecret	No	String	Device login secret
	deviceTag	No	String	User-defined device tag
	deviceUri	No	String	Device URI RTSP stream URL for video camera
	matchingThreshold	No	Double	Match threshold (0-1) Default value 0.5

	streamHeight	No	String	The height of video stream Default value 1920
	streamWidth	No	String	The width of video stream Default value 1080
	faceAttribute	No	Integer	Feature extraction switch (0: on, 1: off) Default off
	livenessCheck	No	Integer	[Deprecated] Liveness check switch (0: on, 1: off)
	ingressQCThreshold	No	String	Quality check threshold (0-1) Default value 0.3
	ingressAngleThreshold	No	String	Detection angle threshold
	frameRate	No	String	Video frame rate
	bklImageStorage	No	String	Background image storing switch (0: on, 1: off) Default off
	boundingBoxSizeThreshold	No	String	Width detection threshold Default value 20
	targetGroup	No	String	Target person groups
	remark	No	String	Remark
	privilege	No	String	Privilege (Department ID)
	validRegionDefinition	No	Array[RegionEntity]	Valid region definition
RegionEntity	regionType	No	String	Region type
	regionPic	No	String	Region image
	validRegion	No	String	Region coordinates

Response Parameters	Name	Required	Type	Description
	data		String	Device unique ID, generated by the system, be used to uniquely identify the device
Sample Request	<pre>{ "deviceSerial": "DeviceSerial0612", "deviceCnName": "Device 00110", "deviceEnName": "Device 00110", "deviceSecret": "", "deviceIP": "10.5.1.15", "devicePort": "", "desc": "", "deviceConnType": "", "deviceProtocol": "0", "deviceTag": "1", "deviceType": "2", "deviceUri": "rtsp://admin:admin123@10.5.1.15:554", "deviceVersion": "1.0", "operatePerson": "1", "productType": "12", "matchingThreshold": "0.6", "streamHeight": "1080", "streamWidth": "1920", "faceAttribute": "0", "ingressQCThreshold": "0.4", "ingressAngleThreshold": "15,20,30", "frameRate": "25", "bklImageStorage": "1", "boundingBoxSizeThreshold": "20", "targetGroup": "9,12,1", "remark": "{\"name\":\"leo\"}", "privilege": "0", "validRegionDefinition": [{ "regionPic": "/images/device/20200612/107/c4ca9b04528af00ecc7d2b10874546fc.jpg", "regionType": "0",</pre>			

	<pre>"validRegion": "[[{"x": 675, "y": 72}, {"x": 447, "y": 1038}, {"x": 1089, "y": 1044}, {"x": 804, "y": 72}, {"x": 693, "y": 75}]]"</pre>
Sample Response	<pre>{ "code": "0000", "msg": "成功", "data": "1271372691009441794", "success": true }</pre>

3.3.2 Deactivate Device

API Endpoint	{{studio_base_url}}/DEVICE/device/deactivate			
Method	POST			
Request Format	JSON			
Request Parameters	Name	Required	Type	Description
	did	Yes	String	Device unique ID
	operatePerson	Yes	String	Operator
Response Parameters	Name	Required	Type	Description
	data		String	Device unique ID
Sample Request	<pre>{ "did": "1271372691009441794", "operatePerson": "1" }</pre>			
Sample Response	<pre>{ "code": "0000", "msg": "成功", "data": "1271372691009441794", "success": true }</pre>			

3.3.3 Update Device

API Endpoint	{{studio_base_url}}/DEVICE/device/update
Method	POST
Request Format	JSON

Request Parameters	Name	Required	Type	Description
	did	Yes	String	Device unique ID
	deviceCnName	No	String	Device name
	deviceSerial	No	String	Device serial
	deviceType	No	String	Device type ID Please refer to 3.2.1 Device Type Definition
	deviceVersion	No	String	Device version
	operatePerson	No	String	Operator
	productType	No	String	Product type ID Please refer to 3.2.2 Product Type Definition
	desc	No	String	Device description
	deviceConnType	No	String	Device connection type 0: Direct connection 1: Transcoding
	deviceEnName	No	String	Device nick name
	deviceIP	No	String	Device IP
	devicePort	No	String	Device port
	deviceProtocol	No	String	Device protocol 0: TCP 1: UDP
	deviceSecret	No	String	Device login secret
	deviceTag	No	String	User-defined device tag
	deviceUri	No	String	Device URI rtsp stream URL for video camera
	sts	No	String	Device status 0: Effective 1: Ineffective 3: Deleted
	matchingThreshold	No	Double	Match threshold (0- 1) Default value 0.5

	streamHeight	No	String	The height of video stream Default value 1920
	streamWidth	No	String	The width of video stream Default value 1080
	faceAttribute	No	Integer	Feature extraction switch (0: on, 1: off) Default off
	livenessCheck	No	Integer	[Deprecated] Liveness check switch (0: on, 1: off)
	ingressQCThreshold	No	String	Quality check threshold (0-1) Default value 0.3
	ingressAngleThreshold	No	String	Detection angle threshold
	frameRate	No	String	Video frame rate
	bklImageStorage	No	String	Background image storing switch (0: on, 1: off) Default off
	boundingBoxSizeThreshold	No	String	Width detection threshold Default value 20
	targetGroup	No	String	Target person groups
	remark	No	String	Remark
	privilege	No	String	Privilege (Department ID)
	validRegionDefinition	No	Array[RegionDef]	Valid region definition
	ExternalDid	No	String	External device ID
RegionDef	regionType	No	String	Region type

	regionPic	No	String	Region image
	validRegion	No	String	Region coordinates
Response Parameters	Name	Required	Type	Description
	data			

Sample Request	<pre>{ "did": "1265185833686667266", "deviceSerial": "DeviceSerial0612", "deviceCnName": "Device 00110 Update", "deviceEnName": "Device 00110", "deviceSecret": "", "deviceIP": "10.5.1.15", "devicePort": "", "desc": "", "deviceConnType": "", "deviceProtocol": "0", "deviceTag": "1", "deviceType": "2", "deviceUri": "rtsp://admin:admin123@10.5.1.15:554", "deviceVersion": "1.0", "operatePerson": "1", "productType": "12", "matchingThreshold": "0.6", "streamHeight": "1080", "streamWidth": "1920", "faceAttribute": "0", "ingressQCThreshold": "0.4", "ingressAngleThreshold": "15,20,30", "frameRate": "25", "bklImageStorage": "1", "boundingBoxSizeThreshold": "20", "targetGroup": "9,12,1", "remark": "{\"name\":\"leo\"}", "privilege": "0", "validRegionDefinition": [{</pre>
----------------	---

	<pre> "regionPic": "/images/device/20200612/107/c4ca9b04528af00ecc7d2b10874546fc.jpg", "regionType": "0", "validRegion": "[[{"x": 675, "y": 72}, {"x": 447, "y": 1038}, {"x": 1089, "y": 1044}, {"x": 804, "y": 72}, {"x": 693, "y": 75}]]"]] } </pre>
Sample Response	<pre> { "code": "0000", "msg": "成功", "data": null, "success": true } </pre>

3.3.4 Query Single Device Information by Device ID

API Endpoint	{{studio_base_url}}/DEVICE/device/info			
Method	POST			
Request Format	JSON			
Request Parameters	Name	Required	Type	Description
	did	Yes	String	Device unique ID
Response Parameters	Name	Required	Type	Description
	data		DeviceInfo	
DeviceInfo	deviceSerial		String	Device serial
	productType		String	Product type ID
	productTypeName		String	Product type name
	deviceType		String	Device type ID
	deviceTypeName		String	Device type name
	devicePort		String	Device port
	deviceIP		String	Device IP
	deviceUri		String	Device URI rtsp stream URL for video camera
	deviceVersion		String	Device version
	deviceConnType		String	Device connection type 0: Direct connection 1: Transcoding

	deviceProtocol		String	Device protocol 0: TCP 1: UDP
	deviceSecret		String	Device login secret
	deviceCnName		String	Device name
	deviceEnName		String	Device nick name
	deviceTag		String	User-defined device tag
	desc		String	Device description
	sts		String	Device status 0: Effective 1: Ineffective 3: Deleted
	matchingThreshold		String	Match threshold
	streamHeight		String	The height of video stream
	streamWidth		String	The width of video stream
	faceAttribute		String	Feature extraction switch (0: on, 1: off)
	livenessCheck		String	[Deprecated] Liveness check switch (0: on, 1: off)
	ingressQCThreshold		String	Quality check threshold
	ingressAngleThreshold		String	Detection angle threshold
	frameRate		String	Video frame rate
	privilege		String	Privilege (Department ID)
	bklImageStorage		String	Background image storing switch (0: on, 1: off)
	boundingBoxSizeThreshold		String	Width detection threshold
	targetGroup		String	Target person groups
	remark		String	Remark
	validRegionDefinition		Array[RegionDef]	Valid region definition
RegionDef	regionType	No	String	Region type

	regionPic	No	String	Region image
	validRegion	No	String	Region coordinates
Sample Request	{ "did": "1257915401329709058" }			
Sample Response	{ "code": "0000", "msg": "成功", "data": { "deviceSerial": "SPS020STDC19G07718_0", "productType": "13", "productTypeName": "sensePass", "deviceType": "15", "deviceTypeName": "sensePass", "devicePort": "0", "deviceIP": "10.161.32.32", "deviceUri": null, "deviceVersion": "1.0.0_E", "deviceConnType": null, "deviceProtocol": "", "deviceSecret": null, "deviceCnName": "SPS020STDC19G07718", "deviceEnName": "SPS020STDC19G07718", "deviceTag": null, "desc": "sample", "sts": "1", "matchingThreshold": null, "streamHeight": null, "streamWidth": null, "faceAttribute": null, "livenessCheck": null, "ingressQCThreshold": null, "ingressAngleThreshold": null, "frameRate": null, "privilege": "0", "bklImageStorage": null, "boundingBoxSizeThreshold": null, "targetGroup": null, }			

```

    "remark": null,
    "validRegionDefinition": null
  },
  "success": true
}

```

3.3.5 Query Multiple Device Information by Device ID

API Endpoint	{{studio_base_url}}/DEVICE/device/deviceInfoByDids			
Method	POST			
Request Format	JSON			
Request Parameters	Name	Required	Type	Description
	did	Yes	Array[String]	Device unique ID
Response Parameters	Name	Required	Type	Description
	data		Array[DeviceInfo]	
DeviceInfo	deviceSerial		String	Device serial
	productType		String	Product type ID
	productTypeName		String	Product type name
	deviceType		String	Device type ID
	deviceTypeName		String	Device type name
	devicePort		String	Device port
	deviceIP		String	Device IP
	deviceUri		String	Device URI rtsp stream URL for video camera
	deviceVersion		String	Device version
	deviceConnType		String	Device connection type 0: Direct connection 1: Transcoding
	deviceProtocol		String	Device protocol 0: TCP 1: UDP
	deviceSecret		String	Device login secret
	deviceCnName		String	Device name
	deviceEnName		String	Device nick name
	deviceTag		String	User-defined device tag
	desc		String	Device description

	sts		String	Device status 0: Effective 1: Ineffective 3: Deleted
	matchingThreshold		String	Match threshold
	streamHeight		String	The height of video stream
	streamWidth		String	The width of video stream
	faceAttribute		String	Feature extraction switch (0: on, 1: off)
	livenessCheck		String	[Deprecated] Liveness check switch (0: on, 1: off)
	ingressQCThreshold		String	Quality check threshold
	ingressAngleThreshold		String	Detection angle threshold
	frameRate		String	Video frame rate
	privilege		String	Privilege (Department ID)
	bklImageStorage		String	Background image storing switch (0: on, 1: off)
	boundingBoxSizeThreshold		String	Width detection threshold
	targetGroup		String	Target person groups
	remark		String	Remark
	validRegionDefinition		Array[RegionEntity]	Valid region definition
RegionEntity	regionType	No	String	Region type
	regionPic	No	String	Region image
	validRegion	No	String	Region coordinates
Sample Request	["1257915401329709058", "1257879674839633922"]			
Sample Response	{ "code": "0000", }			

```
"msg": "成功",
"data": [
  {
    "did": "1257879674839633922",
    "deviceSerial": "SPS020STDC19G05927_0",
    "deviceSecret": null,
    "deviceCnName": "SenseThunder-E Mini",
    "deviceEnName": "SenseThunder-E Mini",
    "devicePort": "0",
    "deviceIP": "10.161.60.242",
    "deviceUri": null,
    "deviceVersion": "1.0.0_E",
    "deviceConnType": null,
    "deviceProtocol": "",
    "last_checked_ts": "2020-06-04T06:01:15.000+0000",
    "productType": "13",
    "deviceType": "15",
    "deviceTag": null,
    "desc": "Office",
    "sts": "1",
    "matchingThreshold": null,
    "streamHeight": null,
    "streamWidth": null,
    "faceAttribute": null,
    "livenessCheck": null,
    "ingressQCThreshold": null,
    "ingressAngleThreshold": null,
    "frameRate": null,
    "privilege": "0",
    "bklImageStorage": null,
    "boundingBoxSizeThreshold": null,
    "targetGroup": null,
    "remark": null,
    "validRegionDefinition": null
  },
  {
    "did": "1257915401329709058",
    "deviceSerial": "SPS020STDC19G07718_0",
```

```
"deviceSecret": null,  
"deviceCnName": "SPS020STDC19G07718",  
"deviceEnName": "SPS020STDC19G07718",  
"devicePort": "0",  
"deviceIP": "10.161.32.32",  
"deviceUri": null,  
"deviceVersion": "1.0.0_E",  
"deviceConnType": null,  
"deviceProtocol": "",  
"last_checked_ts": "2020-05-06T06:32:56.000+0000",  
"productType": "13",  
"deviceType": "15",  
"deviceTag": null,  
"desc": "",  
"sts": "1",  
"matchingThreshold": null,  
"streamHeight": null,  
"streamWidth": null,  
"faceAttribute": null,  
"livenessCheck": null,  
"ingressQCThreshold": null,  
"ingressAngleThreshold": null,  
"frameRate": null,  
"privilege": "0",  
"bklImageStorage": null,  
"boundingBoxSizeThreshold": null,  
"targetGroup": null,  
"remark": null,  
"validRegionDefinition": null  
}  
],  
"success": true  
}
```

3.3.6 List Device

API Endpoint	{{studio_base_url}}/DEVICE/device/list
Method	POST
Request Format	JSON

Request Parameters	Name	Required	Type	Description
	current	Yes	Integer	Query results on page #. Start from 1.
	size	Yes	Integer	Number of results per page
	did	No	String	Device unique ID
	ip	No	String	Device IP
	deviceSerial	No	String	Device serial
	desc	No	String	Device description
	deviceCnName	No	String	Device name
	deviceTag	No	String	Device tag
	deviceType	No	String	Device type ID Please refer to 3.2.1 Device Type Definition
	productType	No	String	Product type ID Please refer to 3.2.2 Product Type Definition
	remark	No	String	Remark
	sts	No	String	Device status 0: Effective 1: Ineffective 3: Deleted
Response Parameters	Name	Required	Type	Description
	data		Array[Device]	
Device	did		String	Device unique ID
	deviceSerial		String	Device serial
	productType		String	Product type ID
	productTypeName		String	Product type name
	productTypeEnName		String	Product type English name
	deviceType		String	Device type ID
	deviceTypeName		String	Device type name
	deviceTypeEnName		String	Device type English name
	deviceCnName		String	Device name
	deviceEnName		String	Device nick name
	deviceTag		String	User-defined device tag
	desc		String	Device description
	sts		String	Device status 0: Effective

				1: Ineffective 3: Deleted
	deviceIp		String	Device IP
	privilege		String	Privilege (Department ID)
	matchingThreshold		String	Match threshold
	faceAttribute		String	Feature extraction switch (0: on, 1: off)
	remark		String	Remark
	livenessCheck		String	[Deprecated] Liveness check switch (0: on, 1: off)
	ingressQCThreshold		String	Quality check threshold
Sample Request	<pre>{ "current": 1, "size": 100, "did": "", "ip": "", "deviceSerial": "", "desc": "", "deviceCnName": "", "deviceTag": "", "deviceType": ["0", "1", "2"], "productType": ["1", "2", "3"], "remark": "", "sts": "0" }</pre>			
Sample Response	<pre>{ "data": [{ "did": "1272374819588935682",</pre>			

```
"deviceSerial": "SPSP-1cdc87f0466cd54940509a90e42f30e0",
"productType": "1",
"productTypeName": "senseKeeper",
"productTypeEnName": "senseKeeper",
"deviceType": "1",
"deviceTypeName": "Keeper",
"deviceTypeEnName": "Keeper",
"deviceCnName": "SPS020PROC19F00272",
"deviceEnName": "Testing00272",
"deviceTag": null,
"desc": "",
"sts": "0",
"deviceIp": "10.161.32.83",
"privilege": "0",
"matchingThreshold": null,
"faceAttribute": "1",
"remark": null,
"livenessCheck": "1",
"ingressQCThreshold": null
    }
  ],
  "size": 100,
  "total": 1,
  "code": "0000",
  "msg": "success"
}
```

3.3.7 List Device Type

API Endpoint	{{studio_base_url}}/DEVICE/deviceType/list			
Method	GET			
Response Parameters	Name	Required	Type	Description
	data		Array[DeviceType]	
DeviceType	deviceTypeId		String	Device type ID
	deviceTypeCnName		String	Chinese device type Name
	deviceTypeEnName		String	English device type name
	registryId		String	Registry ID
	symphonyId		String	Symphony ID

Sample Response

```
{
  "code": "0000",
  "msg": "成功",
  "data": [
    {
      "deviceTypeId": "0",
      "deviceTypeCnName": "Terminal",
      "deviceTypeEnName": "Terminal",
      "registryId": null,
      "symphonyId": null
    },
    {
      "deviceTypeId": "12",
      "deviceTypeCnName": "VirtualMatch",
      "deviceTypeEnName": "VirtualMatch",
      "registryId": null,
      "symphonyId": null
    },
    {
      "deviceTypeId": "14",
      "deviceTypeCnName": "compareCamera",
      "deviceTypeEnName": "compareCamera",
      "registryId": null,
      "symphonyId": null
    },
    {
      "deviceTypeId": "15",
      "deviceTypeCnName": "sensePass",
      "deviceTypeEnName": "sensePass",
      "registryId": null,
      "symphonyId": null
    },
    {
      "deviceTypeId": "16",
      "deviceTypeCnName": "抓拍机-T",
      "deviceTypeEnName": "CaptureCamera-T",
      "registryId": null,
      "symphonyId": null
    }
  ]
}
```

```
},
{
  "deviceTypeId": "17",
  "deviceTypeCnName": "sensePass Pro",
  "deviceTypeEnName": "sensePass Pro",
  "registryId": null,
  "symphonyId": null
},
{
  "deviceTypeId": "18",
  "deviceTypeCnName": "人脸抠图摄像头",
  "deviceTypeEnName": "Face Detecting camera",
  "registryId": null,
  "symphonyId": null
},
{
  "deviceTypeId": "19",
  "deviceTypeCnName": "访客版终端",
  "deviceTypeEnName": "Terminal-Visitor",
  "registryId": null,
  "symphonyId": null
},
{
  "deviceTypeId": "20",
  "deviceTypeCnName": "抓拍机-11",
  "deviceTypeEnName": "CaptureCamera-11",
  "registryId": null,
  "symphonyId": null
},
{
  "deviceTypeId": "21",
  "deviceTypeCnName": "区域结构摄像头",
  "deviceTypeEnName": "Senseeye-Region",
  "registryId": null,
  "symphonyId": null
},
{
  "deviceTypeId": "1",
```

```
"deviceTypeCnName": "Keeper",
"deviceTypeEnName": "Keeper",
"registryId": null,
"symphonyId": null
},
{
  "deviceTypeId": "22",
  "deviceTypeCnName": "结构摄像头",
  "deviceTypeEnName": "Senseye-X",
  "registryId": null,
  "symphonyId": null
},
{
  "deviceTypeId": "23",
  "deviceTypeCnName": "华为抓拍机",
  "deviceTypeEnName": "huawei snapshot camera",
  "registryId": null,
  "symphonyId": null
},
{
  "deviceTypeId": "24",
  "deviceTypeCnName": "dlc-D",
  "deviceTypeEnName": "dlc-D",
  "registryId": null,
  "symphonyId": null
},
{
  "deviceTypeId": "25",
  "deviceTypeCnName": "海康抓拍机",
  "deviceTypeEnName": "haikang snapshot camera",
  "registryId": null,
  "symphonyId": null
},
{
  "deviceTypeId": "26",
  "deviceTypeCnName": "sensegate-H",
  "deviceTypeEnName": "sensegate-H",
  "registryId": null,
```

```
"symphonyId": null
},
{
  "deviceTypeId": "27",
  "deviceTypeCnName": "大华抓拍机",
  "deviceTypeEnName": "dahua snapshot camera",
  "registryId": null,
  "symphonyId": null
},
{
  "deviceTypeId": "28",
  "deviceTypeCnName": "senseeyeApp",
  "deviceTypeEnName": "senseeyeApp",
  "registryId": null,
  "symphonyId": null
},
{
  "deviceTypeId": "2",
  "deviceTypeCnName": "摄像机",
  "deviceTypeEnName": "Camera",
  "registryId": null,
  "symphonyId": null
},
{
  "deviceTypeId": "3",
  "deviceTypeCnName": "抓拍机",
  "deviceTypeEnName": "GrabMachine",
  "registryId": null,
  "symphonyId": null
},
{
  "deviceTypeId": "6",
  "deviceTypeCnName": "GB2818",
  "deviceTypeEnName": "GB2818",
  "registryId": null,
  "symphonyId": null
},
{
```

```
"deviceTypeId": "9",
"deviceTypeCnName": "前置服务器",
"deviceTypeEnName": "Server",
"registryId": null,
"symphonyId": null
},
{
  "deviceTypeId": "7",
  "deviceTypeCnName": "Pad",
  "deviceTypeEnName": "Pad",
  "registryId": null,
  "symphonyId": null
},
{
  "deviceTypeId": "10",
  "deviceTypeCnName": "VirtualGate",
  "deviceTypeEnName": "VirtualGate",
  "registryId": null,
  "symphonyId": null
},
{
  "deviceTypeId": "11",
  "deviceTypeCnName": "VirtualId",
  "deviceTypeEnName": "VirtualId",
  "registryId": null,
  "symphonyId": null
}
},
"success": true
}
```

3.3.8 List Product Type

API Endpoint	{{studio_base_url}}/DEVICE/productType/list			
Method	GET			
Response Parameters	Name	Required	Type	Description
	data		Array[ProductType]	
ProductType	productTypeId		String	Product type ID
	productTypeCnName		String	Chinese product type Name

	productTypeEnName		String	English product type name
	registryId		String	Registry ID
	symphonyId		String	Symphony ID
Sample Response	<pre>{ "code": "0000", "msg": "成功", "data": [{ "productId": "0", "productCnName": "senseID", "productEnName": "senseID", "registryId": null, "symphonyId": null }, { "productId": "9", "productCnName": "senseHello", "productEnName": "senseHello", "registryId": null, "symphonyId": null }, { "productId": "10", "productCnName": "ViperImage", "productEnName": "ViperImage", "registryId": null, "symphonyId": null }, { "productId": "11", "productCnName": "SenseNebula", "productEnName": "SenseNebula", "registryId": null, "symphonyId": null }, { "productId": "12", "productCnName": "SenseEye",</pre>			

```
"productTypeEnName": "SenseEye",
"registryId": null,
"symphonyId": null
},
{
  "productId": "13",
  "productTypeCnName": "sensePass",
  "productTypeEnName": "sensePass",
  "registryId": null,
  "symphonyId": null
},
{
  "productId": "14",
  "productTypeCnName": "SenseClass",
  "productTypeEnName": "SenseClass",
  "registryId": null,
  "symphonyId": null
},
{
  "productId": "15",
  "productTypeCnName": "ViperFace",
  "productTypeEnName": "ViperFace",
  "registryId": null,
  "symphonyId": null
},
{
  "productId": "1",
  "productTypeCnName": "senseKeeper",
  "productTypeEnName": "senseKeeper",
  "registryId": null,
  "symphonyId": null
},
{
  "productId": "2",
  "productTypeCnName": "senseFace",
  "productTypeEnName": "senseFace",
  "registryId": null,
  "symphonyId": null
}
```

```
    },
    {
      "productId": "3",
      "productCnName": "senseCity",
      "productEnName": "senseCity",
      "registryId": null,
      "symphonyId": null
    },
    {
      "productId": "5",
      "productCnName": "senseCrowd",
      "productEnName": "senseCrowd",
      "registryId": null,
      "symphonyId": null
    },
    {
      "productId": "6",
      "productCnName": "DLC",
      "productEnName": "DLC",
      "registryId": null,
      "symphonyId": null
    },
    {
      "productId": "7",
      "productCnName": "senseGo",
      "productEnName": "senseGo",
      "registryId": null,
      "symphonyId": null
    }
  ],
  "success": true
}
```

3.4 BI Device Management

3.4.1 Set BI Device Configuration

API Endpoint	{{studio_base_url}}/DEVICE/biDeviceInfo/setBIDeviceConfig
Method	POST
Request Format	JSON

Request Parameters	Name	Required	Type	Description
	deviceId	Yes	String	Device unique ID
	deviceProductTypeUuid	Yes	String	Product type uuid
	deviceTypeUuid	Yes	String	Device type uuid
	operatePerson	Yes	String	Operator
	serverVerify	No	String	Backend recognition 0: off; 1: on; 2: compatible
	serverAddress	No	String	Recognition server address
	ntpServerAddress	No	String	NTP server address
	otherServer	No	String	Third-party service configuration true: on; false: off
	otherServerAddress	No	String	Third-party upload address
	networkRelayAddress	No	String	Network relay address
	faceUpdateThreshold	No	String	Face alignment threshold
	yaw	No	String	Acceptable yaw degree
	pitch	No	String	Acceptable pitch degree
	roll	No	String	Acceptable roll degree
	faceWidth	No	String	Face width (pixel)
	wakeUpDistance	No	String	Device awake distance (cm)
	recognitionDistance	No	String	Recognition distance (m)
	liveness	No	String	Liveness detection true: on; false: off
	livenessThreshold	No	String	Liveness detection threshold
	verifyThreshold	No	String	FR match threshold
	logo	No	String	Logo display (true/false)
	videoTip	No	String	Video message
	welcomeTip	No	String	Welcome message
	showName	No	String	Name display (true/false)
	showAvatarType	No	String	Avatar type display
	verifyFaultRoleTipe	No	String	Role tag display when face not recognized
	livenessSuccessRoleTip	No	String	Role tag display when liveness detection success
	livenessSuccessTip	No	String	Message display when liveness detection success

	verifySuccessTip	No	String	Message display when recognition success
	verifyFaultTip	No	String	Message display when face not recognized
	useShowAvatar	No	String	[SensePassPro] Person avatar display (true/false)
	showUserName	No	String	[SensePassPro] Name display (true/false)
	showUserInfo	No	String	[SensePass] User info display when recognition success 0: false; 1: true 1 st digit – Job number 2 nd digit – Department 3 rd digit – Position 4 th digit – ID number 5 th digit – Custom message E.g. "01000" means show department only
	autoLed	No	String	Auto fill light true: on; false: off
	fillLight	No	String	[SensePassPro] Auto fill light true: on; false: off
	forcedOpenUpload	No	String	Forced open gate true: on; false: off
	dismantleUpload	No	String	Tamper alarm true: on; false: off
	guestNotification	No	String	Welcome notice true: on; false: off
	voice	No	String	Voice cast true: on; false: off
	voiceBroadcast	No	String	[SensePassPro] Voice cast true: on; false: off
	infraredProbe	No	String	Infrared sensor true: on; false: off

	startTime	No	String	Start time (Unix timestamp)
	endTime	No	String	End time (Unix timestamp)
	buzzerStatus	No	String	[SensePass] Buzzer status true: on; false: off
	gpioA	No	String	GPIO A output
	gpioB	No	String	GPIO B input
	gpioC	No	String	GPIO C input
	openDoorTime	No	String	Gate open delay time (s)
	openInterval	No	String	Gate open interval
	keepDoorOpenDuration	No	String	Gate open duration (s)
	wiganInput	No	String	Wiegand input 1: N.A. 2: Wiegand26(8+16bit ID) 3: Wiegand26(24bit ID) 4: Wiegand32 5: Wiegand34
	openDoorType	No	String	Open mode 0: Local relay 1: Wiegand26(8+16bit ID) 2: Wiegand32 3: Wiegand34 4: Network relay 5: Wiegand26(24bit ID) 8: Bluetooth
	standbyOpen	No	String	Standby switch true: on; false: off
	faceNum	No	String	Capture face number
	noActionThreshold	No	String	Repeated times when no respond

	noMatchThreshold	No	String	Repeated times when unmatched
	configDefaultFlag	No	String	[SensePass] Reset to default settings (true/false)
	deviceRunType	No	String	[SensePass] Device state 1: Running 2: Stopped
	autoReboot	No	String	[SensePass] Auto reboot true: on; false: off
	rebootTime	No	String	[SensePass] Device reboot time HH:mm:ss
	waitTime	No	String	Standby time (min)
	simplifyModel	No	String	Lite mode true: on; false: off
	adminPwd	No	String	Admin password
	dormancyType	No	String	Sleep mode 0: Sleep 1: Multimedia
	languageType	No	String	[SensePass] Display language 1: Chinese 2: English
	useMode	No	String	[SensePass] Device mode 1: Gate control 2: Door security
	proUseMode	No	String	[SensePassPro] Device mode 3: Single-person 4: Multi-person
	mode	No	String	Verification mode

	blackListTip	No	String	Blacklist message
	blackListOpen	No	String	Gate open for blacklist true: on; false: off
	deviceRebootTime	No	String	Device reboot time
	bluetoothMac	No	String	Bluetooth MAC address
	changeServerUrl	No	String	Change server URL
	deviceButtonName1	No	String	Name of custom button 1
	deviceButtonValue1	No	String	Value of custom button 1
	deviceButtonName2	No	String	Name of custom button 2
	deviceButtonValue2	No	String	Value of custom button 2
	deviceButtonName3	No	String	Name of custom button 3
	deviceButtonValue3	No	String	Value of custom button 3
	deviceButtonName4	No	String	Name of custom button 4
	deviceButtonValue4	No	String	Value of custom button 4
	deviceButtonName5	No	String	Name of custom button 5
	deviceButtonValue5	No	String	Value of custom button 5
	deviceButtonName6	No	String	Name of custom button 6
	deviceButtonValue6	No	String	Value of custom button 6
	certificateThreshold	No	String	Certification match threshold
	strongHint	No	String	Strong prompt true: on; false: off
	saveElecMode	No	String	Power saving mode true: on; false: off
	touchRecognition	No	String	Touch recognition true: on; false: off

	touchRecognitionCustomPicture	No	String	Touch recognition custom picture
	touchRecognitionReturnTime	No	String	Touch recognition return time
	touchRecognitionTimeout	No	String	Touch recognition timeout
	doorSensorTimeout	No	String	Door sensor timeout time
	wakeDistance	No	String	Wakeup distance detection
	wifiSsid	No	String	WIFI SSID
	wifiPwd	No	String	WIFI password
	temperatureDetect	No	String	Temperature detection true: on; false: off
	feverTemperature	No	String	Temperature detection threshold (°C)
	noAccessToFevers	No	String	No admittance with fever true: on; false: off
	showThermalScreen	No	String	Thermograph display true: on; false: off
	indoorTemperature	No	String	Indoor temperature (°C)
	temperatureDetectDistance	No	String	Temperature detection distance (m)
	quickDetectMode	No	String	Fast temperature detection mode true: on; false: off
	maskDetect	No	String	Mask detection true: on; false: off
	noAccessWithoutMask	No	String	No admittance without mask true: on; false: off
	accessControlStatus	No	String	Access control status
	accessControlTimetable	No	String	Access control timetable
	timetableUpdTz	No	String	Timetable last updated time
	Name	Required	Type	Description

Response Parameters	data		String	Device unique ID
Sample Request	{ "deviceUuid": "1257915401329709058", "deviceProductTypeUuid": "13", "deviceTypeUuid": "15", "operatePerson": "1", "serverVerify": "0", "serverAddress": "127.0.0.1", "ntpServerAddress": "127.0.0.1", "otherServer": "true", "otherServerAddress": "127.0.0.1", "networkRelayAddress": "127.0.0.1", "faceUpdateThreshold": "0", "yaw": "18", "pitch": "18", "roll": "18", "faceWidth": "60", "wakeUpDistance": "6", "recognitionDistance": "1.5", "liveness": "true", "livenessThreshold": "0.98", "verifyThreshold": "0.9", "logo": "true", "videoTip": "video message", "welcomeTip": "", "showName": "true", "showAvatarType": "1", "verifyFaultRoleTip": "failed role tag", "livenessSuccessRoleTip": "role", "livenessSuccessTip": "success", "verifySuccessTip": "", "verifyFaultTip": "", "useShowAvatar": "true", "showUserName": "true", "showUserInfo": "00000", "autoLed": "true", "fillLight": "true", }			

```
"forcedOpenUpload": "false",
"dismantleUpload": "false",
"guestNotification": "false",
"voice": "true",
"voiceBroadcast": "true",
"infraredProbe": "true",
"startTime": "1545647713",
"endTime": "1545687713",
"buzzerStatus": "true",
"gpioA": "1",
"gpioB": "1",
"gpioC": "1",
"openDoorTime": "5",
"openInterval": "5",
"keepDoorOpenDuration": "6",
"wiganInput": "1",
"openDoorType": "0",
"standbyOpen": "true",
"faceNum": "1",
"noActionThreshold": "10",
"noMatchThreshold": "3",
"configDefaultFlag": "false",
"deviceRunType": "1",
"autoReboot": "true",
"rebootTime": "02:00:00",
"waitTime": "5",
"simplifyModel": "false",
"adminPwd": "admin",
"dormancyType": "0",
"languageType": "2",
"useMode": "2",
"proUseMode": "2",
"mode": "9",
"blackListTip": "",
"blackListOpen": "false",
"deviceRebootTime": "",
"bluetoothMac": "",
"changeServerUrl": "",
```

	<pre>"deviceButtonName1": "", "deviceButtonValue1": "", "deviceButtonName2": "", "deviceButtonValue2": "", "deviceButtonName3": "", "deviceButtonValue3": "", "deviceButtonName4": "", "deviceButtonValue4": "", "deviceButtonName5": "", "deviceButtonValue5": "", "deviceButtonName6": "", "deviceButtonValue6": "", "certificateThreshold": "0.6", "strongHint": "false", "saveElecMode": "false", "touchRecognition": "false", "touchRecognitionCustomPicture": "", "touchRecognitionReturnTime": "5", "touchRecognitionTimeout": "5", "doorSensorTimeout": "3", "wakeDistance": "150", "wifiSsid": "", "wifiPwd": "", "temperatureDetect": "", "feverTemperature": "", "noAccessToFever": "", "showThermalScreen": "", "indoorTemperature": "", "temperatureDetectDistance": "", "quickDetectMode": "", "maskDetect": "", "noAccessWithoutMask": "", "accessControlStatus": "", "accessControlTimetable": "", "timetableUpdTs": "" }</pre>
Sample Response	<pre>{ "code": "0000",</pre>

```
"msg": "成功",  
"data": "1257915401329709058",  
"success": true  
}
```

3.4.2 List BI Device Configuration (Pagination)

API Endpoint	{{studio_base_url}}/DEVICE/biDeviceInfo/getBIDeviceConfigPageList			
Method	POST			
Request Format	JSON			
Request Parameters	Name	Required	Type	Description
	current	Yes	Integer	Query results on page #. Start from 1.
	size	Yes	Integer	Number of results per page
	deviceUuid	No	String	Device unique ID
Response Parameters	Name	Required	Type	Description
	data		Array[BIDeviceConfig]	
BIDeviceConfig	deviceUuid		String	Device unique ID
	serverVerify		String	Backend recognition 0: off; 1: on; 2: compatible
	serverAddress		String	Recognition server address
	ntpServerAddress		String	NTP server address
	otherServer		String	Third-party service configuration true: on; false: off
	otherServerAddress		String	Third-party upload address
	networkRelayAddress		String	Network relay address
	faceUpdateThreshold		String	Face alignment threshold

	yaw		String	Acceptable yaw degree
	pitch		String	Acceptable pitch degree
	roll		String	Acceptable roll degree
	faceWidth		String	Face width (pixel)
	wakeUpDistance		String	Device awake distance (cm)
	recognitionDistance		String	Recognition distance (m)
	liveness		String	Liveness detection true: on; false: off
	livenessThreshold		String	Liveness detection threshold
	verifyThreshold		String	FR match threshold
	logo		String	Logo display (true/false)
	videoTip		String	Video message
	welcomeTip		String	Welcome message
	showName		String	Name display (true/false)
	showAvatarType		String	Avatar type display
	verifyFaultRoleTip		String	Role tag display when face not recognized
	livenessSuccessRoleTip		String	Role tag display when liveness detection success
	livenessSuccessTip		String	Message display when liveness detection success
	verifySuccessTip		String	Message display when recognition success

	verifyFaultTip		String	Message display when face not recognized
	useShowAvatar		String	[SensePassPro] Person avatar display (true/false)
	showUserName		String	[SensePassPro] Name display (true/false)
	showUserInfo		String	[SensePass] User info display when recognition success 0: false; 1: true 1 st digit – Job number 2 nd digit – Department 3 rd digit – Position 4 th digit – ID number 5 th digit – Custom message E.g. "01000" means show department only
	autoLed		String	Auto fill light true: on; false: off
	fillLight		String	[SensePassPro] Auto fill light true: on; false: off
	forcedOpenUpload		String	Forced open gate true: on; false: off
	dismantleUpload		String	Tamper alarm true: on; false: off
	guestNotification		String	Welcome notice true: on; false: off
	voice		String	Voice cast

			true: on; false: off
voiceBroadcast		String	[SensePassPro] Voice cast true: on; false: off
infraredProbe		String	Infrared sensor true: on; false: off
startTime		String	Start time (Unix timestamp)
endTime		String	End time (Unix timestamp)
buzzerStatus		String	[SensePass] Buzzer status true: on; false: off
gpioA		String	GPIO A output
gpioB		String	GPIO B input
gpioC		String	GPIO C input
openDoorTime		String	Gate open delay time (s)
openInterval		String	Gate open interval
keepDoorOpenDuration		String	Gate open duration (s)
wiganInput		String	Wiegand input 1: N.A. 2: Wiegand26(8+16bit ID) 3: Wiegand26(24bit ID) 4: Wiegand32 5: Wiegand34
openDoorType		String	Open mode 0: Local relay 1: Wiegand26(8+16bit ID) 2: Wiegand32 3: Wiegand34 4: Network relay

				5: Wiegand26(24bit ID) 8: Bluetooth
	standbyOpen		String	Standby switch true: on; false: off
	faceNum		String	Capture face number
	noActionThreshold		String	Repeated times when no respond
	noMatchThreshold		String	Repeated times when unmatched
	configDefaultFlag		String	[SensePass] Reset to default settings (true/false)
	deviceRunType		String	[SensePass] Device state 1: Running 2: Stopped
	autoReboot		String	[SensePass] Auto reboot true: on; false: off
	rebootTime		String	[SensePass] Device reboot time HH:mm:ss
	waitTime		String	Standby time (min)
	simplifyModel		String	Lite mode true: on; false: off
	adminPwd		String	Admin password
	dormancyType		String	Sleep mode 0: Sleep 1: Multimedia
	languageType		String	[SensePass] Display language 1: Chinese 2: English
	useMode		String	[SensePass] Device mode 1: Gate control

				2: Door security
	proUseMode		String	[SensePassPro] Device mode 3: Single-person 4: Multi-person
	mode		String	Verification mode
	deviceProductTypeUuid		String	Product type uuid
	deviceTypeUuid		String	Device type uuid
	blackListTip		String	Blacklist message
	blackListOpen		String	Gate open for blacklist true: on; false: off
	deviceRebootTime		String	Device reboot time
	bluetoothMac		String	Bluetooth MAC address
	changeServerUrl		String	Change server URL
	deviceButtonName1		String	Name of custom button 1
	deviceButtonValue1		String	Value of custom button 1
	deviceButtonName2		String	Name of custom button 2
	deviceButtonValue2		String	Value of custom button 2
	deviceButtonName3		String	Name of custom button 3
	deviceButtonValue3		String	Value of custom button 3
	deviceButtonName4		String	Name of custom button 4
	deviceButtonValue4		String	Value of custom button 4
	deviceButtonName5		String	Name of custom button 5
	deviceButtonValue5		String	Value of custom button 5

	deviceButtonName6		String	Name of custom button 6
	deviceButtonValue6		String	Value of custom button 6
	certificateThreshold		String	Certification match threshold
	strongHint		String	Strong prompt true: on; false: off
	saveElecMode		String	Power saving mode true: on; false: off
	touchRecognition		String	Touch recognition true: on; false: off
	touchRecognitionCustomPicture		String	Touch recognition custom picture
	touchRecognitionReturnTime		String	Touch recognition return time
	touchRecognitionTimeout		String	Touch recognition timeout
	doorSensorTimeout		String	Door sensor timeout time
	wakeDistance		String	Wakeup distance detection
	wifiSsid		String	WIFI SSID
	wifiPwd		String	WIFI password
	temperatureDetect		String	Temperature detection true: on; false: off
	feverTemperature		String	Temperature detection threshold (°C)
	noAccessToFevers		String	No admittance with fever true: on; false: off
	showThermalScreen		String	Thermograph display true: on; false: off
	indoorTemperature		String	Indoor temperature (°C)

	temperatureDetectDistance		String	Temperature detection distance (m)
	quickDetectMode		String	Fast temperature detection mode true: on; false: off
	maskDetect		String	Mask detection true: on; false: off
	noAccessWithoutMask		String	No admittance without mask true: on; false: off
	accessControlStatus		String	Access control status
	accessControlTimetable		String	Access control timetable
	timetableUpdTs		String	Timetable last updated time
Sample Request	<pre>{ "current": 1, "size": 10, "deviceUuid": "1257915401329709058" }</pre>			
Sample Response	<pre>{ "data": [{ "deviceUuid": "1257915401329709058", "serverVerify": "0", "serverAddress": "127.0.0.1", "ntpServerAddress": "127.0.0.1", "otherServer": "true", "otherServerAddress": "127.0.0.1", "networkRelayAddress": "127.0.0.1", "faceUpdateThreshold": "0", "yaw": "18", "pitch": "18", "roll": "18", "faceWidth": "60", "wakeUpDistance": "6", }] }</pre>			

```
"recognitionDistance": "1.5",
"liveness": "true",
"livenessThreshold": "0.98",
"verifyThreshold": "0.9",
"logo": "true",
"videoTip": "video message",
"welcomeTip": "",
"showName": "true",
"showAvatarType": "1",
"verifyFaultRoleTip": "failed role tag",
"livenessSuccessRoleTip": "role",
"livenessSuccessTip": "success",
"verifySuccessTip": "",
"verifyFaultTip": "",
"useShowAvatar": "true",
"showUserName": "true",
"showUserInfo": "00000",
"autoLed": "true",
"fillLight": "true",
"forcedOpenUpload": "false",
"dismantleUpload": "false",
"guestNotification": "false",
"voice": "true",
"voiceBroadcast": "true",
"infraredProbe": "true",
"startTime": "1545647713",
"endTime": "1545687713",
"buzzerStatus": "true",
"gpioA": "1",
"gpioB": "1",
"gpioC": "1",
"openDoorTime": "5",
"openInterval": "5",
"keepDoorOpenDuration": "6",
"wiganInput": "1",
"openDoorType": "0",
"standbyOpen": "true",
"faceNum": "1",
```

```
"noActionThreshold": "10",
"noMatchThreshold": "3",
"configDefaultFlag": "false",
"deviceRunType": "1",
"autoReboot": "true",
"rebootTime": "02:00:00",
"waitTime": "5",
"simplifyModel": "false",
"adminPwd": "admin",
"dormancyType": "0",
"languageType": "2",
"useMode": "2",
"proUseMode": "2",
"mode": "9",
"deviceProductTypeUuid": "13",
"deviceTypeUuid": "15",
"blackListTip": "",
"blackListOpen": "false",
"deviceRebootTime": "",
"bluetoothMac": "",
"changeServerUrl": "",
"deviceButtonName1": "",
"deviceButtonValue1": "",
"deviceButtonName2": "",
"deviceButtonValue2": "",
"deviceButtonName3": "",
"deviceButtonValue3": "",
"deviceButtonName4": "",
"deviceButtonValue4": "",
"deviceButtonName5": "",
"deviceButtonValue5": "",
"deviceButtonName6": "",
"deviceButtonValue6": "",
"certificateThreshold": "0.6",
"strongHint": "false",
"saveElecMode": "false",
"touchRecognition": "false",
"touchRecognitionCustomPicture": "",
```

```

        "touchRecognitionReturnTime": "5",
        "touchRecognitionTimeout": "5",
        "doorSensorTimeout": "3",
        "wakeDistance": "150",
        "wifiSsid": "",
        "wifiPwd": "",
        "temperatureDetect": "",
        "feverTemperature": "",
        "noAccessToFevers": "",
        "showThermalScreen": "",
        "indoorTemperature": "",
        "temperatureDetectDistance": "",
        "quickDetectMode": "",
        "maskDetect": "",
        "noAccessWithoutMask": "",
        "accessControlStatus": "",
        "accessControlTimetable": "",
        "timetableUpdTs": ""
    }
],
    "size": 10,
    "total": 1,
    "code": "0000",
    "msg": "success"
}

```

3.4.3 Query BI Device Configuration by Device ID

API Endpoint	{{studio_base_url}}/DEVICE/biDeviceInfo/getBIDeviceConfig			
Method	POST			
Request Format	JSON			
Request Parameters	Name	Required	Type	Description
	deviceUuid	Yes	String	Device unique ID
Response Parameters	Name	Required	Type	Description
	data		BIDeviceConfig	
BIDeviceConfig	deviceUuid		String	Device unique ID
	serverVerify		String	Backend recognition

			0: off; 1: on; 2: compatible
serverAddress		String	Recognition server address
ntpServerAddress		String	NTP server address
otherServer		String	Third-party service configuration true: on; false: off
otherServerAddress		String	Third-party upload address
networkRelayAddress		String	Network relay address
faceUpdateThreshold		String	Face alignment threshold
yaw		String	Acceptable yaw degree
pitch		String	Acceptable pitch degree
roll		String	Acceptable roll degree
faceWidth		String	Face width (pixel)
wakeUpDistance		String	Device awake distance (cm)
recognitionDistance		String	Recognition distance (m)
liveness		String	Liveness detection true: on; false: off
livenessThreshold		String	Liveness detection threshold
verifyThreshold		String	FR match threshold
logo		String	Logo display (true/false)
videoTip		String	Video message
welcomeTip		String	Welcome message
showName		String	Name display (true/false)
showAvatarType		String	Avatar type display

	verifyFaultRoleTipe		String	Role tag display when face not recognized
	livenessSuccessRoleTip		String	Role tag display when liveness detection success
	livenessSuccessTip		String	Message display when liveness detection success
	verifySuccessTip		String	Message display when recognition success
	verifyFaultTip		String	Message display when face not recognized
	useShowAvatar		String	[SensePassPro] Person avatar display (true/false)
	showUserName		String	[SensePassPro] Name display (true/false)
	showUserInfo		String	[SensePass] User info display when recognition success 0: false; 1: true 1 st digit – Job number 2 nd digit – Department 3 rd digit – Position 4 th digit – ID number 5 th digit – Custom message E.g. "01000" means show department only

	autoLed		String	Auto fill light true: on; false: off
	fillLight		String	[SensePassPro] Auto fill light true: on; false: off
	forcedOpenUpload		String	Forced open gate true: on; false: off
	dismantleUpload		String	Tamper alarm true: on; false: off
	guestNotification		String	Welcome notice true: on; false: off
	voice		String	Voice cast true: on; false: off
	voiceBroadcast		String	[SensePassPro] Voice cast true: on; false: off
	infraredProbe		String	Infrared sensor true: on; false: off
	startTime		String	Start time (Unix timestamp)
	endTime		String	End time (Unix timestamp)
	buzzerStatus		String	[SensePass] Buzzer status true: on; false: off
	gpioA		String	GPIO A output
	gpioB		String	GPIO B input
	gpioC		String	GPIO C input
	openDoorTime		String	Gate open delay time (s)
	openInterval		String	Gate open interval
	keepDoorOpenDuration		String	Gate open duration (s)
	wiganInput		String	Wiegand input 1: N.A. 2: Wiegand26(8+16bit ID)

				3: Wiegand26(24bit ID) 4: Wiegand32 5: Wiegand34
	openDoorType		String	Open mode 0: Local relay 1: Wiegand26(8+16bit ID) 2: Wiegand32 3: Wiegand34 4: Network relay 5: Wiegand26(24bit ID) 8: Bluetooth
	standbyOpen		String	Standby switch true: on; false: off
	faceNum		String	Capture face number
	noActionThreshold		String	Repeated times when no respond
	noMatchThreshold		String	Repeated times when unmatched
	configDefaultFlag		String	[SensePass] Reset to default settings (true/false)
	deviceRunType		String	[SensePass] Device state 1: Running 2: Stopped
	autoReboot		String	[SensePass] Auto reboot true: on; false: off
	rebootTime		String	[SensePass] Device reboot time HH:mm:ss
	waitTime		String	Standby time (min)
	simplifyModel		String	Lite mode

			true: on; false: off
adminPwd		String	Admin password
dormancyType		String	Sleep mode 0: Sleep 1: Multimedia
languageType		String	[SensePass] Display language 1: Chinese 2: English
useMode		String	[SensePass] Device mode 1: Gate control 2: Door security
proUseMode		String	[SensePassPro] Device mode 3: Single-person 4: Multi-person
mode		String	Verification mode
deviceProductTypeUuid		String	Product type uuid
deviceTypeUuid		String	Device type uuid
blackListTip		String	Blacklist message
blackListOpen		String	Gate open for blacklist true: on; false: off
deviceRebootTime		String	Device reboot time
bluetoothMac		String	Bluetooth MAC address
changeServerUrl		String	Change server URL
deviceButtonName1		String	Name of custom button 1
deviceButtonValue1		String	Value of custom button 1
deviceButtonName2		String	Name of custom button 2
deviceButtonValue2		String	Value of custom button 2

	deviceButtonName3		String	Name of custom button 3
	deviceButtonValue3		String	Value of custom button 3
	deviceButtonName4		String	Name of custom button 4
	deviceButtonValue4		String	Value of custom button 4
	deviceButtonName5		String	Name of custom button 5
	deviceButtonValue5		String	Value of custom button 5
	deviceButtonName6		String	Name of custom button 6
	deviceButtonValue6		String	Value of custom button 6
	certificateThreshold		String	Certification match threshold
	strongHint		String	Strong prompt true: on; false: off
	saveElecMode		String	Power saving mode true: on; false: off
	touchRecognition		String	Touch recognition true: on; false: off
	touchRecognitionCustomPicture		String	Touch recognition custom picture
	touchRecognitionReturnTime		String	Touch recognition return time
	touchRecognitionTimeout		String	Touch recognition timeout
	doorSensorTimeout		String	Door sensor timeout time
	wakeDistance		String	Wakeup distance detection
	wifiSsid		String	WIFI SSID
	wifiPwd		String	WIFI password
	temperatureDetect		String	Temperature detection

				true: on; false: off
	feverTemperature		String	Temperature detection threshold (°C)
	noAccessToFever		String	No admittance with fever true: on; false: off
	showThermalScreen		String	Thermograph display true: on; false: off
	indoorTemperature		String	Indoor temperature (°C)
	temperatureDetectDistance		String	Temperature detection distance (m)
	quickDetectMode		String	Fast temperature detection mode true: on; false: off
	maskDetect		String	Mask detection true: on; false: off
	noAccessWithoutMask		String	No admittance without mask true: on; false: off
	accessControlStatus		String	Access control status
	accessControlTimetable		String	Access control timetable
	timetableUpdT		String	Timetable last updated time
Sample Request	<pre>{ "deviceUuid": "1257915401329709058" }</pre>			
Sample Response	<pre>{ "code": "0000", "msg": "成功", "data": { "deviceUuid": "1257915401329709058", "serverVerify": "0", </pre>			

```
"serverAddress": "127.0.0.1",
"ntpServerAddress": "127.0.0.1",
"otherServer": "true",
"otherServerAddress": "127.0.0.1",
"networkRelayAddress": "127.0.0.1",
"faceUpdateThreshold": "0",
"yaw": "18",
"pitch": "18",
"roll": "18",
"faceWidth": "60",
"wakeUpDistance": "6",
"recognitionDistance": "1.5",
"liveness": "true",
"livenessThreshold": "0.98",
"verifyThreshold": "0.9",
"logo": "true",
"videoTip": "video message",
"welcomeTip": "",
"showName": "true",
"showAvatarType": "1",
"verifyFaultRoleTip": "failed role tag",
"livenessSuccessRoleTip": "role",
"livenessSuccessTip": "success",
"verifySuccessTip": "",
"verifyFaultTip": "",
"useShowAvatar": "true",
"showUserName": "true",
"showUserInfo": "00000",
"autoLed": "true",
"fillLight": "true",
"forcedOpenUpload": "false",
"dismantleUpload": "false",
"guestNotification": "false",
"voice": "true",
"voiceBroadcast": "true",
"infraredProbe": "true",
"startTime": "1545647713",
"endTime": "1545687713",
```

```
"buzzerStatus": "true",  
"gpioA": "1",  
"gpioB": "1",  
"gpioC": "1",  
"openDoorTime": "5",  
"openInterval": "5",  
"keepDoorOpenDuration": "6",  
"wiganInput": "1",  
"openDoorType": "0",  
"standbyOpen": "true",  
"faceNum": "1",  
"noActionThreshold": "10",  
"noMatchThreshold": "3",  
"configDefaultFlag": "false",  
"deviceRunType": "1",  
"autoReboot": "true",  
"rebootTime": "02:00:00",  
"waitTime": "5",  
"simplifyModel": "false",  
"adminPwd": "admin",  
"dormancyType": "0",  
"languageType": "2",  
"useMode": "2",  
"proUseMode": "2",  
"mode": "9",  
"deviceProductTypeUuid": "13",  
"deviceTypeUuid": "15",  
"blackListTip": "",  
"blackListOpen": "false",  
"deviceRebootTime": "",  
"bluetoothMac": "",  
"changeServerUrl": "",  
"deviceButtonName1": "",  
"deviceButtonValue1": "",  
"deviceButtonName2": "",  
"deviceButtonValue2": "",  
"deviceButtonName3": "",  
"deviceButtonValue3": "",
```

```
"deviceButtonName4": "",
"deviceButtonValue4": "",
"deviceButtonName5": "",
"deviceButtonValue5": "",
"deviceButtonName6": "",
"deviceButtonValue6": "",
"certificateThreshold": "0.6",
"strongHint": "false",
"saveElecMode": "false",
"touchRecognition": "false",
"touchRecognitionCustomPicture": "",
"touchRecognitionReturnTime": "5",
"touchRecognitionTimeout": "5",
"doorSensorTimeout": "3",
"wakeDistance": "150",
"wifiSsid": "",
"wifiPwd": "",
"temperatureDetect": "",
"feverTemperature": "34.5",
"noAccessToFevers": "",
"showThermalScreen": "",
"indoorTemperature": "5",
"temperatureDetectDistance": "1",
"quickDetectMode": "",
"maskDetect": "",
"noAccessWithoutMask": "",
"accessControlStatus": "",
"accessControlTimetable": "",
"timetableUpdTs": ""
},
"success": true
}
```

3.4.4 Delete BI Device Configuration

API Endpoint	{{studio_base_url}}/DEVICE/biDeviceInfo/deleteBIDeviceConfig			
Method	POST			
Request Format	JSON			
	Name	Required	Type	Description

Request Parameters	deviceUuid	Yes	String	Device unique ID
	operatePerson	Yes	String	Operator
Response Parameters	Name	Required	Type	Description
	data		String	null
Sample Request	{ "deviceUuid": "1257915401329709058", "operatePerson": "1" }			
Sample Response	{ "code": "0000", "msg": "成功", "data": null, "success": true }			

4 Gate Control Policy API

4.1 Gate Control Policy API Return Code

Return Code	Return Code Description
0000	Success
0001	Failed
03003000	The data has already existed
03003001	Data does not exist
03003002	the caller is associated with authorization
03003003	There is a relationship
03003004	policyId is Required
03003005	policyGroups is Required
03003006	policyDevices is Required
03003007	controlDuration of noshowpolicy is required
03003008	controlDurationOff/on of workshiftpolicy is required
03003009	policyActions is Required
03003010	File upload failure
03003011	Delete file failure
03003012	file does not exist
03003013	File format is incorrect
03003014	Failure to create gate control policy
03003015	The basis for generating special personnel policy does not exist
03010001	[ActionInfo] actionType is required
03010001	[ActivateControlPolicyReq] operatePerson is required
03010001	[ActivateControlPolicyReq] policyIds is required
03010001	[AddDeviceInControlPolicyReq] policyIds is required
03010001	[AddDeviceInControlPolicyReq] deviceInfos is required
03010001	[AddDeviceInControlPolicyReq] operatePerson is required
03010001	[ControlPolicyBaseReq] policyName is required
03010001	[ControlPolicyBaseReq] policyType is required
03010001	[ControlPolicyBaseReq] operatePerson is required
03010001	[ControlPolicyBaseReq] policyActions is required
03010001	[DeactivateControlPolicyReq] operatePerson is required
03010001	[DeactivateControlPolicyReq] policyIds is required
03010001	[IsDeviceInControlPolicyReq] deviceIds is required
03010001	[RemoveDeviceInPolicyReq] deviceInfos is required
03010001	[RemoveTargetInPolicyReq] policyTargets is required

03990001	Service cannot be accessed
03003009	Exceed the maximum file limit

4.2 Parameter Definition

4.2.1 Gate Control Policy Type Definition

Gate Control Policy Type ID	Gate Control Policy Type
0	General
1	Work overtime
2	Absent
4	Backend recognition

4.2.2 Access Control Type Definition

Access Control Type ID	Access Control Type
0	Block
1	Allow

4.2.3 Notification Type Definition

Notification Type ID	Notification Type	Notification Target
2	HTTP push	HTTP push address
3	SMS	Phone number
4	Email	Email address

4.3 Gate Control Policy Management

4.3.1 Create Gate Control Policy

API Endpoint	{{studio_base_url}}/POLICYCONTROL/gateControlPolicy/create			
Method	POST			
Request Format	JSON			
Request Parameters	Name	Required	Type	Description
	policyName	Yes	String	Gate control policy name
	policyActions	Yes	Array[PolicyAction]	Access control and policy notification
	policyType	Yes	String	Gate control policy type Please refer to 4.2.1 Gate Control Policy Type Definition
	policyTargets	Yes	Array[PolicyTarget]	Target groups

	operatePerson	Yes	String	Operator
	policyDevices	No	Array[PolicyDevice]	Target devices
	policyZones	No	Array[PolicyZone]	Target zones
	validPeriodStart	No	String	Effective date (yyyyMMdd)
	validPeriodEnd	No	String	Expiration date (yyyyMMdd)
	controlPeriodWeekday	No	String	Active days of week (1-7), separated by commas For example, if the policy only takes effects on Sat and Sun, then put controlPeriodWeekday = "6,7"
	monitorPeriodStart	No	String	Effective time (HH:mm:ss)
	monitorPeriodEnd	No	String	Expiration time (HH:mm:ss)
	controlDurationOn	No	String	Mandatory when policyType=1 Continuous working days
	controlDurationOff	No	String	Mandatory when policyType=1 Rest days
	controlDuration	No	String	Mandatory when policyType=2 Absent days
PolicyAction	desc	No	String	Description
	actionType	Yes	String	Access control type / Notification Type Please refer to 4.2.2 Access Control Type Definition & 4.2.3 Notification Type Definition
	actionTarget	Yes	String	Notification target Please refer to 4.2.3 Notification Type Definition
	actionContent	No	String	Message
PolicyTarget	operator	Yes	String	2: Equal to 3: Not equal to
	targetId	Yes	String	Target type ID 0: Person group
	targetValue	Yes	String	Person group unique ID
	classId	No	String	The target groups that need to be matched at the same time share the same classId. Start from 0.

PolicyDevice	deviceId	Yes	String	Device unique ID
	deviceSerial	No	String	Device serial
	deviceType	No	String	Device type
PolicyZone	zoneId	Yes	String	Zone ID
Response Parameters	Name	Required	Type	Description
	data		String	Gate control policy unique ID, generated by the system, be used to uniquely identify the gate control policy

Sample Request	<pre>{ "policyName": "Policy name", "policyActions": [{ "actionContent": "Allow Access", "actionTarget": "", "actionType": "1" }, { "actionContent": "", "actionTarget": "http://pushserver", "actionType": "2" }], "policyType": "0", "policyTargets": [{ "targetValue": "1", "operator": "2", "classId": "0", "targetId": "0" }], "operatePerson": "9", "policyDevices": [{ "deviceId": "1257915401329709058", "deviceSerial": "", "deviceType": "" }] }</pre>			
----------------	--	--	--	--

	<pre> }], "policyZones": [{ "zoneId": "1242648468460146690" }], "validPeriodStart": "20200622", "validPeriodEnd": "20200630", "controlPeriodWeekday": "6,7", "monitorPeriodStart": "09:30:00", "monitorPeriodEnd": "18:30:00", "controlDurationOn": "", "controlDurationOff": "", "controlDuration": "", "desc": "Description" }</pre>
Sample Response	<pre>{ "code": "0000", "msg": "成功", "data": "1272815419639468034", "success": true }</pre>

4.3.2 Deactivate Gate Control Policy

API Endpoint	{{studio_base_url}}/POLICYCONTROL/gateControlPolicy/deactivate			
Method	POST			
Request Format	JSON			
Request Parameters	Name	Required	Type	Description
	policyIds	Yes	Array[String]	Policy unique IDs
	operatePerson	Yes	String	Operator
Response Parameters	Name	Required	Type	Description
	data		String	null
Sample Request	<pre>{ "policyIds": ["1272815419639468034"] }</pre>			

	<pre>], "operatePerson": "1" } </pre>
Sample Response	<pre> { "code": "0000", "msg": "成功", "data": null, "success": true } </pre>

4.3.3 Update Gate Control Policy

API Endpoint	{{studio_base_url}}/POLICYCONTROL/gateControlPolicy/update			
Method	POST			
Request Format	JSON			
Request Parameters	Name	Required	Type	Description
	policyId	Yes	String	Gate control policy unique ID
	policyName	Yes	String	Gate control policy name
	policyActions	Yes	Array[PolicyAction]	Access control and policy notification
	policyType	Yes	String	Gate control policy type Please refer to 4.2.1 Gate Control Policy Type Definition
	policyTargets	Yes	Array[PolicyTarget]	Target groups
	operatePerson	Yes	String	Operator
	policyDevices	No	Array[PolicyDevice]	Target devices
	policyZones	No	Array[PolicyZone]	Target zones
	validPeriodStart	No	String	Effective date (yyyyMMdd)
	validPeriodEnd	No	String	Expiration date (yyyyMMdd)
	controlPeriodWeekday	No	String	Active days of week (1-7), separated by commas For example, if the policy only takes effects on Sat and Sun, then put controlPeriodWeekday = "6,7"
	monitorPeriodStart	No	String	Effective time (HH:mm:ss)

	monitorPeriodEnd	No	String	Expiration time (HH:mm:ss)
	controlDurationOn	No	String	Mandatory when policyType=1 Continuous working days
	controlDurationOff	No	String	Mandatory when policyType=1 Rest days
	controlDuration	No	String	Mandatory when policyType=2 Absent days
	desc	No	String	Description
PolicyAction	actionType	Yes	String	Access control type / Notification Type Please refer to 4.2.2 Access Control Type Definition & 4.2.3 Notification Type Definition
	actionTarget	Yes	String	Notification target Please refer to 4.2.3 Notification Type Definition
	actionContent	No	String	Message
PolicyTarget	operator	Yes	String	2: Equal to 3: Not equal to
	targetId	Yes	String	Target type ID 0: Person group
	targetValue	Yes	String	Person group unique ID
	classId	No	String	The target groups that need to be matched at the same time share the same classId. Start from 0.
PolicyDevice	deviceId	Yes	String	Device unique ID
	deviceSerial	No	String	Device serial
	deviceType	No	String	Device type
PolicyZone	zoneId	Yes	String	Zone ID
Response Parameters	Name	Required	Type	Description
	data		String	null
Sample Request	<pre>{ "policyId": "1272817325392793602", "policyName": "Policy name", "policyActions": [{ "actionContent": "Allow Access",</pre>			

```
        "actionTarget": "",
        "actionType": "1"
    },
    {
        "actionContent": "",
        "actionTarget": "http://pushserver",
        "actionType": "2"
    }
],
"policyType": "0",
"policyTargets": [
    {
        "targetValue": "1",
        "operator": "2",
        "classId": "0",
        "targetId": "0"
    }
],
"operatePerson": "9",
"policyDevices": [
    {
        "deviceId": "1257915401329709058",
        "deviceSerial": "",
        "deviceType": ""
    }
],
"policyZones": [
    {
        "zoneId": "1242648468460146690"
    }
],
"validPeriodStart": "20200622",
"validPeriodEnd": "20200630",
"controlPeriodWeekday": "6,7",
"monitorPeriodStart": "09:30:00",
"monitorPeriodEnd": "18:30:00",
"controlDurationOn": "",
"controlDurationOff": "",
```

	<pre>"controlDuration": "", "desc": "Description" }</pre>
Sample Response	<pre>{ "code": "0000", "msg": "成功", "data": null, "success": true }</pre>

4.3.4 List Gate Control Policy (Pagination)

API Endpoint	{{studio_base_url}}/POLICYCONTROL/gateControlPolicy/getPageList			
Method	POST			
Request Format	JSON			
Request Parameters	Name	Required	Type	Description
	current	Yes	Integer	Query results on page #. Start from 1.
	size	Yes	Integer	Number of results per page
	policyType	No	String	Gate control policy type Please refer to 4.2.1 Gate Control Policy Type Definition
	privilege	No	String	Privilege (Department ID)
Response Parameters	Name	Required	Type	Description
	data		Array[Policy]	null
Policy	policyName		String	Gate control policy name
	policyType		String	Gate control policy type Please refer to 4.2.1 Gate Control Policy Type Definition
	policyTargets		Array[PolicyTarget]	Target groups
	policyDevices		Array[PolicyDevice]	Target devices
	monitorPeriodStart		String	Effective time (HH:mm:ss)
	monitorPeriodEnd		String	Expiration time (HH:mm:ss)
	validPeriodStart		String	Effective date (yyyyMMdd)
	validPeriodEnd		String	Expiration date (yyyyMMdd)
	controlDuration		String	Mandatory when policyType=2

				Absent days
	controlDurationOn		String	Mandatory when policyType=1 Continuous working days
	controlDurationOff		String	Mandatory when policyType=1 Rest days
	controlPeriodWeekday		String	Active days of week (1-7), separated by commas For example, if the policy only takes effects on Sat and Sun, then put controlPeriodWeekday = "6,7"
	actionInfoResList		Array[ActionInfo]	
	privilege		String	Privilege (Department ID)
	sts		String	Gate control policy status 0: Effective 1: Ineffective 2: Suspended 3: Deleted 9: Created
PolicyTarget	privilege		String	null
	targetId		String	Target type ID 0: Person group
	targetValue		String	Person group unique ID
	targetCnName		String	null
	targetEnName		String	null
	operator		String	2: Equal to 3: Not equal to
	classId		String	The target groups that need to be matched at the same time share the same classId
PolicyDevice	deviceId		String	Device unique ID
	deviceCnName		String	null
	deviceEnName		String	null
	deviceType		String	Device type
	deviceSerial		String	Device serial
	importSource		String	Device source 0: Portal, 1: Map service

ActionInfo	actionType		String	Access control type / Notification Type Please refer to 4.2.2 Access Control Type Definition & 4.2.3 Notification Type Definition
	actionTarget		String	Notification target Please refer to 4.2.3 Notification Type Definition
	actionContent		String	Message
Sample Request	<pre>{ "current": 1, "size": 1, "policyType": "", "privilege": "" }</pre>			
Sample Response	<pre>{ "data": [{ "policyId": "1272817325392793602", "policyName": "Policy name", "policyType": "0", "policyTargets": [{ "privilege": null, "targetId": "0", "targetValue": "1", "targetCnName": null, "targetEnName": null, "operator": "2", "classId": "0" }], "policyDevices": [{ "deviceId": "1257915401329709058", "deviceCnName": null, "deviceEnName": null, "deviceType": null, </pre>			

```
        "deviceSerial": null,
        "importSource": "0"
      }
    ],
    "monitorPeriodStart": "09:30:00",
    "monitorPeriodEnd": "18:30:00",
    "validPeriodStart": "20200622",
    "validPeriodEnd": "20200630",
    "controlDuration": null,
    "controlDurationOn": null,
    "controlDurationOff": null,
    "controlPeriodWeekday": "6,7",
    "actionInfoResList": [
      {
        "actionType": "1",
        "actionTarget": "",
        "actionContent": "Allow Access"
      },
      {
        "actionType": "2",
        "actionTarget": "http://pushserver",
        "actionContent": ""
      }
    ],
    "privilege": "0",
    "sts": "9"
  }
],
"size": 1,
"total": 13,
"code": "0000",
"msg": "success"
}
```

4.3.5 Query Gate Control Policy Information by Policy ID

API Endpoint	{{studio_base_url}}/POLICYCONTROL/gateControlPolicy/info
Method	POST

Request Format	form-data			
Request Parameters	Name	Required	Type	Description
	policyId	Yes	String	Gate control policy unique ID
Response Parameters	Name	Required	Type	Description
	data		PolicyInfo	
PolicyInfo	policyId		String	Gate control policy unique ID
	policyName		String	Gate control policy name
	policyType		String	Gate control policy type Please refer to 4.2.1 Gate Control Policy Type Definition
	policyTargets		Array[PolicyTarget]	Target groups
	policyTargetNames		String	null
	policyDevices		Array[PolicyDevice]	Target devices
	monitorPeriodStart		String	Effective time (HH:mm:ss)
	monitorPeriodEnd		String	Expiration time (HH:mm:ss)
	validPeriodStart		String	Effective date (yyyyMMdd)
	validPeriodEnd		String	Expiration date (yyyyMMdd)
	controlDuration		String	Mandatory when policyType=2 Absent days
	controlDurationOn		String	Mandatory when policyType=1 Continuous working days
	controlDurationOff		String	Mandatory when policyType=1 Rest days
	controlPeriodWeekday		String	Active days of week (1-7), separated by commas For example, if the policy only takes effects on Sat and Sun, then put controlPeriodWeekday = "6,7"
	desc		String	Description
	policyActions		Array[PolicyAction]	
	sts		String	null
	privilege		String	Privilege (Department ID)
PolicyTarget	privilege		String	null
	targetId		String	Target type ID 0: Person group

	targetValue		String	Person group unique ID
	targetCnName		String	null
	targetEnName		String	null
	operator		String	2: Equal to 3: Not equal to
	classId		String	The target groups that need to be matched at the same time share the same classId
PolicyDevice	deviceId		String	Device unique ID
	deviceCnName		String	Device name
	deviceEnName		String	Device name
	deviceType		String	Device type
	deviceSerial		String	Device serial
	importSource		String	Device source 0: Portal, 1: Map service
PolicyAction	actionType		String	Access control type / Notification Type Please refer to 4.2.2 Access Control Type Definition & 4.2.3 Notification Type Definition
	actionTarget		String	Notification target Please refer to 4.2.3 Notification Type Definition
	actionContent		String	Message
Sample Request	--form 'policyId=1272817325392793602'			
Sample Response	<pre>{ "code": "0000", "msg": "成功", "data": { "policyId": "1272817325392793602", "policyName": "Policy name", "policyType": "0", "policyTargets": [{ "privilege": null, "targetId": "0", </pre>			

```
        "targetValue": "1",
        "targetCnName": "Staff",
        "targetEnName": null,
        "operator": "2",
        "classId": "0"
    }
],
"policyTargetNames": null,
"policyDevices": [
    {
        "deviceId": "1257915401329709058",
        "deviceCnName": "SPS020STDC19G07718",
        "deviceEnName": "SPS020STDC19G07718",
        "deviceType": "15",
        "deviceSerial": "SPS020STDC19G07718_0",
        "importSource": "0"
    }
],
"monitorPeriodStart": "09:30:00",
"monitorPeriodEnd": "18:30:00",
"validPeriodStart": "20200622",
"validPeriodEnd": "20200630",
"controlDuration": null,
"controlDurationOn": null,
"controlDurationOff": null,
"controlPeriodWeekday": "6,7",
"desc": "Description",
"policyActions": [
    {
        "actionType": "1",
        "actionTarget": "",
        "actionContent": "Allow Access"
    },
    {
        "actionType": "2",
        "actionTarget": "http://pushserver",
        "actionContent": ""
    }
]
```

```
    ],  
    "sts": null,  
    "privilege": "0"  
  },  
  "success": true  
}
```

4.3.6 Add Devices to Gate Control Policy

API Endpoint	{{studio_base_url}}/POLICYCONTROL/gateControlPolicy/addDeviceInControlPolicy			
Method	POST			
Request Format	JSON			
Request Parameters	Name	Required	Type	Description
	deviceInfos	Yes	Array[DeviceInfo]	Devices to add to policies
	policyIds	Yes	Array[String]	Policies to be added to
	operatePerson	Yes	String	Operator
	privilege	No	String	Privilege (Department ID)
DeviceInfo	deviceId	Yes	String	Device unique ID
	importSource	Yes	String	Device source 0: Portal, 1: Map service
Response Parameters	Name	Required	Type	Description
	data		String	null
Sample Request	{ "deviceInfos": [{ "deviceId": "1272444628489998337", "importSource": "0" }], "policyIds": ["1272817325392793602", "1272445532513505282"], "operatePerson": "1" }			
Sample Response	{ "code": "0000", 			

```
"msg": "成功",  
"data": null,  
"success": true  
}
```

4.3.7 Remove Devices from Gate Control Policy

API Endpoint	{{studio_base_url}}/POLICYCONTROL/gateControlPolicy/removeDeviceInControlPolicy			
Method	POST			
Request Format	JSON			
Request Parameters	Name	Required	Type	Description
	deviceInfos	Yes	Array[DeviceInfo]	Devices to remove from policy
	policyId	Yes	String	Policy to be removed from
	operatePerson	Yes	String	Operator
DeviceInfo	deviceId	Yes	String	Device unique ID
	importSource	Yes	String	Device source 0: Portal, 1: Map service
Response Parameters	Name	Required	Type	Description
	data		String	null
Sample Request	{ "deviceInfos": [{ "deviceId": "1272444628489998337", "importSource": "0" }], "policyId": "1272817325392793602", "operatePerson": "1" }			
Sample Response	{ "code": "0000", "msg": "成功", "data": null, "success": true }			

4.3.8 Remove Target Groups from Gate Control Policy

API Endpoint	{{studio_base_url}}/POLICYCONTROL/gateControlPolicy/removeTargetInControlPolicy
--------------	---

Method	POST			
Request Format	JSON			
Request Parameters	Name	Required	Type	Description
	policyTargets	Yes	Array[PolicyTarget]	Person groups to remove from policy
	policyId	Yes	String	Policy to be removed from
	privilege	No	String	Privilege (Department ID)
PolicyTarget	operator	No	String	2: Equal to 3: Not equal to
	targetId	No	String	Target type ID 0: Person group
	targetValue	No	String	Person group unique ID
	classId	No	String	The target groups that need to be matched at the same time share the same classId. Start from 0.
Response Parameters	Name	Required	Type	Description
	data		String	null
Sample Request	<pre>{ "policyTargets": [{ "operator": "2", "targetId": "0", "targetValue": "1", "classId": "0" }], "policyId": "1270932137394049025" }</pre>			
Sample Response	<pre>{ "code": "0000", "msg": "成功", "data": null, "success": true }</pre>			

4.3.9 Query Gate Control Policy of Device

API Endpoint	{{studio_base_url}}/POLICYCONTROL/gateControlPolicy/isDeviceInControlPolicy			
Method	POST			
Request Format	JSON			
Request Parameters (in JSON array)	Name	Required	Type	Description
	deviceId	Yes	Array[String]	Device unique ID
Response Parameters	Name	Required	Type	Description
	data		Array[DevicePolicy]	
DevicePolicy	deviceId		String	Device unique ID
	policyId		String	Gate control policy unique ID
	policyName		String	Gate control policy name
	policyType		String	Gate control policy type Please refer to 4.2.1 Gate Control Policy Type Definition
Sample Request	["1257915401329709058","1257879674839633922"]			
Sample Response	<pre>{ "code": "0000", "msg": "成功", "data": [{ "deviceId": "1257915401329709058", "policyId": "1257916436144197633", "policyName": "SPS020STDC19G07718", "policyType": "0" }, { "deviceId": "1257879674839633922", "policyId": "1266270965512278018", "policyName": "SenseThunder 05927", "policyType": "0" }, { "deviceId": "1257915401329709058", "policyId": "1272817325392793602", "policyName": "Policy name", </pre>			

```
"policyType": "0"

    }

  ],

  "success": true

}
```

4.3.10 Query Gate Control Policy of Person Group

API Endpoint	{{studio_base_url}}/POLICYCONTROL/gateControlPolicy/isTargetInControlPolicy			
Method	POST			
Request Format	JSON			
Request Parameters	Name	Required	Type	Description
	targetId	Yes	String	Target type ID 0: Person group
	targetValue	Yes	String	Person group unique ID
Response Parameters	Name	Required	Type	Description
	data		Array[String]	Gate control policy unique IDs
Sample Request	{ "targetId": "0", "targetValue": "1" }			
Sample Response	{ "code": "0000", "msg": "成功", "data": ["1243431058267967490", "1245608535987851265", "1272817325392793602"], "success": true }			

4.3.11 List Gate Control Policy (Pagination)

API Endpoint	{{studio_base_url}}/POLICYCONTROL/gateControlPolicy/PageList
Method	POST
Request Format	JSON

	size	Yes	Integer	Number of results per page
	privilege	No	String	Privilege (Department ID)
	policyName	No	String	Gate monitor policy name
	policyType	No	String	Gate control policy type Please refer to 4.2.1 Gate Control Policy Type Definition
	policyId	No	String	Monitor policy unique ID
	desc	No	String	Description
	monitorPeriodFrom	No	String	Effective time (HH:mm:ss)
	monitorPeriodTo	No	String	Expiration time (HH:mm:ss)
	validPeriodFrom	No	String	Effective date (yyyyMMdd)
	validPeriodTo	No	String	Expiration date (yyyyMMdd)
Response Parameters	Name	Required	Type	Description
	data		Array[Policy]	null
Policy	policyName		String	Gate control policy name
	policyType		String	Gate control policy type Please refer to 4.2.1 Gate Control Policy Type Definition
	policyTargets		Array[PolicyTarget]	Target groups
	policyDevices		Array[PolicyDevice]	Target devices
	monitorPeriodStart		String	Effective time (HH:mm:ss)
	monitorPeriodEnd		String	Expiration time (HH:mm:ss)
	validPeriodStart		String	Effective date (yyyyMMdd)
	validPeriodEnd		String	Expiration date (yyyyMMdd)
	controlDuration		String	Mandatory when policyType=2 Absent days
	controlDurationOn		String	Mandatory when policyType=1 Continuous working days
	controlDurationOff		String	Mandatory when policyType=1 Rest days
	controlPeriodWeekday		String	Active days of week (1-7), separated by commas For example, if the policy only takes effects on Sat and Sun, then put controlPeriodWeekday = "6,7"
	actionInfoResList		Array[ActionInfo]	

	privilege		String	Privilege (Department ID)
	sts		String	Gate control policy status 0: Effective 1: Ineffective 2: Suspended 3: Deleted 9: Created
Sample Request	<pre>{ "current": 1, "size": 10, "policyName": "", "policyType": "", "desc": "", "monitorPeriodFrom": "", "monitorPeriodTo": "", "validPeriodFrom": "", "validPeriodTo": "" }</pre>			
Sample Response	<pre>{ "data": [{ "policyId": "1437362081983631362", "externalPolicyId": null, "policyName": "Access Policy 1", "policyType": "0", "policyDeviceNames": null, "validPeriodStart": "", "validPeriodEnd": "", "monitorPeriodStart": "15:00:00", "monitorPeriodEnd": "18:00:00", "controlPeriodWeekday": "1,2,3,4,5", "desc": "Access Policy 1", "sts": "9", "stsDesc": "建立", "lastModTs": "2021-09-14 15:31:47" }], "size": 10, }</pre>			

	<pre>"total": 1, "code": "0000", "msg": "success" }</pre>
--	---

5 Monitor Policy API

5.1 Monitor Policy API Return Code

Return Code	Return Code Description	Possible Causes	Solution Suggestion
0000	Success		
04001000	monitor policy has already exist	Duplicate monitor policy	Monitor policy name should be unique.
04001020	monitor policy doesn't exist	Monitor policy doesn't exist	Check the database if the policy has been deleted. Check the operator and deletion time, to determine whether it is deleted normally.
04001021	gate monitor policy doesn't exist	Gate monitor policy doesn't exist	Check the database if the policy has been deleted. Check the operator and deletion time, to determine whether it is deleted normally.
04001022	face monitor policy doesn't exist	Face monito policy doesn't exist	Check the database if the policy has been deleted. Check the operator and deletion time, to determine whether it is deleted normally.
04001023	crowd monitor policy doesn't exist	Crowd monitor policy doesn't exist	Check the database if the policy has been deleted. Check the operator and deletion time, to determine whether it is deleted normally.
04001024	image monitor policy doesn't exist	Attribute monitor policy doesn't exist	Check the database if the policy has been deleted. Check the operator and deletion time, to determine whether it is deleted normally.
04001040	create monitor policy failed	Monitor policy creation failed	Check the status of DB and monitor policy service. Check if input parameters are valid.

04001041	create gate monitor policy failed	Gate monitor policy creation failed.	Check the status of DB and monitor policy service. Check if input parameters are valid.
04001042	create monitor policy and notification relation failed	Policy-notification mapping creation failed	Check the status of DB and monitor policy service. Check if input parameters are valid.
04001043	create monitor policy and device relation failed	Policy-device mapping creation failed	Check the status of DB and monitor policy service. Check if input parameters are valid.
04001044	create monitor policy and target relation failed	Policy-target mapping creation failed	Check the status of DB and monitor policy service. Check if input parameters are valid.
04001045	create face monitor policy failed	Face monitor policy creation failed	Check the status of DB and monitor policy service. Check if input parameters are valid.
04001046	add device in monitor policy failed	Add Device to monitor policy failed	Check the status of DB and monitor policy service. Check if input parameters are valid.
04001051	create crowd monitor policy failed	Crowd monitor policy creation failed	Check the status of DB and monitor policy service. Check if input parameters are valid.
04001052	create image monitor policy failed	Attribute monitor policy creation failed	Check the status of DB and monitor policy service. Check if input parameters are valid.
04011056	create roi monitor policy failed"	Region monitor policy creation failed	Check the status of DB and monitor policy service. Check if input parameters are valid.
04011057	create monitoring roi policy failed	Cross-line monitor policy creation failed	Check the status of DB and monitor policy service.

			Check if input parameters are valid.
04001060	update monitor policy failed	Monitor policy update failed	Check the status of DB and monitor policy service. Check if input parameters are valid.
04001061	update gate monitor policy failed	Gate monitor policy update failed	Check the status of DB and monitor policy service. Check if input parameters are valid.
04001062	update crowd monitor policy failed	Crowd monitor policy update failed	Check the status of DB and monitor policy service. Check if input parameters are valid.
04001063	update face monitor policy failed	Face monitor policy update failed	Check the status of DB and monitor policy service. Check if input parameters are valid.
04001064	update image monitor policy failed	Attribute monitor policy update failed	Check the status of DB and monitor policy service. Check if input parameters are valid.
04001065	update monitor policy and notification relation failed	Policy-notification mapping update failed	Check the status of DB and monitor policy service. Check if input parameters are valid.
04001066	update monitor policy and device relation failed	Policy-device mapping update failed	Check the status of DB and monitor policy service. Check if input parameters are valid.
04001067	update monitor policy and target relation failed	Policy-target mapping update failed	Check the status of DB and monitor policy service. Check if input parameters are valid.
04001068	update target definition info failed	Target definition info update failed	Check the status of DB and monitor policy service. Check if input parameters are valid.

04001069	update target dictionary info failed	Target dictionary info update failed	Check the status of DB and monitor policy service. Check if input parameters are valid.
04001070	failed!data already exists!	Data already exists	Check if the data already exists
04001071	update roi monitor policy failed	Region monitor policy update failed	Check the status of DB and monitor policy service. Check if input parameters are valid.
04001072	update monitoring roi policy failed	Cross-line monitor policy update failed	Check the status of DB and monitor policy service. Check if input parameters are valid.
04001080	activate monitor policy failed	Attribute monitor policy update failed	Check the status of DB and monitor policy service. Check if input parameters are valid.
04001081	deactivate monitor policy failed	Monitor policy deletion failed	Check the status of DB and monitor policy service. Check if input parameters are valid.
04011090	target value is required	Missing target attributes	Input target attributes
04011091	event type is required	Missing control result	Input control result
04011092	device is required	Missing target devices	Input target devices
04011093	notification is required	Missing notification type	Input notification type
04011094	monitor policy name is required	Missing monitor policy name	Input monitor policy name
04011095	monitor policy type is required	Missing monitor policy type	Input monitor policy type
0401999	deptId can not be empty	Missing column deptId	Input column deptId
04011098	tDid is required	Missing column tDid	Input column tDid
04011099	targetOptionIdentifier is required	Missing column targetOptionIdentifier	Input column targetOptionIdentifier
04011100	targetOptionEn is required	Missing column targetOptionEn	Input column targetOptionEn
04011101	targetOption is required	Missing column targetOption	Input column targetOption

04011102	operatePerson is required	Missing column operatePerson	Input column operatePerson
04011103	dlcid is required	Missing column dlcid	Input column dlcid
04011104	trackLevel is required	Missing column trackLevel	Input column trackLevel

5.2 Parameter Definition

5.2.1 Monitor Policy Type Definition

Monitor Policy Type ID	Monitor Policy Type
0	Face monitor
1	Attribute monitor
2	Crowd monitor
3	Gate monitor
4	Region monitor
5	Cross-line monitor
6	X-model
7	Crowd Counting

5.2.2 Notification Type Definition

Notification Type ID	Notification Type	Notification Target
2	HTTP push	HTTP push address
3	SMS	Phone number
4	Email	Email address

5.2.3 Control Result Definition

eventType Value	Control Result
0	Deny
1	Allow
2	All
-1	Stranger
-2	Non-living
-3	Tamper alarm
-4	Forced opening
-5	Gate magnetic timeout
-6	Password attack
-7	Dirty lens
-8	Fire alarm

-9	Bluetooth in low battery
-10	Thermographic camera connection error
-11	Feature extraction failed

5.2.4 Attribute Feature Definition

5.2.4.1 Person Group

Feature ID	Feature Name	Feature Value	Feature Value Name
0	Person Group	N/A(Group unique ID) -99 for Strangers	

5.2.4.2 Intelligent Device

Feature ID	Feature Name	Feature Value	Feature Value Name
75	AbnormalType	0	None
		1	FaceIDMismatch
		2	FaceCardMismatch
		3	FaceCodeMismatch
		4	InvalidVisitingPeriod
		5	InvalidAccessibleTime
		6	InvalidIDCard
		7	InvalidICCard
		8	InvalidQRCode
		9	AbnormalBodyTemperature

5.2.4.3 Pedestrian

Feature ID	Feature Name	Feature Value	Feature Value Name
-1	All(Human)	0	No
		1	Yes
4	Age	0	Adult
		1	Old
		2	Child
5	Gender	0	Male
		1	Female
6	Angle	0	Front
		1	Side
		2	Back
7	WithUmbrella	0	NoUmbrella
		1	WithUnbrella

8	BagType	0	HandBag
		1	ShoulderBag
		2	Backpack
		3	Trolley
		4	NoBag
		8	WaistPack
9	SleevesType	0	ShortSleeve
		1	LongSleeve
		2	Shirtless
10	BottomsType	0	Trousers
		1	Shorts
		2	Skirt
11	TopsColor	0	Black
		1	White
		2	Gray
		3	Red
		4	Yellow
		5	Blue
		6	Green
		7	Purple
		8	Orange
		9	Brown
		10	Cyan
		11	Pink
		13	TRANSPARENT
12	BottomsColor	0	Black
		1	White
		2	Gray
		3	Red
		4	Yellow
		5	Blue
		6	Green
		7	Purple
		8	Orange
		9	Brown
		10	Cyan
		11	Pink

		13	TRANSPARENT
13	TopsPattern	0	Solid
		1	Stripes
		2	Graphic
		3	Joint
		4	Checks
14	UmbrellaColor	1	Black
		2	White
		3	Gray
		4	Red
		5	Blue
		6	Yellow
		7	Orange
		8	Brown
		9	Green
		10	Purple
		11	Cyan
		12	Pink
		13	Transparent
15	ShoesType	2	LeatherShoes
		5	Sandals
		6	CasualShoes
		7	Boots
16	HatsType	0	NoHat
		5	BucketHat
		6	Bonnet
		7	Cap
		9	Helmet
17	BottomsPattern	0	Solid
		1	Stripes
		2	Graphic
		3	Joint
		4	Checks
18	HoldingThings	0	HoldObject
22	TopsType	1	BusinessSuit
		3	T-Shirt
		4	Shirt

		6	Jacket
		8	LongCoat
		10	SWEATER
		12	DownCoat
		13	SportsWear
		17	Dress
		99	Others
23	HairStyle	6	Bald
		11	Long
		100	Short
24	HairColor	1	Black
		2	White
		3	GRAY
		4	RED
		5	BLUE
		6	Yellow
		7	ORANGE
		8	BROWN
		9	GREEN
		10	PURPLE
		11	CRAN
		12	PINK
		13	TRANSPARENT
25	ShoesColor	1	Black
		2	White
		3	Gray
		4	Red
		5	Blue
		6	Yellow
		7	Orange
		8	Brown
		9	Green
		10	Purple
		11	Cyan
		12	Pink
		13	TRNASPARENT
26	BagColor	1	Black

		2	White
		3	Gray
		4	Red
		5	Blue
		6	Yellow
		7	Orange
		8	Brown
		9	Green
		10	Purple
		11	Cyan
		12	Pink
		13	TRANSPARENT
27	WithMask	0	Unknown
		1	Unrecognized
		2	NoMask
		3	WithMask
		4	WithAbnormalMask
28	MaskColor	1	Black
		2	White
		3	Gray
		4	RED
		5	Blue
		6	YELLOW
		7	ORANGE
		8	BROWN
		9	GREEN
		10	PURPLE
		11	CRAN
		12	PINK
		13	TRANSPARENT
29	HatsColor	1	Black
		2	White
		3	Gray
		4	RED
		5	Blue
		6	YELLOW
		7	ORANGE

		8	BROWN
		9	GREEN
		10	PURPLE
		11	CRAN
		12	PINK
		13	TRNASPARENT
55	SkinColor	2	Yellow
		3	Black
		4	White
56	ExpressionType	1	Glasses
		2	OrdinaryGlasses
		3	Anger
		4	Sad
		5	Disgust
		6	Fear
		7	Surprise
		8	Normal
		9	Calm
		10	NoGlasses
		11	Joy
		12	Confusion
		13	Scream
		14	SunGlasses
		15	Depression
		16	Yawn
57	AgeEstimation	N/A(Estimated Age)	
58	UniformType	0	GeneralUniform
		1	OfficeUniform
		2	WorkerUniform
		3	ChefClothes
		4	MedicalClothes
		5	PoliceUniform
		6	FireFightingCoat
		8	ElemeUniform
		10	MeituanUniform
59	PedestrianBehavior	0	Normal
		1	HoldingThings

		2	PlayingWithSmartPhone
		3	MakingPhoneCall
		5	UnDetected
60	WithGlove	0	WithGlove
		1	NoGlove
61	Pose	0	Stand
		1	Sit
		2	LieDown
		3	SleepOnTable
62	WithReflectiveVest	0	ReflectiveVest
		1	NoReflectiveVest
63	Smoking	0	IsSmoking
		1	NoSmoking
		2	Unknown
70	WithBag	0	WithBag
71	WithHat	0	WithHat
72	AgeUpperLimit	0	90
		1	59
		2	14
73	AgeLowerLimit	0	60
		1	18
		2	0
74	BodyTemperature	N/A(Body Temperature)	
76	WithBeard	0	Unknown
		1	Unrecognized
		2	noMoustache
		3	moustache

5.2.4.4 Vehicle

Feature ID	Feature Name	Feature Value	Feature Value Name
-3	All(Vehicle)	0	No
		1	Yes
1	vehicleColor	0	Gray
		1	White
		2	Red
		3	Black
		4	Blue

		5	Green
		6	Brown
		7	Yellow
		8	Purple
		9	Pink
2	vehicleType	0	Car
		1	Van
		2	SmallTruck
		3	BigTruck
		4	SUV
		5	LargeBus
		6	MediumBus
3	CarSubBrand	Name of the sub brand	
79	CarBrand	Name of the brand	
80	CarPlate	Car plate number	
81	VehicleCategory	0	GeneralVehicle
		1	Ambulance
		2	FireTruck
		3	GuardCar
		4	DumpTruck
		5	MixerTruck
		6	Taxi
82	CarPlateColor	0	None
		1	Black
		2	Blue
		3	Green
		4	Yellow
		5	YellowGreen
		6	White
83	CarPlateType	0	None
		1	LargeVehicle
		2	Trailer
		3	LargeNewEnergyVehicle
		4	SmallVehicle
		5	SmallNewEnergyVehicle
		6	EmbassyVehicle
		7	ConsularVehicle

		8	HK/MacaoVehicle
		9	TrainingVehicle
		10	PolicyVehicle
		11	FireRescueVehicle
		12	ArmedPoliceVehicle
		13	ArmyVehicle
		14	HK/MacaoLocalVehicle

5.2.4.5 Non-motor Vehicle

Feature ID	Feature Name	Feature Value	Feature Value Name
-2	All(Moped)	0	No
		1	Yes
77	MopedColor	0	Gray
		1	White
		2	Red
		3	Black
		4	Blue
		5	Green
		6	Brown
		7	Yellow
		8	Purple
		9	Pink
78	MopedType	0	e-bike
		1	motor
		2	bicycle
		3	tricycle
		4	pram
		5	wheelchair
		6	other moped

5.2.4.6 Crowd

Feature ID	Feature Name	Feature Value	Feature Value Name
21	CrowdAlarm	3	Lingering
54	PeopleCount	N/A(People count)	

5.3 Gate Monitor Policy Management

5.3.1 Create Gate Monitor Policy

API Endpoint	{{studio_base_url}}/POLICYMONITOR/gateMonitoringPolicy/create			
Method	POST			
Request Format	JSON			
Request Parameters	Name	Required	Type	Description
	policyName	Yes	String	Gate monitor policy name
	policyActions	Yes	Array[PolicyAction]	Notification
	policyType	Yes	String	Monitor policy type Please refer to 5.2.1 Monitor Policy Type Definition "3" for gate monitory policy
	eventType	Yes	String	Control result Please refer to 5.2.3 Control Result Definition
	policyDevices	Yes	Array[PolicyDevice]	Target devices
	deviceScope	Yes	String	Device aggregation "": No device aggregation "1": Single device calculation "N": Compound device calculation
	operatePerson	Yes	String	Operator
	validPeriodStart	No	String	Effective date (yyyyMMdd)
	validPeriodEnd	No	String	Expiration date (yyyyMMdd)
	monitorPeriodWeekday	No	String	Active days of week (1-7), separated by commas For example, if the policy only takes effects on Sat and Sun, then put monitorPeriodWeekday = "6,7"
	monitorPeriodStart	No	String	Effective time (HH:mm:ss)
	monitorPeriodEnd	No	String	Expiration time (HH:mm:ss)
	targetScope	No	String	Feature aggregation 1: Single feature calculation N: Compound feature calculation

	rollingWindow	No	String	Rolling interval Format: {Days}:{Hours}:{Minutes}
	rollingCount	No	String	Rolling accumulations
	consecutiveCount	No	String	Continuous interval Format: {Days}:{Hours}:{Minutes}
	consecutiveWindow	No	String	Continuous accumulation
	policyTargets	No	Array[PolicyTarget]	Target attribute features
	desc	No	String	Description
PolicyAction	actionType	Yes	String	Notification type Please refer to 5.2.2 Notification Type Definition
	actionTarget	Yes	String	Notification target Please refer to 5.2.2 Notification Type Definition
	actionContent	No	String	Message
PolicyDevice	deviceId	Yes	String	Device unique ID
	deviceSerial	No	String	Device serial
	deviceType	No	String	Device type
PolicyTarget	operator	Yes	String	0: Greater than 1: Less than 2: Equal to 3: Not equal to 4: Not exists 5: Exists
	targetId	Yes	String	Attribute feature ID Please refer to 5.2.4 Attribute Feature Definition - 5.2.4.1 Person Group & 5.2.4.2 Intelligent Device
	targetValue	Yes	String	Attribute feature value Please refer to 5.2.4 Attribute Feature Definition - 5.2.4.1 Person Group & 5.2.4.2 Intelligent Device
	classId	Yes	String	The target attributes that need to be monitored

				together share the same classId. Start from 0.
Response Parameters	Name	Required	Type	Description
	data		String	Monitor policy unique ID, generated by the system, be used to uniquely identify the monitor policy
Sample Request	<pre>{ "policyName": "Gate Monitor Policy", "policyActions": [{ "actionType": "2", "actionTarget": "http://pushserver", "actionContent": "" }], "policyType": "3", "eventType": "2", "policyDevices": [{ "deviceId": "1257879674839633922", "deviceSerial": "", "deviceType": "" }], "deviceScope": "N", "operatePerson": "9", "validPeriodStart": "", "validPeriodEnd": "", "monitorPeriodWeekday": "", "monitorPeriodEnd": "", "monitorPeriodStart": "", "targetScope": "1", "rollingWindow": "1:0:0", "rollingCount": "2", "policyTargets": [{ "operator": "0",</pre>			

	<pre>"targetId": "74", "targetValue": "38", "classId": "0" }, { "operator": "2", "targetId": "75" "targetValue": "9", "classId": "1", }], "desc": "Description" }</pre>
Sample Response	<pre>{ "code": "0000", "msg": "成功", "data": "1272815419639468034", "success": true }</pre>

5.3.2 Update Gate Monitor Policy

API Endpoint	{{studio_base_url}}/POLICYMONITOR/gateMonitoringPolicy/update			
Method	POST			
Request Format	JSON			
Request Parameters	Name	Required	Type	Description
	policyId	Yes	String	Monitor policy unique ID
	policyName	Yes	String	Gate monitor policy name
	policyActions	Yes	Array[PolicyAction]	Policy notification
	policyType	Yes	String	Monitor policy type Please refer to 5.2.1 Monitor Policy Type Definition
	eventType	Yes	String	Control result Please refer to 5.2.3 Control Result Definition
	policyDevices	Yes	Array[PolicyDevice]	Target devices
	operatePerson	Yes	String	Operator
	validPeriodStart	No	String	Effective date (yyyyMMdd)

	validPeriodEnd	No	String	Expiration date (yyyyMMdd)
	monitorPeriodWeekday	No	String	Active days of week (1-7), separated by commas For example, if the policy only takes effects on Sat and Sun, then put monitorPeriodWeekday = "6,7"
	monitorPeriodStart	No	String	Effective time (HH:mm:ss)
	monitorPeriodEnd	No	String	Expiration time (HH:mm:ss)
	targetScope	No	String	Feature aggregation 1: Single feature calculation N: Compound feature calculation
	deviceScope	No	String	Device aggregation 1: Single device calculation N: Compound device calculation
	rollingWindow	No	String	Rolling interval Format: {Days}:{Hours}:{Minutes}
	rollingCount	No	String	Rolling accumulations
	consecutiveCount	No	String	Continuous interval Format: {Days}:{Hours}:{Minutes}
	consecutiveWindow	No	String	Continuous accumulation
	policyTargets	No	Array[PolicyTarget]	Target attribute features
	desc	No	String	Description
PolicyAction	actionType	Yes	String	Notification type Please refer to 5.2.2 Notification Type Definition
	actionTarget	Yes	String	Notification target Please refer to 5.2.2 Notification Type Definition
	actionContent	No	String	Message
PolicyDevice	deviceId	Yes	String	Device unique ID
	deviceSerial	No	String	Device serial
	deviceType	No	String	Device type

PolicyTarget	operator	Yes	String	0: Greater than 1: Less than 2: Equal to 3: Not equal to 4: Not exists 5: Exists
	targetId	Yes	String	Attribute feature ID Please refer to 5.2.4 Attribute Feature Definition - 5.2.4.1 Person Group & 5.2.4.2 Intelligent Device
	targetValue	Yes	String	Attribute feature value Please refer to 5.2.4 Attribute Feature Definition - 5.2.4.1 Person Group & 5.2.4.2 Intelligent Device
	classId	Yes	String	The target attributes that need to be monitored together share the same classId. Start from 0.
Response Parameters	Name	Required	Type	Description
	data		String	Monitor policy unique ID
Sample Request	<pre>{ "policyId": "1272860662661386242", "policyName": "Gate Monitor Policy", "policyActions": [{ "actionType": "2", "actionTarget": "http://pushserver", "actionContent": "" }], "policyType": "3", "eventType": "2", "policyDevices": [{ "deviceId": "1257879674839633922", "deviceSerial": "", </pre>			

	<pre>"deviceType": "" }], "deviceScope": "N", "operatePerson": "9", "validPeriodStart": "", "validPeriodEnd": "", "monitorPeriodWeekday": "", "monitorPeriodEnd": "", "monitorPeriodStart": "", "targetScope": "1", "rollingWindow": "1:0:0", "rollingCount": "2", "policyTargets": [{ "operator": "0", "targetId": "74", "targetValue": "38", "classId": "0" }, { "operator": "2", "targetId": "75", "targetValue": "9", "classId": "1", }], "desc": "Description" }</pre>
Sample Response	<pre>{ "code": "0000", "msg": "成功", "data": "1272815419639468034", "success": true }</pre>

5.3.3 Query Gate Monitor Policy Information by Policy ID

API Endpoint	{{studio_base_url}}/POLICYMONITOR/gateMonitoringPolicy/info
--------------	---

Method	POST			
Request Format	form-data			
Request Parameters	Name	Required	Type	Description
	policyId	Yes	String	Monitor policy unique ID
Response Parameters	Name	Required	Type	Description
	data		PolicyInfo	
PolicyInfo	policyId		String	Monitor policy unique ID
	policyType		String	Monitor policy type Please refer to 5.2.1 Monitor Policy Type Definition
	policyName		String	Monitor policy name
	monitorPeriodStart		String	Effective time (HH:mm:ss)
	monitorPeriodEnd		String	Expiration time (HH:mm:ss)
	validPeriodStart		String	Effective date (yyyyMMdd)
	validPeriodEnd		String	Expiration date (yyyyMMdd)
	monitorPeriodWeekday		String	Active days of week (1-7), separated by commas For example, if the policy only takes effects on Sat and Sun, then put monitorPeriodWeekday = "6,7"
	sts		String	Monitor policy status 0: Effective 1: Ineffective 2: Suspended 3: Deleted 9: Created
	desc		String	Description
	policyActions		Array[PolicyAction]	Notification
	policyTargets		Array[PolicyTarget]	Target attribute features
	policyDevices		Array[PolicyDevice]	Target devices
	privilege		String	Privilege (Department ID)
	deviceScope		String	Device aggregation 1: Single device calculation

				N: Compound device calculation
	targetScope		String	Feature aggregation 1: Single feature calculation N: Compound feature calculation
	rollingWindow		String	Rolling interval Format: {Days}:{Hours}:{Minutes}
	rollingCount		String	Rolling accumulations
	consecutiveWindow		String	Continuous accumulation
	consecutiveCount		String	Continuous interval Format: {Days}:{Hours}:{Minutes}
	eventType		String	Control result Please refer to 5.2.3 Control Result Definition
PolicyAction	actionType		String	Notification type Please refer to 5.2.2 Notification Type Definition
	actionTarget		String	Notification target Please refer to 5.2.2 Notification Type Definition
	actionContent		String	Message
PolicyTarget	targetId		String	Feature ID Please refer to 5.2.4 Attribute Feature Definition - 5.2.4.1 Person Group & 5.2.4.2 Intelligent Device
	targetEnName		String	Feature English name
	targetCnName		String	Feature Chinese name
	targetCnOption		String	Feature Chinese option
	targetEnOption		String	Feature English option
	targetValue		String	Feature value Please refer to 5.2.4 Attribute Feature Definition - 5.2.4.1 Person Group & 5.2.4.2 Intelligent Device

	operator		String	0: Greater than 1: Less than 2: Equal to 3: Not equal to 4: Not exists 5: Exists
	classId		String	The target attributes that need to be monitored together share the same classId. Start from 0.
	matchingThreshold		String	Feature threshold
PolicyDevice	deviceId		String	Device unique ID
	deviceEnName		String	Device name
	deviceCnName		String	Device name
	deviceType		String	Device type
	deviceSerial		String	Device serial
	importSource		String	Device source 0: Portal, 1: Map service
Sample Request	--form 'policyId=1243410060239966209'			
Sample Response	{ "code": "0000", "msg": "成功", "data": { "policyId": "1243410060239966209", "policyType": "3", "policyName": "Abnormal Temperature", "monitorPeriodStart": "", "monitorPeriodEnd": "", "validPeriodStart": "", "validPeriodEnd": "", "monitorPeriodWeekday": "", "sts": "0", "desc": "", "policyActions": [{ "actionType": "2", "actionTarget": "http://pushaddress", }] } }			

```
        "actionContent": ""
      }
    ],
    "policyTargets": [
      {
        "targetId": "74",
        "targetEnName": "BodyTemperature",
        "targetCnName": "体温",
        "targetCnOption": "37.3",
        "targetEnOption": null,
        "targetValue": "37.3",
        "operator": "0",
        "classId": "0",
        "matchingThreshold": null
      }
    ],
    "policyDevices": [
      {
        "deviceId": "1243403012861333506",
        "deviceEnName": "SenseThunder-E_1",
        "deviceCnName": "SenseThunder-E_1",
        "deviceType": "17",
        "deviceSerial": "SPS020PROC19H04459_0",
        "importSource": "0"
      }
    ],
    "privilege": "0",
    "deviceScope": "",
    "targetScope": "",
    "rollingWindow": "",
    "rollingCount": "",
    "eventType": "2"
  },
  "success": true
}
```

5.4 Face Monitor Policy Management

5.4.1 Create Face Monitor Policy

API Endpoint	{{studio_base_url}}/POLICYMONITOR/faceMonitoringPolicy/create			
Method	POST			
Request Format	JSON			
Request Parameters	Name	Required	Type	Description
	policyName	Yes	String	Face monitor policy name
	policyActions	Yes	Array[PolicyAction]	Notification
	policyType	Yes	String	Monitor policy type Please refer to 5.2.1 Monitor Policy Type Definition "0" for face monitor policy
	policyTargets	Yes	Array[PolicyTarget]	Target attribute features
	policyDevices	Yes	Array[PolicyDevice]	Target devices
	operatePerson	Yes	String	Operator
	targetScope	Yes	String	Feature aggregation "": No feature aggregation "1": Single feature calculation "N": Compound feature calculation
	validPeriodStart	No	String	Effective date (yyyyMMdd)
	validPeriodEnd	No	String	Expiration date (yyyyMMdd)
	monitorPeriodWeekday	No	String	Active days of week (1-7), separated by commas For example, if the policy only takes effects on Sat and Sun, then put monitorPeriodWeekday = "6,7"
	monitorPeriodStart	No	String	Effective time (HH:mm:ss)
	monitorPeriodEnd	No	String	Expiration time (HH:mm:ss)
	deviceScope	No	String	Device aggregation 1: Single device calculation N: Compound device calculation

	rollingWindow	No	String	Rolling interval Format: {Days}:{Hours}:{Minutes}
	rollingCount	No	String	Rolling accumulation
	consecutiveCount	No	String	Continuous interval Format: {Days}:{Hours}:{Minutes}
	consecutiveWindow	No	String	Continuous accumulation
	desc	No	String	Description
PolicyAction	actionType	Yes	String	Notification type Please refer to 5.2.2 Notification Type Definition
	actionTarget	Yes	String	Notification target Please refer to 5.2.2 Notification Type Definition
	actionContent	No	String	Message
PolicyTarget	operator	Yes	String	0: Greater than 1: Less than 2: Equal to 3: Not equal to 4: Not exists 5: Exists
	targetId	Yes	String	Attribute feature ID Please refer to 5.2.4 Attribute Feature Definition - 5.2.4.3 Pedestrian, 5.2.4.4 Vehicle & 5.2.4.5 Non-motor Vehicle
	targetValue	Yes	String	Attribute feature value Please refer to 5.2.4 Attribute Feature Definition - 5.2.4.3 Pedestrian, 5.2.4.4 Vehicle & 5.2.4.5 Non-motor Vehicle
	classId	Yes	String	The target attributes that need to be monitored together share the same classId. Start from 0.
PolicyDevice	deviceId	Yes	String	Device unique ID
	deviceSerial	No	String	Device serial

	deviceType	No	String	Device type
Response Parameters	Name	Required	Type	Description
	data		String	Monitor policy unique ID, generated by the system, be used to uniquely identify the monitor policy
Sample Request	<pre>{ "policyName": "Face Monitor Policy", "policyActions": [{ "actionType": "2", "actionTarget": "https://10.2.355:23", "actionContent": "" }], "policyType": "0", "policyTargets": [{ "targetValue": "1", "operator": "2", "classId": "0", "targetId": "0" }], "policyDevices": [{ "deviceId": "1265185833686667266", "deviceSerial": "", "deviceType": "" }], "operatePerson": "1", "targetScope": "", "validPeriodStart": "20200619", "validPeriodEnd": "20200630", "monitorPeriodWeekday": "1,2,3,4,5", "monitorPeriodStart": "16:41:38", "monitorPeriodEnd": "23:56:44", }</pre>			

	<pre>"deviceScope": "N", "rollingWindow": "4:0:0", "rollingCount": "5", "consecutiveCount": "", "consecutiveWindow": "" }</pre>
Sample Response	<pre>{ "code": "0000", "msg": "成功", "data": "1273903813576626178", "success": true }</pre>

5.4.2 Update Face Monitor Policy

API Endpoint	{{studio_base_url}}/POLICYMONITOR/faceMonitoringPolicy/update			
Method	POST			
Request Format	JSON			
Request Parameters	Name	Required	Type	Description
	policyId	Yes	String	Monitor policy unique ID
	policyName	Yes	String	Face monitor policy name
	policyActions	Yes	Array[PolicyAction]	Notification
	policyType	Yes	String	Monitor policy type Please refer to 5.2.1 Monitor Policy Type Definition "0" for face monitor policy
	policyTargets	Yes	Array[PolicyTarget]	Target attribute features
	policyDevices	Yes	Array[PolicyDevice]	Target devices
	operatePerson	Yes	String	Operator
	validPeriodStart	No	String	Effective date (yyyyMMdd)
	validPeriodEnd	No	String	Expiration date (yyyyMMdd)
	monitorPeriodWeekday	No	String	Active days of week (1-7), separated by commas For example, if the policy only takes effects on Sat and Sun, then put

				monitorPeriodWeekday = "6,7"
	monitorPeriodStart	No	String	Effective time (HH:mm:ss)
	monitorPeriodEnd	No	String	Expiration time (HH:mm:ss)
	targetScope	No	String	Feature aggregation 1: Single feature calculation N: Compound feature calculation
	deviceScope	No	String	Device aggregation 1: Single device calculation N: Compound device calculation
	rollingWindow	No	String	Rolling interval Format: {Days}:{Hours}:{Minutes}
	rollingCount	No	String	Rolling accumulation
	consecutiveCount	No	String	Continuous interval Format: {Days}:{Hours}:{Minutes}
	consecutiveWindow	No	String	Continuous accumulation
	desc	No	String	Description
PolicyAction	actionType	Yes	String	Notification type Please refer to 5.2.2 Notification Type Definition
	actionTarget	Yes	String	Notification target Please refer to 5.2.2 Notification Type Definition
	actionContent	No	String	Message
PolicyTarget	operator	Yes	String	0: Greater than 1: Less than 2: Equal to 3: Not equal to 4: Not exists 5: Exists
	targetId	Yes	String	Attribute feature ID Please refer to 5.2.4 Attribute Feature Definition - 5.2.4.3

				Pedestrian, 5.2.4.4 Vehicle & 5.2.4.5 Non-motor Vehicle
	targetValue	Yes	String	Attribute feature value Please refer to 5.2.4 Attribute Feature Definition - 5.2.4.3 Pedestrian, 5.2.4.4 Vehicle & 5.2.4.5 Non-motor Vehicle
	classId	Yes	String	The target attributes that need to be monitored together share the same classId. Start from 0.
PolicyDevice	deviceId	Yes	String	Device unique ID
	deviceSerial	No	String	Device serial
	deviceType	No	String	Device type
Response Parameters	Name	Required	Type	Description
	data		String	Monitor policy unique ID
Sample Request	<pre>{ "policyId": "1273903813576626178", "policyName": "Face Monitor Policy Name", "policyActions": [{ "actionType": "2", "actionTarget": "https://10.2.355:23", "actionContent": "" }], "policyType": "0", "policyTargets": [{ "targetValue": "1", "operator": "2", "classId": "0", "targetId": "0" }], "policyDevices": [{ "deviceId": "1265185833686667266",</pre>			

	<pre>"deviceSerial": "", "deviceType": "" }], "operatePerson": "1", "targetScope": "", "validPeriodStart": "20200619", "validPeriodEnd": "20200630", "monitorPeriodWeekday": "1,2,3,4,5", "monitorPeriodStart": "16:41:38", "monitorPeriodEnd": "23:56:44", "deviceScope": "N", "rollingWindow": "4:0:0", "rollingCount": "5", "consecutiveCount": "", "consecutiveWindow": "" }</pre>
Sample Response	<pre>{ "code": "0000", "msg": "成功", "data": "1273903813576626178", "success": true }</pre>

5.4.3 Query Face Monitor Policy Information by Policy ID

API Endpoint	{{studio_base_url}}/POLICYMONITOR/faceMonitoringPolicy/info			
Method	POST			
Request Format	form-data			
Request Parameters	Name	Required	Type	Description
	policyId	Yes	String	Monitor policy unique ID
Response Parameters	Name	Required	Type	Description
	data		PolicyInfo	
PolicyInfo	policyId		String	Monitor policy unique ID
	policyType		String	Monitor policy type Please refer to 5.2.1 Monitor Policy Type Definition
	policyName		String	Monitor policy name

	monitorPeriodStart		String	Effective time (HH:mm:ss)
	monitorPeriodEnd		String	Expiration time (HH:mm:ss)
	validPeriodStart		String	Effective date (yyyyMMdd)
	validPeriodEnd		String	Expiration date (yyyyMMdd)
	monitorPeriodWeekday		String	Active days of week (1-7), separated by commas For example, if the policy only takes effects on Sat and Sun, then put monitorPeriodWeekday = "6,7"
	sts		String	Monitor policy status 0: Effective 1: Ineffective 2: Suspended 3: Deleted 9: Created
	desc		String	Description
	policyActions		Array[PolicyAction]	Notification
	policyTargets		Array[PolicyTarget]	Target attribute features
	policyDevices		Array[PolicyDevice]	Target devices
	privilege		String	Privilege (Department ID)
	deviceScope		String	Device aggregation 1: Single device calculation N: Compound device calculation
	targetScope		String	Feature aggregation 1: Single feature calculation N: Compound feature calculation
	rollingWindow		String	Rolling interval Format: {Days}:{Hours}:{Minutes}
	rollingCount		String	Rolling accumulations
	consecutiveWindow		String	Continuous accumulation
	consecutiveCount		String	Continuous interval

				Format: {Days}:{Hours}:{Minutes}
PolicyAction	actionType		String	Notification type Please refer to 5.2.2 Notification Type Definition
	actionTarget		String	Notification target Please refer to 5.2.2 Notification Type Definition
	actionContent		String	Message
PolicyTarget	targetId		String	Feature ID Please refer to 5.2.4 Attribute Feature Definition - 5.2.4.3 Pedestrian, 5.2.4.4 Vehicle & 5.2.4.5 Non-motor Vehicle
	targetEnName		String	Feature English name
	targetCnName		String	Feature Chinese name
	targetCnOption		String	Feature Chinese option
	targetEnOption		String	Feature English option
	targetValue		String	Feature value Please refer to 5.2.4 Attribute Feature Definition - 5.2.4.3 Pedestrian, 5.2.4.4 Vehicle & 5.2.4.5 Non-motor Vehicle
	operator		String	0: Greater than 1: Less than 2: Equal to 3: Not equal to 4: Not exists 5: Exists
	classId		String	The target attributes that need to be monitored together share the same classId. Start from 0.
	matchingThreshold		String	Feature threshold
PolicyDevice	deviceId		String	Device unique ID
	deviceEnName		String	Device name
	deviceCnName		String	Device name
	deviceType		String	Device type

	deviceSerial		String	Device serial
	importSource		String	Device source 0: Portal, 1: Map service
Sample Request	--form 'policyId=1273903813576626178'			
Sample Response	<pre>{ "code": "0000", "msg": "成功", "data": { "policyId": "1273903813576626178", "policyType": "0", "policyName": "Face Monitor Policy", "monitorPeriodStart": null, "monitorPeriodEnd": null, "validPeriodStart": null, "validPeriodEnd": null, "monitorPeriodWeekday": null, "sts": "0", "desc": "", "policyActions": [{ "actionType": "2", "actionTarget": "https://10.2.355:23", "actionContent": "" }], "policyTargets": [{ "targetId": "0", "targetEnName": "PersonGroup", "targetCnName": "人员群组", "targetCnOption": "Staff", "targetEnOption": null, "targetValue": "1", "operator": "2", "classId": "0", "matchingThreshold": null }] } }</pre>			

```
    ],  
    "policyDevices": [  
      {  
        "deviceId": "1265185833686667266",  
        "deviceEnName": "Device 00110",  
        "deviceCnName": "Device 00110 Update",  
        "deviceType": "2",  
        "deviceSerial": "DeviceSerial0612",  
        "importSource": "0"  
      }  
    ],  
    "privilege": "0",  
    "deviceScope": null,  
    "targetScope": "",  
    "rollingWindow": null,  
    "rollingCount": null,  
    "consecutiveWindow": null,  
    "consecutiveCount": null  
  },  
  "success": true  
}
```

5.5 Monitor Policy Management

5.5.1 Activate Monitor Policy

API Endpoint	{{studio_base_url}}/POLICYMONITOR/monitoringPolicy/activate			
Method	POST			
Request Format	JSON			
Request Parameters	Name	Required	Type	Description
	policyIds	Yes	Array[String]	Monitor policy unique ID
	operatePerson	Yes	String	Operator
Response Parameters	Name	Required	Type	Description
	data		String	null
Sample Request	{ "policyIds": ["1272860662661386242", "1272843964122140673"], "operatePerson": "1" }			

	}
Sample Response	{ "code": "0000", "msg": "成功", "data": null, "success": true }

5.5.2 Deactivate Monitor Policy

API Endpoint	{{studio_base_url}}/POLICYMONITOR/monitoringPolicy/deactivate			
Method	POST			
Request Format	JSON			
Request Parameters	Name	Required	Type	Description
	policyIds	Yes	Array[String]	Monitor policy unique ID
	operatePerson	Yes	String	Operator
Response Parameters	Name	Required	Type	Description
	data		String	null
Sample Request	{ "policyIds": ["1272860662661386242", "1272843964122140673"], "operatePerson": "1" }			
Sample Response	{ "code": "0000", "msg": "成功", "data": null, "success": true }			

5.5.3 Add Device to Monitor Policy

API Endpoint	{{studio_base_url}}/POLICYMONITOR/monitoringPolicy/addDeviceInMonitorPolicy			
Method	POST			
Request Format	JSON			
Request Parameters	Name	Required	Type	Description
	deviceInfos	Yes	Array[DeviceInfo]	Device information

	policyId	Yes	String	Monitor policy unique ID
	operatePerson	Yes	String	Operator
DeviceInfo	deviceId	Yes	String	Device unique ID
	importSource	Yes	String	Device source 0: Portal, 1: Map service
Response Parameters	Name	Required	Type	Description
	data		Array[String]	
Sample Request	<pre>{ "deviceInfos": [{ "deviceId": "1257879674839633922", "importSource": "0" }, { "deviceId": "1243403012861333506", "importSource": "0" }], "policyId": "1272844240195424257", "operatePerson": "1" }</pre>			
Sample Response	<pre>{ "code": "0000", "msg": "成功", "data": [], "success": true }</pre>			

5.5.4 Remove Device from Monitor Policy

API Endpoint	{{studio_base_url}}/POLICYMONITOR/monitoringPolicy/removeDeviceInMonitorPolicy			
Method	POST			
Request Format	JSON			
Request Parameters	Name	Required	Type	Description
	deviceInfos	Yes	Array[DeviceInfo]	Device information
	policyId	Yes	String	Monitor policy unique ID
	operatePerson	Yes	String	Operator
DeviceInfo	deviceId	Yes	String	Device unique ID

	importSource	Yes	String	Device source 0: Portal, 1: Map service
Response Parameters	Name	Required	Type	Description
	data		String	null
Sample Request	<pre>{ "deviceInfos": [{ "deviceId": "1257879674839633922", "importSource": "0" }, { "deviceId": "1243403012861333506", "importSource": "0" }], "policyId": "1272844240195424257", "operatePerson": "1" }</pre>			
Sample Response	<pre>{ "code": "0000", "msg": "成功", "data": null, "success": true }</pre>			

5.5.5 Remove Target from Monitor Policy

API Endpoint	{{studio_base_url}}/POLICYMONITOR/monitoringPolicy/removeTargetInMonitorPolicy			
Method	POST			
Request Format	JSON			
Request Parameters	Name	Required	Type	Description
	policyId	Yes	String	Monitor policy unique ID
	policyTargets	Yes	Array[PolicyTarget]	Target attribute features
PolicyTarget	targetId	Yes	String	Attribute feature ID
	targetValue	Yes	String	Attribute feature value
Response Parameters	Name	Required	Type	Description
	data		String	null

Sample Request	<pre>{ "policyId": "1272844240195424257", "policyTargets": [{ "targetId": "74", "targetValue": "38" }, { "targetId": "75", "targetValue": "9" }] }</pre>
Sample Response	<pre>{ "code": "0000", "msg": "成功", "data": null, "success": true }</pre>

5.5.6 List Monitor Policy (Pagination)

API Endpoint	{{studio_base_url}}/POLICYMONITOR/monitoringPolicy/list			
Method	POST			
Request Format	JSON			
Request Parameters	Name	Required	Type	Description
	current	Yes	Integer	Query results on page #. Start from 1.
	size	Yes	Integer	Number of results per page
	policyName	No	String	Monitor policy name
	monitorPeriodStart	No	String	Effective time (HH:mm:ss)
	monitorPeriodEnd	No	String	Expiration time (HH:mm:ss)
	validPeriodStart	No	String	Effective date (yyyyMMdd)
	validPeriodEnd	No	String	Expiration date (yyyyMMdd)
	sts	No	String	Monitor policy status

				0: Effective 1: Ineffective 2: Suspended 3: Deleted 9: Created
	desc	No	String	Description
	policyType	No	String	Monitor policy type Please refer to 5.2.1 Monitor Policy Type Definition
Response Parameters	Name	Required	Type	Description
	data		Policy	null
Policy	policyId		String	Monitor policy unique ID
	policyName		String	Monitor policy name
	monitorPeriodStart		String	Effective time (HH:mm:ss)
	monitorPeriodEnd		String	Expiration time (HH:mm:ss)
	validPeriodStart		String	Effective date (yyyyMMdd)
	validPeriodEnd		String	Expiration date (yyyyMMdd)
	sts		String	Monitor policy status 0: Effective 1: Ineffective 2: Suspended 3: Deleted 9: Created
	desc		String	Description
	policyType		String	Monitor policy type Please refer to 5.2.1 Monitor Policy Type Definition
	monitorPeriodWeekday		String	Active days of week (1-7), separated by commas For example, if the policy only takes effects on Sat and Sun, then put

				monitorPeriodWeekday = "6,7"
	privilege		String	Privilege (Department ID)
Sample Request	<pre>{ "current": 1, "size": 10, "policyName": "Temperature", "monitorPeriodStart": "", "monitorPeriodEnd": "", "validPeriodStart": "", "validPeriodEnd": "", "sts": "", "desc": "", "policyType": "3" }</pre>			
Sample Response	<pre>{ "data": [{ "policyId": "1272844240195424257", "policyName": "Abnormal Temperature", "monitorPeriodStart": "", "monitorPeriodEnd": "", "validPeriodStart": "", "validPeriodEnd": "", "sts": "0", "desc": "", "policyType": "3", "monitorPeriodWeekday": "", "privilege": "0" }], "size": 10, "total": 1, "code": "0000", "msg": "success" }</pre>			

5.5.7 Query Monitor Policy of Device

API Endpoint	{{studio_base_url}}/POLICYCONTROL/monitoringPolicy/isDeviceInMonitorPolicy			
Method	POST			
Request Format	JSON			
Request Parameters (in JSON array)	Name	Required	Type	Description
	deviceId	Yes	Array[String]	Device unique ID
Response Parameters	Name	Required	Type	Description
	data		Array[DevicePolicy]	
DevicePolicy	deviceId		String	Device unique ID
	policyId		String	Gate control policy unique ID
	policyName		String	Gate control policy name
	privilege		String	Privilege (Department ID)
	policyType		String	Gate control policy type Please refer to 4.2.1 Gate Control Policy Type Definition
Sample Request	["1257879674839633922", "1243403012861333506"]			
Sample Response	{ "code": "0000", "msg": "成功", "data": [{ "deviceId": "1243403012861333506", "policyId": "1272844240195424257", "policyName": "Abnormal Temperature", "privilege": null, "policyType": "3" }, { "deviceId": "1257879674839633922", "policyId": "1257881270512259073", "policyName": "Contact Tracing", "privilege": null, "policyType": "3" }] }			

```
}  
],  
"success": true  
}
```

5.5.8 Query Monitor Policy of Target

API Endpoint	{{studio_base_url}}/POLICYCONTROL/gateControlPolicy/isTargetInControlPolicy			
Method	POST			
Request Format	JSON			
Request Parameters	Name	Required	Type	Description
	targetId	Yes	String	Target type ID 0: Person group
	targetValue	Yes	String	Person group unique ID
	operator	Yes	String	0: Greater than 1: Less than 2: Equal to 3: Not equal to 4: Not exists 5: Exists
Response Parameters	Name	Required	Type	Description
	data		Array[String]	Monitor policy unique IDs
Sample Request	{ "targetId": "74", "targetValue": "37.5", "operator": "0" }			
Sample Response	{ "code": "0000", "msg": "成功", "data": ["1272844240195424257 "], "success": true }			

6 Event Handler API

6.1 Event Handler API Return Code

Return Code	Return Code Description	Possible Causes	Solution Suggestion
0000	Success		
06999999	SystemError	System exception	Investigate the cause by checking the backend logs
06011001	ParameterException	Parameter exception	Check input parameters
06011002	DateFormat error	Date format is fault	Check input date parameter
06022000	No permission	Without permission	Check API permission
06999000	MySQL error	MySQL exception	Check MySQL database
06999001	Cassandra error	Cassandra exception	Check Cassandra database

6.2 Parameter Definition

6.2.1 Access Control Type Definition

Access Control Type ID	Access Control Type
0	Block
1	Allow

6.2.2 Notification Type Definition

Notification Type ID	Notification Type	Notification Target
2	HTTP push	HTTP push address
3	SMS	Phone number
4	Email	Email address

6.3 Event Record Management

6.3.1 Search Event Record

API Endpoint	{{studio_base_url}}/EVENTHANDLER/triggerEventHistory/eventRecordQuery			
Method	POST			
Request Format	JSON			
Request Parameters	Name	Required	Type	Description
	current	Yes	Integer	Query results on page #. Start from 1.
	size	Yes	Integer	Number of results per page

	startDate	No	String	Search start date Format: yyyyMMdd_HH:mm:ss
	endDate	No	String	Search end date Format: yyyyMMdd_HH:mm:ss
	policyId	No	Array[String]	Policy unique ID
	deviceId	No	Array[String]	Device unique ID
	personId	No	Array[String]	Person unique ID
Response Parameters	Name	Required	Type	Description
	data		Array[EventRecord]	null
EventRecord	uuid		String	Event record unique ID
	eventType		String	Event type
	policyId		String	Policy unique ID
	policyType		String	Gate control policy type Please refer to 4.2.1 Gate Control Policy Type Definition
	policyName		String	Policy name
	triggerDeviceId		String	Triggered device unique ID
	triggerDeviceSerial		String	Triggered device serial
	triggerTime		String	Triggered time Format: yyyy-MM-dd HH:mm:ss.SSS
	triggerTargets		Array[TriggerTarget]	Triggered target (person group)
	triggerValues		String	Triggered value
	triggerPersonId		String	Triggered person unique ID
	triggerImgUrl		String	Triggered image URL
	targetImgUrl		String	Matched base image URL
	bkImgUrl		String	Background image URL
	triggerActionType		String	Access control type and notification type, separated by commas Please refer to 6.2.1 Access Control Type Definition & 6.2.2 Notification Type Definition

	triggerActionTarget		String	Notification target, separated by commas Please refer to 6.2.2 Notification Type Definition
	triggerActionContent		String	Message Separated by commas
	privilege		String	Privilege (Department ID)
	idNumber		String	Person ID number
	icNumber		String	Person document number
	createUser		String	Create user
	createTs		String	Create time
	lastModUser		String	Last modified user
	lastModTs		String	Last modified time
	detect		String	Face coordinates
	triggerImageRoild		String	Triggered region ID
	triggerImageType		String	Triggered structural type
	desc		String	Event description
	attributes		String	null
	score		String	Matching score
TriggerTarget	targetId		String	Feature ID
	targetValue		String	Feature value
Sample Request	<pre>{ "current": 1, "size": 2, "startDate": "", "endDate": "", "eventType": "", "policyId": [], "deviceId": [], "personId": [], "uuid": "" }</pre>			
Sample Response	<pre>{ "code": "0000", "msg": "success", "data": [{</pre>			

```
"uuid": "Xn-D1e4nQc6dRuqZrvQdcw",
"eventType": "gateControlTriggerEvent",
"policyId": "1266270965512278018",
"policyType": "0",
"policyName": "SenseThunder 05927",
"triggerDeviceId": "1257879674839633922",
"triggerDeviceSerial": "SPS020STDC19G05927_0",
"triggerTime": "2020-06-03 16:53:54.979",
"triggerTargets": "[{\\"targetId\\":\\"0\\",\\"targetValue\\":\\"1\\"}]",
"triggerValues": "",
"triggerPersonId": "1255",
"triggerImgUrl":
"http://hostserver/intersense//images/sensepass/20200603/20200603_2_f4742079256
24105b2a4162b3d1a3838.jpeg",
"targetImgUrl":
"http://hostserver/intersense//images/person/20200603/27/b3d87561d0382cea03b7e5
72000425ba.jpg",
"bkImgUrl": null,
"triggerActionType": "1,2",
"triggerActionTarget": "defaultActionTarget,defaultActionTarget",
"triggerActionContent": "defaultActionContent,defaultActionContent",
"privilege": "0",
"idNumber": "12345",
"icNumber": "",
"createUser": "Alice",
"createTs": "2020-06-03 16:48:48",
"lastModUser": "Alice",
"lastModTs": "2020-06-03 16:48:48",
"detect": null,
"triggerImageRoild": "",
"triggerImageType": "",
"desc": "{\\"name\\":\\"Peter\\",\\"ic\\":\\"12345\\",\\"mobile\\":\\"\\\"}",
"attributes": null,
"score": "0.9787836"
},
{
"uuid": "UGitaMRCQdGbtX_T1bGoXg",
"eventType": "gateControlTriggerEvent",
```

```
"policyId": "1266270965512278018",
"policyType": "0",
"policyName": "SenseThunder 05927",
"triggerDeviceId": "1257879674839633922",
"triggerDeviceSerial": "SPS020STDC19G05927_0",
"triggerTime": "2020-06-03 16:41:38.088",
"triggerTargets": "[{"targetId": "0", "targetValue": "1"}]",
"triggerValues": "",
"triggerPersonId": "1255",
"triggerImgUrl":
"http://hostserver/intersense//images/sensepass/20200603/20200603_2_610f6932272
64023a029233ff08d35dd.jpeg",
"targetImgUrl":
"http://hostserver/intersense//images/person/20200603/27/b3d87561d0382cea03b7e5
72000425ba.jpg",
"bkImgUrl": null,
"triggerActionType": "1,2",
"triggerActionTarget": "defaultActionTarget,defaultActionTarget",
"triggerActionContent": "defaultActionContent,defaultActionContent",
"privilege": "0",
"idNumber": "12345",
"icNumber": "",
"createUser": "Alice",
"createTs": "2020-06-03 16:36:32",
"lastModUser": "Alice",
"lastModTs": "2020-06-03 16:36:32",
"detect": null,
"triggerImageRoild": "",
"triggerImageType": "",
"desc": "{\"name\":\"Peter\",\"ic\":\"12345\",\"mobile\":\"\"}",
"attributes": null,
"score": "0.9690521"
}
],
"size": 2,
"total": 41
}
```

6.4 Advanced Search Management

6.4.1 Search Person by Image

API Endpoint	{{studio_base_url}}/EVENTHANDLER/realTime/searchFaceByImage			
Method	POST			
Request Format	form-data			
Request Parameters	Name	Required	Type	Description
	multifile	Yes	MultipartFile	Image file
	personType	Yes	String	Person type Target: Person list Passer: Stranger list
Response Parameters	Name	Required	Type	Description
	data		Array[Face]	
	Face			
	score		String	Matched score
	personID		String	Matched person unique ID
	targetType		String	Matched person type Target: Person list Passer: Stranger list
Sample Request	--form 'multifile=@/D:/sample.jpg' \ --form 'personType=Target'			
Sample Response	{ "msg": "成功", "code": "0000", "data": [{ "score": 0.9967434, "personID": "304", "targetType": "Target" }] }			

6.4.2 Search Structured Record by Attribute

API Endpoint	{{studio_base_url}}/EVENTHANDLER/realTime/searchImageByAttribute
Method	POST

Request Format	JSON			
Request Parameters	Name	Required	Type	Description
	attributes	Yes	Array[Attribute]	Target attributes
	deviceId	Yes	String	Device unique IDs, separated by commas
	startDate	Yes	String	Search start date Format: yyyyMMdd_HH:mm:ss
	endDate	Yes	String	Search end date Format: yyyyMMdd_HH:mm:ss
	current	Yes	Integer	Query results on page #. Start from 1.
	size	Yes	Integer	Number of results per page
Attribute	targetId	Yes	String	Feature ID
	targetValue	Yes	String	Feature option/value
Response Parameters	Name	Required	Type	Description
	data		Array[StructuralRecord]	
Structural Record	deviceId		String	Device unique ID
	trackId		String	Track ID
	deviceType		String	Device type
	dt		String	Timestamp Format: yyyy-MM-dd HH:mm:ss
	event_type		String	Event type
	triggerImgUrl		String	Triggered image URL
	bkImgUrl		String	Background image URL
	ts		String	Timestamp in GMT Format: yyyy-MM-ddTHH:mm:ss.SSSZ
	attributes		Array[Attribute]	
	privilege		String	Privilege (Department ID)
	eventValidity		String	False recognition mark null: unmark 1: marked
	eventRemark		String	Event record remark
	triggerImageRoild		String	Triggered region ID
	triggerImageType		String	Triggered structural type

	description		String	Event record description
Attribute	conf		String	Confidence score
	value		String	Attribute feature value
	key		String	Attribute feature ID
Sample Request	<pre>{ "attributes": [{ "targetId": "-3", "targetValue": "1" }], "deviceId": "'1273487133813772290', '1272390159005913090'", "startDate": "20200623_00:00:00", "endDate": "20200623_23:59:59", "current": 1, "size": 10 }</pre>			
Sample Response	<pre>{ "code": "0000", "msg": "success", "data": [{ "deviceId": "1273134705075556354", "trickId": "2", "deviceType": null, "dt": "2020-06-23 16:59:21", "eventType": "-3", "triggerImgUrl": "http://hostserver/intersense/images/structural/20200623/1273134705075556354 1275352706738819073/event/1273134705075556354 1275352706738819073_1275352707317633025_210_0_0_1592902761628.jpg", "bkImgUrl": "http://hostserver/intersense/images/structural/20200623/1273134705075556354 1275352706738819073/event/1273134705075556354 1275352706738819073_1275352707317633025_210_0_0_1592902761628.jpg", "ts": "2020-06-23T08:59:21.000+0000", "attributes": [{ </pre>			

```
"key": "80",
"value": "",
"conf": "0.0"
},
{
  "key": "1",
  "value": "1",
  "conf": "0.9999985694885254"
},
{
  "key": "2",
  "value": "0",
  "conf": "0.9999833106994629"
},
{
  "key": "Type",
  "value": "Car",
  "conf": "0.999937117099762"
}
{
  "key": "81",
  "value": "3",
  "conf": "0.9999940395355225"
},
{
  "key": "-3",
  "value": "1",
  "conf": "0.9406358003616333"
}
],
"privilege": "0",
"eventValidity": null,
"eventRemark": null,
"triggerImageRoild": "1275352707317633025",
"triggerImageType": "-3",
"description": null,
"detect":
"{\"left\":\"19\",\"top\":\"186\",\"right\":\"270\",\"bottom\":\"29\"}"
```

```
    },
    {
      "deviceId": "1273134705075556354",
      "trickId": "18",
      "deviceType": null,
      "dt": "2020-06-23 16:59:29",
      "eventType": "-3",
      "triggerImgUrl":
"http://hostserver/intersense/images/structural/20200623/1273134705075556354|1275352706738819073/event/1273134705075556354|1275352706738819073_1275352707317633025_210_1_0_1592902769154.jpg",
      "bkImgUrl":
"http://hostserver/intersense/images/structural/20200623/1273134705075556354|1275352706738819073/event/1273134705075556354|1275352706738819073_1275352707317633025_210_1_0_1592902769154.jpg",
      "ts": "2020-06-23T08:59:29.000+0000",
      "attributes": [
        {
          "key": "80",
          "value": "",
          "conf": "0.0"
        },
        {
          "key": "1",
          "value": "5",
          "conf": "0.999639630317688"
        },
        {
          "key": "2",
          "value": "0",
          "conf": "0.9999822378158569"
        },
        {
          "key": "Type",
          "value": "SUV",
          "conf": "0.9981290698051453"
        }
      ]
    }
  ]
}
```

```
        "key": "81",
        "value": "6",
        "conf": "0.9986579418182373"
    },
    {
        "key": "-3",
        "value": "1",
        "conf": "0.8059031963348389"
    }
],
"privilege": "0",
"eventValidity": null,
"eventRemark": null,
"triggerImageRoild": "1275352707317633025",
"triggerImageType": "-3",
"description": null,
"detect":
"{\"left\":\"423\",\"top\":\"352\",\"right\":\"466\",\"bottom\":\"296\"}"
    }
],
"size": 2,
"total": 2
}
```

6.5 HTTP Push

If a HTTP endpoint is set in policy, SenseStudio will push the event record to it when the policy is triggered.

* Policy name

Description

Notification

☐ SMS

☐ Email

☒ Push

6.5.1 HTTP Push Notification

API Endpoint	Provided by third-party system.		
Method	POST		
Request Format	JSON		
Request Parameters	Name	Type	Description
	msgSource	String	Message source
	eventType	String	Event type E.g. gateControlTriggerEvent , gateMonitorTriggerEvent
	policyId	String	Policy unique ID
	policyType	String	Gate control policy type / Monitor policy type Please refer to 4.2.1 Gate Control Policy Type Definition & 5.2.1 Monitor Policy Type Definition
	policyName	String	Policy name
	triggerTime	String	Event triggered time
	deviceId	String	Device unique ID
	deviceType	String	Device type ID
	deviceSerial	String	Device serial

	deviceName	String	Device name
	deviceTag	String	Device tag
	personId	String	Person unique ID
	triggerImgUrl	String	Triggered image URL
	targetImgUrl	String	Matched base image URL
	bkImageUrl	String	Background image URL
	similarity	String	Matching score
	triggerTargets	Array[TriggerTarget]	Triggered targets
	triggerValues	String	Triggered values
	personName	String	Person name
	personTag	String	Person tag
	actionType	String	Notification type 2: HTTP push
	actionTarget	String	HTTP push address
	actionContent	String	Notification message
	desc	String	Event description
	idNumber	String	Person ID number
	icNumber	String	Person document number
	attributes	Array[Attribute]	
	privilege	String	Privilege (Department ID)
	triggerImageRoild	String	Triggered region ID
	triggerImageType	String	Triggered structural type
	detect	String	Face coordinates
	uuid	String	Event unique ID
	classId	String	The target groups that need to be matched at the same time share the same classId. Start from 0.
TriggerTarget	targetId	String	Target type ID 0: Person group
	canCleanMatchingThreshHold	Boolean	
	targetValue	String	Person group unique ID
	matchingThreshold	String	Matching threshold
	operator	String	2: Equal to 3: Not equal to
Attribute	conf	String	Confidence score

	value	String	Attribute feature value
	key	String	Attribute feature ID
Sample Request	<pre>{ "msgSource": "bizTrigger", "eventType": "gateMonitorTriggerEvent", "policyId": "1260521300258983937", "policyType": "3", "policyName": "Temperature Screening", "triggerTime": "2020-05-13 18:52:40.868", "deviceId": "1260520595813044226", "deviceType": "15", "deviceSerial": "SPS020STDC19E02393_0", "deviceName": "Lobby", "deviceTag": null, "personId": "1", "triggerImgUrl": "http://hostserver/intersense//images/sensepass/20200513/20200513_2_1cce5b0fc33d4e0f9dea06827d2de67e.jpeg", "targetImgUrl": "http://hostserver/intersense//images/person/20200402/32/091e6a1bb66d720e51a71832cc904875.jpg", "bklImageUrl": "http://hostserver/intersense//images/sensepass/20200513/20200513_2_abda4f0639e445c3baa74a8f1d1ef0bf.jpeg", "similarity": "0.9314568", "triggerTargets": [{ "classId": "0", "targetId": "74", "canCleanMatchingThreshHold": true, "targetValue": "32", "matchingThreshold": "", "operator": "0" }], "triggerValues": null, "personName": "Allen", "personTag": "" }</pre>		

```
"actionType": "2",
"actionTarget": "http://hostserver:1880/httppush",
"actionContent": "defaultActionContent",
"desc": "",
"idNumber": "1234",
"icNumber": "",
"attributes": [
  {
    "conf": "1",
    "value": "36.582077",
    "key": "74"
  },
  {
    "conf": "1",
    "value": "0",
    "key": "75"
  }
],
"privilege": "0",
"triggerImageRoild": "",
"triggerImageType": "",
"detect": null,
"uuid": "_Mz-lhTyReO2FK_rVpZ8TA"
}
```

7 Utility API

7.1 Utility API Return Code

Return Code	Return Code Description	Possible Causes	Solution Suggestion
0000	Success		
07001000	SEND KAFKA FAILED	Send message to Kafka failed	
07001001	SEND SMS FAILED	Send SMS failed	
07001002	SEND MAIL FAILED	Send email failed	
07001003	PUSH MESSAGE FAILED	Send HTTP push notification failed	
07002001	UPLOAD FILE FAILED	Update file failed	
07002101	VERIFY IMAGE QUALITY FAILED	Image quality detection failed	
07002102	BAD IMAGE QUALITY	Image quality does not meet the requirements	Image quality does not meet the requirements, change the image
07002103	SAVE FILE FAILED	Save file failed	
07002104	ILLEGAL IMAGE TYPE	Illegal image service type	Check image service type parameter
07002105	ILLEGAL FILE NAME	Illegal filename	Check if the file name has a format suffix
07002106	ILLEGAL FILE TYPE	Illegal file service type	Check file service type parameter
07002107	FILE OVER SIZE	File size exceeds the limit	
07002108	FILE IS BROKEN	File is broken	Re-upload
07002109	DETECT FACE FAILED	Calling the cog interface to detect a face failure	The image has no clear face, change the image
07002110	NO FACE DETECTED	No face detected	The image has no clear face, change the image
07003100	EXISTED JOB	Duplicate job	
07003101	NOT FOUND JOB	Job not found	
07003102	CREATE JOB FAILED	Create job failed	
07003103	UPDATE JOB FAILED	Update job failed	
07003104	REMOVE JOB FAILED	Delete job failed	
07003204	REMOVE MESSAGE EVENT HISTORY FAILED	Purge event record failed	

07004100	EXISTED TASK	Duplicate task	
07004101	NOT FOUND TASK	Task not found	
07004102	CREATE TASK FAILED	Create task failed	
07004103	UPDATE TASK FAILED	Update task failed	
07004104	REMOVE TASK FAILED	Remove task failed	
07004201	EXTRACT COMPRESS FILE FAILED	Tar file extract failed	Re-archive the file and upload again.
07004202	BATCH IMPORT FAILED	Batch import failed	
07004203	OVER MAXIMUN LIMIT	Batch import file exceed file limit	Separate the images to multiple files

7.2 File Management

7.2.1 Upload Image in Form-data

API Endpoint	{{studio_base_url}}/UTILITY/files/uploadImage			
Method	POST			
Request Format	form-data			
Request Parameters	Name	Required	Type	Description
	image	Yes	File	Image file
	serviceType	No	String	Image type person: face image device: device image map: map image temp: temp image (default value, this type of images will be purged periodically)
	checkFace	No	Boolean	Whether to do face detection (true/false) Only take effect when serviceType=person
	faceEnlargeRate	No	Double	Face frame enlarge rate Only take effect when serviceType=person
	checkQuality	No	Boolean	Whether to check photo quality (true/false) Only take effect when serviceType=person
Response Parameters	Name	Required	Type	Description
	data		String	Image URI
Sample Request	--form 'image=@/D:/test.jpg' \			
Sample Response	{ "code": "0000",			

	<pre>"msg": "success", "data": "/images/person/20200311/93/eae3d34d7071a6bc54dbc4e44a47c643.jpg", "success": true }</pre>
--	---

7.2.2 Upload Image in Base64

API Endpoint	{{studio_base_url}}/UTILITY/files/uploadImageBase64			
Format	JSON			
Method	POST			
Request Parameters	Name	Required	Type	Description
	imageContent	Yes	String	Image content in Base64
	imageName	Yes	String	Image file name(With a suffix, such as .jpg, .png)
	serviceType	No	String	Image type person: face image device: device image map: map image temp: temp image (default value, this type of images will be cleaned up periodically)
Response Parameters	Name	Required	Type	Description
	data		String	Image URI
Sample Request	<pre>{ "imageContent ": "{{image in Base64}}", "imageName": "example.jpg", "serviceType": "person" }</pre>			
Sample Response	<pre>{ "code": "0000", "msg": "success", "data": "/images/person/20200311/93/eae3d34d7071a6bc54dbc4e44a47c643.jpg", "success": true }</pre>			

7.2.3 Purge Image

API Endpoint	{{studio_base_url}}/UTILITY/files/purgeImage
Format	JSON
Method	POST

Request Parameters	Name	Required	Type	Description
	beforeDays	Yes	String	Clean images that last modified time is before "beforeDays"
	folders	Yes	Array[String]	Image folders "/images/person": person images "/images/map": map images "/images/sensedlc": images captured by SenseDLC "/images/sensehello": images captured by SenseHello "/images/senseid": images captured by SenseID "/images/sensenebula": images captured by SenseNebula "/images/sensepass": images captured by SensePass "/images/skeeper": images captured by SenseKeeper "/images/cognitivesvc": images captured for facial analysis "/images/structural ": images captured for structured analysis
Response Parameters	Name	Required	Type	Description
	data		String	null
Sample Request	{ "folders": ["/images/sensepass"], "beforeDays": "180" }			
Sample Response	{ "code": "0000", "msg": "成功", "data": null, "success": true }			

7.2.4 Upload File in Form-data

API Endpoint	{{studio_base_url}}/UTILITY/files/uploadFile
Method	POST

Request Format	form-data			
Request Parameters	Name	Required	Type	Description
	file	Yes	File	File
	serviceType	No	String	File type log: log file video: video file ota: OTA upgrade pack temp: temp file (default value, this type of files will be purged periodically)
Response Parameters	Name	Required	Type	Description
	data		String	Uploaded file path
Sample Request	<pre>--form 'file=@/D:/sample.log' \ --form 'serviceType=log'</pre>			
Sample Response	<pre>{ "code": "0000", "msg": "成功", "data": "/files/log/20200624/0/sample.log", "success": true }</pre>			

7.2.5 Upload File in Base64

API Endpoint	{{studio_base_url}}/UTILITY/files/uploadFileBase64			
Method	POST			
Request Format	JSON			
Request Parameters	Name	Required	Type	Description
	fileName	Yes	String	File name with filename extension
	fileContent	Yes	String	File content in Base64
	serviceType	No	String	File type log: log file video: video file ota: OTA upgrade pack temp: temp file (default value, this type of files will be purged periodically)
Response Parameters	Name	Required	Type	Description
	data		String	Uploaded file path

Sample Request	<pre>{ "fileContent": "c2FtcGxLMxvZw==", "fileName": "example.log", "serviceType": "log" }</pre>
Sample Response	<pre>{ "code": "0000", "msg": "成功", "data": "/files/log/20200624/192/example.log", "success": true }</pre>

8 Cognitive API

8.1 Face Management

8.1.1 Extract Feature from Image Base64

API Endpoint	{{studio_base_url}}/COGNITIVESVC/cognitive/face/getFeatureByImageBase64			
Method	POST			
Request Format	form-data			
Request Parameters	Name	Required	Type	Description
	figureImageBase64	Yes	String	Image in Base64
Response Parameters	Name	Required	Type	Description
	data		Feature	
Feature	face		Array[Face]	
Face	feature		String	Feature
	detect		Array[Array[Integer]]	Face coordinates
	id		Integer	Face ID
Sample Request	--form 'figureImageBase64={{image in Base64}}'			
Sample Response	<pre>{ "code": "0000", "msg": "成功", "data": { "face": [{ "feature": "Ph%2Bdp72yKKGK9vLRGPZNCIj29JpC8jBnwvWlmWz1fsle88xeWvJoAiD2PLtc%2BBuCXPQm vnr06Yf2%2BCu38PJHwd70i8qS%2BMA%2FPvb24OL0l1vC9DaFkPXJRjz2SlqG9%2FOMtPeYc nbwnZzk889AGPODMUrxT2c09pIKmPYLPgixm64S9nUEXPf17jL000eG9DI%2F9O7zomzvLi 29je%2B2PUjVNz0MSOS81QmavEQsiDzjU7W98GUUPYKapD2ByMK7XOHIO0joUTy4Xhy9IG O7vblBEbwy2Vq98q9cvYraJr3xsJm8JZOGPgEc0D0K1gq59uoovL3hmj3AUyU9IxXsPDIYrD12e Gc9a3AOvW6PGj2C17y8fR1%2BPRDBot1FglU9DdvNvSvnkzz5Gme74R2PvQkq2z2x7HA%2 BA%2B%2F%2FPQJhD72Udsk9KHQgvlbbgr3Z%2FNU%2BFVKkPTx80z3lwKI9HUPpPZ0q6Tw h2C%2B9Tt9EvQjprzzl6ai7giCTvYrDpL12MFu9NUPYvVm%2Fir2t2bg9UTkNPWVFZj3aTL27w 9OKPZtRsb3BgNg9OLdNPKAXtT1gcPK86UPSO62Ksz2PgJm9jtmUvaErTTybNjG9QTgfPOTeZL yNPBO89LRwvT4flr36K9I9r901PPNkELyveLu9Aeq9OtOUb701PK09y5MoPcmi2D2lWca%2B HBa6vSgztr2hR8c8xfCePYORHD1W2Ls9XKBfPXgyeb24xOu9M4PdvZebZj0Qmeg9D9dJvhLZB D4K4%2BI96IWPJQq9Ub0PPxy7JiWNPTeYzj3MbE69CnTyvADRBLzbyGc91vbtPY7fK74BJSm</pre>			

```

9TWIRvYHICby%2FFrY9tBgIPaXLLT4C2eg8hzGXPbDzmrzohbG9ELw6vY1WfjqzMB%2B82yv8
PSbOfTxi57O9JLddPae%2F%2Bzy%2BZu482v17PPfQcD1MArW9FUX9PDImCL4JDiW8yYU%2
BPHGnlj2yZ1k9J82gPaiQZ70pTqM8%2FkL7PQpnADzDBtW9sBNXPUTf5L1Leuq8iGIFvAD6ez
y9M3E9r6AUvSAwn72Tq5e8YsVvvJFmx72Cjvw78EuevHwtwjxbdtQ8UWMGO4yNvj4F0V87
guV2PVndPDuc%2BAY8s02HvDAfbj4LhY69E8S%2FvLISsb13hzW8KIJDvXM9Mz0ZfjG71FyxO
2eF1z0tva69iwkCvMOVvb29Qnm5IrtqPMJTPj2yEBy8eUeiPPXUeLt1f4667F4RPXUZuTy0NN
A9dbNfvD9dnz0V8eK8nuG4vWQv71nknU9Ue8FvQz1kL0ffiy8tiEPvHocFj1PXhg9S7i6PErF8
D0snCW9g5xhvPVH%2Fjs8yh68l%2BuZPQCCsDz%2Bc4q7IE4%2BvbDukg%3D%3D",
    "detect": [
        [
            64,
            103
        ],
        [
            200,
            240
        ]
    ],
    "id": 0
}
},
"success": true
}

```

8.1.2 Extract Feature from Image Path

API Endpoint	{{studio_base_url}}/COGNITIVESVC/cognitive/face/getFeatureByImagePath			
Method	POST			
Request Format	form-data			
Request Parameters	Name	Required	Type	Description
	figureImageUrl	Yes	String	Image path
Response Parameters	Name	Required	Type	Description
	data		Feature	
Feature	face		Array[Face]	
Face	feature		String	Feature
	detect		Array[Array[Integer]]	Face coordinates

	id		Integer	Face ID
Sample Request	<pre>--form 'figureImageUrl=/images/person/20200624/93/763b1bcc79e658beb572d86f28544caa.jpg'</pre>			
Sample Response	<pre>{ "code": "0000", "msg": "成功", "data": { "face": [{ "feature": "PilulL2M9ye932ldPdJwmj4OcFu80%2FhKvRDIEj10oJm8qxx5uicIPj2QH7c%2BC%2FsUO%2F8zhr279CC%2BBchfPQ3CV7yhF8S%2BOU6AvaW1aryG3fO9OLLGPRF3nT21vYO9puOYPe4B0rwzoGI8UdWUPQhrpLwl1SU9U8Q5PWGADj1NSia9x%2BjaPc4iOb1DVGO9MwPSuw47Xry39i%2B9nPIMPRJIVTyeO229OAE1vL0zbDzvoEC903thPN1mQT2Fc%2F88l0QkvKlz8z0pNPq9kGVVve85pbybf8q%2BBLOgvZL2TL3dAPc7AfVtPgpd6rwlva28%2Fe%2BuvVGK8z3jJec8%2FdWOPCbF8j2R%2BCK9JYq2vcvm8j221cg8m00API8Laj1j3049p3QkvPYU6Dypd989Dt1ivWpCcD2mDIM9%2FHEPPOAArL0vzcs8q9nNO5s2872VvP495HumPPk3Hj30SqU83xKHPUMIEDxW%2Fm%2B8n79YvUgFKj0KR6s8K4lJvavWO71Qgh%2B9PnpnvUwWwb2iomo9PRZ5PWNC%2Bz36NzC9kjkPclx673bjus9TarcPT64vD2WDzm8SmXGPODWkD2EkPO9mqrnvXsxuzy2T8a8akhfPTr697zhYjW9G%2FDKvUE9cL4YYbA9zltfPQ4sU70YLyy9rVsHu72%2BYL2GZ1o9RdtXP7Y5z4LiDS%2BIL7uvCXmK73IfKo8ue3KPZCQAT2Cni88ToYTPV27BL3XN7C9HNhNvZS%2FLz0sLo48Jeg9vbNPUD3IXRg9ePStPBz2RL1ls9U8qxNRPXNyiT2Sowy9AclXvH1GLLzMb8k9yNg5PbOgPL38m%2F%2B9b3eUvbqpajxadsQ9u%2F5kPe%2F2Fz3cZ0s8hVHJPaLQWzuoTGu9XdGavVKs5z0bcwM7LqqAOxPayrJsPe9RudTPVakiT0wooa8iVyGvI4xFD1FAkQ8svnhPLUT9r3k6ia7caO4PLEGAD1p6wU9KWIBPY0agjsLWjE9ZTHLPF6XbT1HCfC9hS5IPYqZjBwKp7%2B9BMKjPSiu8Dysy3A9pspivWJ9Zb2A%2FmC8AQ2PPQLMwL2ZOOe7x7thPMjQSjwqG8S8kitMu66aBT2u9ws8LaPOPUNyszWuUf882hs3PUcUQj3Fw069mKVou9Yi6708TXq9G22ZvaTPozsEHY88mfWMPOHIBz1wqBK9dKaFPQhWkr2pN2y64NoyPSy4zj2QcFm8lWhqPRwBU7xobc28%2BZWYPX6B5Tz0FEI9B26jPC57aTyXkeS86MqPvVFGt72MQ%2Bs9QsPDu7b7Qb0T46c85hb4vLAu4D1gbII9gEOvPNMfsT0q%2F%2FO9MgCPvJDle7v%2F2kA8QbQbPHPX2TxB5Zq8td1zvYoKAg%3D%3D", "detect": [[27, 59], [457,</pre>			

```
477
    ]
    ],
    "id": 0
  }
]
},
"success": true
}
```

8.1.3 Extract Attribute from Image Base64

API Endpoint	{{studio_base_url}}/COGNITIVESVC/cognitive/face/getAttributeByImageBase64			
Method	POST			
Request Format	form-data			
Request Parameters	Name	Required	Type	Description
	figureImageBase64	Yes	String	Image in Base64
Response Parameters	Name	Required	Type	Description
	data		AttributeInfo	
AttributeInfo	id		Integer	Face ID
	imageId		Integer	Image ID
	label		Integer	Label
	hunter		Array[Array[Integer]]	Face coordinates
	attribute		Attribute	
Sample Request	--form 'figureImageBase64={{image in Base64}}'			
Sample Response	{ "code": "0000", "msg": "成功", "data": [{ "id": 0, "imageId": 0, "label": 37017, "hunter": [64,			

```
103
],
[
    200,
    240
]
],
"attribute": {
    "age_lower_limit": {
        "age_lower_limit": 17
    },
    "age_up_limit": {
        "age_up_limit": 27
    },
    "respirator_color": {
        "color_type_other": 0.000023760961994412355,
        "color_type_none": 0.99997627735137939
    },
    "st_expression": {
        "st_angry": 0.007050276268273592,
        "st_sorrow": 0.014857389032840729,
        "st_calm": 0.9416508674621582,
        "st_scared": 0.0036149132065474987,
        "st_yawn": 0.003620693227276206,
        "st_surprised": 0.0053753373213112354,
        "st_happy": 0.020214738324284554,
        "st_disgust": 0.0036157472059130669
    },
    "ethic_code": {
        "uygur": 0.020860686898231506,
        "ethic_code_type_other": 0.97913932800292969
    },
    "black_type": {
        "st_brown": 0.22751863300800323,
        "st_black": 0.772481381893158
    },
    "skin_color": {
        "white": 0.028056677430868149,
```

```
        "yellow": 0.96879482269287109,  
        "black": 0.0031484670471400023  
    },  
    "glass_style": {  
        "sunglasses": 0.000071419060986954719,  
        "glasses_style_type_none": 0.99962866306304932,  
        "transparent_glasses": 0.00029991695191711187  
    },  
    "st_age_value": {  
        "st_age_value": 22  
    },  
    "gender_code": {  
        "female": 0.99997609853744507,  
        "male": 0.000023927606889628805  
    },  
    "st_age": {  
        "st_adult": 1,  
        "st_child": 0,  
        "st_old": 0  
    },  
    "cap_style": {  
        "cap": 0.0017324171494692564,  
        "hat_style_type_none": 0.99826759099960327  
    },  
    "st_helmet_style": {  
        "st_helmet_style_type_none": 0.99993318319320679,  
        "st_helmet": 0.00006682171078864485  
    },  
    "mustache_style": {  
        "whiskers": 0.015038252808153629,  
        "mustache_style_type_none": 0.98496168851852417  
    }  
    }  
    },  
    "success": true  
}
```

8.1.4 Extract Attribute from Image Path

API Endpoint	{{studio_base_url}}/COGNITIVESVC/cognitive/face/getAttributeByImagePath			
Method	POST			
Request Format	form-data			
Request Parameters	Name	Required	Type	Description
	figureImageUrl	Yes	String	Image in Base64
Response Parameters	Name	Required	Type	Description
	data		Attribute	
Attribute	id		Integer	Face ID
	imageId		Integer	Image ID
	label		Integer	Label
	hunter		Array[Array[Integer]]	Face coordinates
	attribute		Attribute	
Sample Request	--form 'figureImageUrl=/images/person/20200624/93/763b1bcc79e658beb572d86f28544caa.jpg'			
Sample Response	{ "code": "0000", "msg": "成功", "data": [{ "id": 0, "imageId": 0, "label": 37017, "hunter": [[27, 59], [457, 477]], "attribute": { "age_lower_limit": { "age_lower_limit": 15 } } }] }			

```
},
"age_up_limit": {
  "age_up_limit": 25
},
"respirator_color": {
  "color_type_other": 0.00027386393048800528,
  "color_type_none": 0.99972611665725708
},
"st_expression": {
  "st_angry": 0.00069544912548735738,
  "st_sorrow": 0.00010877022577915341,
  "st_calm": 0.99858838319778442,
  "st_scared": 0.00010238654795102775,
  "st_yawn": 0.0001024908633553423,
  "st_surprised": 0.00017854649922810495,
  "st_happy": 0.00012150364636909217,
  "st_disgust": 0.0001024030934786424
},
"ethic_code": {
  "uygur": 0.952105700969696,
  "ethic_code_type_other": 0.04789423942565918
},
"black_type": {
  "st_brown": 0.0079759117215871811,
  "st_black": 0.99202406406402588
},
"skin_color": {
  "white": 0.074003152549266815,
  "yellow": 0.89875364303588867,
  "black": 0.027243111282587051
},
"glass_style": {
  "sunglasses": 0.017305755987763405,
  "glasses_style_type_none": 0.90273618698120117,
  "transparent_glasses": 0.079958021640777588
},
"st_age_value": {
  "st_age_value": 20
}
```

```
    },
    "gender_code": {
      "female": 0.9937012791633606,
      "male": 0.0062987585552036762
    },
    "st_age": {
      "st_adult": 1,
      "st_child": 0,
      "st_old": 0
    },
    "cap_style": {
      "cap": 0.001062926952727139,
      "hat_style_type_none": 0.99893707036972046
    },
    "st_helmet_style": {
      "st_helmet_style_type_none": 0.99924498796463013,
      "st_helmet": 0.00075507804285734892
    },
    "mustache_style": {
      "whiskers": 0.017389640212059021,
      "mustache_style_type_none": 0.98261034488677979
    }
  }
},
"success": true
}
```

8.1.5 Extract Feature and Attribute from Image Base64

API Endpoint	{{studio_base_url}}/COGNITIVESVC/cognitive/face/getFeatureAndAttributeByImageBase64			
Method	POST			
Request Format	form-data			
Request Parameters	Name	Required	Type	Description
	figureImageBase64	Yes	String	Image in Base64
Response Parameters	Name	Required	Type	Description
	data		ExtractInfo	

ExtractInfo	face		Array[Face]	
Face	feature		String	Feature
	detect		Array[Array[Integer]]	Face coordinates
	attributes		Attribute	
	id		Integer	Face ID
Sample Request	--form 'figureImageBase64={{image in Base64}}'			
Sample Response	<pre>{ "code": "0000", "msg": "成功", "data": { "face": [{ "feature": "Ph%2Bdp72yKGK9vLRGPZNCIj29JpC8jBnwvWlmWz1fsle88xeWvJoAiD2PLtc%2BBuCXPQm vnr06Yf2%2BCu38PIHwd70i8qS%2BMA%2FPvb24OL0l1vC9DaFkPXJRjZ2SlqG9%2FOMtPeYc nbwnZzk889AGPODMUrxT2c09plKmPYLPgJxm64S9nUEXPf17jL000eG9DI%2F9O7zomzvwLi 29je%2B2PUjVNz0MSOS81QmavEQsiDzjU7W98GUUPYKapD2ByMK7XOHio0joUTy4Xhy9IG O7vbIBebwy2Vq98q9cvYraJr3xsJm8JZOGPgEc0D0K1gq59uooVL3hmj3AUyU9IxXsPDIYrD12e Gc9a3AOvW6PGj2C17y8fR1%2BPRDBot1FglU9DdvNvSvnkzz5Gme74R2PvQkq2z2x7HA%2 BA%2B%2F%2FPQJhD72Udsk9KHQgvlbbgr3Z%2FNU%2BFVKkPTx80z3lwKI9HUPpPZ0q6Tw h2C%2B9Tt9EvQjprzzl6ai7giCTvYrDpL12MFu9NUPyVvVm%2Fir2t2bg9UTkNPWVFZj3aTL27w 9OKPZtRsb3BgNg9OLdNPKAXtT1gcPK86UPSO62Ksz2PgJm9jtmUvaErTTybNjG9QTgfPOTeZL yNPBO89LRwvT4flr36K9I9r901PPNkELyveLu9Aeq9OtOUb701PK09y5MoPcmi2D2lWca%2B HBa6vSgztr2hR8c8xfCePY0RHD1W2Ls9XKBfPXgyeb24xOu9M4PdvZebZj0Qmeg9D9dJvhLZB D4K4%2BI96lWJPQq9Ub0PPxy7JiWNPTeYZj3MbE69CnTyvADRBzbyGc91vbtPY7fK74BJSm 9TWIRvYHICby%2FFrY9tBgIPaXLLT4C2eg8hzGXPbDzmrzohbG9ELw6vY1WfjqzMB%2B82yv8 PSb0fTxi57O9JLddPae%2F%2Bzy%2BZu482v17PPfQcD1MArW9FUX9PDImCL4JDiW8yYU%2 BPHGnlj2yZ1k9J82gPaiQZ70pTqM8%2FkL7PQpnADzDBtW9sBNXPUTf5L1Leuq8iGIFvAD6ez y9M3E9r6AUvSAwn72Tq5e8YsVvvJFmx72Cjvw78EuevHwtwjxbdtQ8UWMGO4yNvj4F0V87 guV2PVndPDuc%2BAY8s02HvDAfbj4LhY69E8S%2FvLISsb13hzW8KIJDvXM9Mz0ZfjG71FyxO 2eF1z0tva69iwkCvMOVvb29Qnm5lRtqPMJTPj2yEBY8eUeiPPXUeLt1f4667F4RPXUZuTy0NN A9dbNfvD9dnz0V8eK8nuG4vWQv71nknU9Ue8FvQz1kL0ffiy8tiEPvHocFj1PXhg9S7i6PErF8 D0snCW9g5xhvPVH%2Fjs8yh68l%2BuZPQCCsDz%2Bc4q7IE4%2BvbDukg%3D%3D", "detect": [[64, 103]] }] } }</pre>			

```
],  
[  
    200,  
    240  
]  
],  
"attributes": {  
    "age_lower_limit": {  
        "age_lower_limit": 17  
    },  
    "age_up_limit": {  
        "age_up_limit": 27  
    },  
    "respirator_color": {  
        "color_type_other": 0.000023760961994412355,  
        "color_type_none": 0.99997627735137939  
    },  
    "st_expression": {  
        "st_angry": 0.007050276268273592,  
        "st_sorrow": 0.014857389032840729,  
        "st_calm": 0.9416508674621582,  
        "st_scared": 0.0036149132065474987,  
        "st_yawn": 0.003620693227276206,  
        "st_surprised": 0.0053753373213112354,  
        "st_happy": 0.020214738324284554,  
        "st_disgust": 0.0036157472059130669  
    },  
    "ethic_code": {  
        "uygur": 0.020860686898231506,  
        "ethic_code_type_other": 0.97913932800292969  
    },  
    "black_type": {  
        "st_brown": 0.22751863300800323,  
        "st_black": 0.772481381893158  
    },  
    "skin_color": {  
        "white": 0.028056677430868149,  
        "yellow": 0.96879482269287109,
```

```
        "black": 0.0031484670471400023
      },
      "glass_style": {
        "sunglasses": 0.000071419060986954719,
        "glasses_style_type_none": 0.99962866306304932,
        "transparent_glasses": 0.00029991695191711187
      },
      "st_age_value": {
        "st_age_value": 22
      },
      "gender_code": {
        "female": 0.99997609853744507,
        "male": 0.000023927606889628805
      },
      "st_age": {
        "st_adult": 1,
        "st_child": 0,
        "st_old": 0
      },
      "cap_style": {
        "cap": 0.0017324171494692564,
        "hat_style_type_none": 0.99826759099960327
      },
      "st_helmet_style": {
        "st_helmet_style_type_none": 0.99993318319320679,
        "st_helmet": 0.00006682171078864485
      },
      "mustache_style": {
        "whiskers": 0.015038252808153629,
        "mustache_style_type_none": 0.98496168851852417
      }
    },
    "id": 0
  }
]
},
"success": true
}
```

8.1.6 Extract Feature and Attribute from Image Path

API Endpoint	{{studio_base_url}}/COGNITIVESVC/cognitive/face/getFeatureAndAttributeByImagePath			
Method	POST			
Request Format	form-data			
Request Parameters	Name	Required	Type	Description
	figureImageUrl	Yes	String	Image path
Response Parameters	Name	Required	Type	Description
	data		ExtractInfo	
ExtractInfo	face		Array[Face]	
Face	feature		String	Feature
	detect		Array[Array[Integer]]	Face coordinates
	attributes		Attribute	
	id		Integer	Face ID
Sample Request	--form 'figureImageUrl=/images/person/20200624/93/763b1bcc79e658beb572d86f28544caa.jpg'			
Sample Response	<pre>{ "code": "0000", "msg": "成功", "data": { "face": [{ "feature": "PilulL2M9ye932ldPdJwmj4OcFu80%2FhKvRDIEj10oJm8qxx5uicIPj2QH7c%2BC%2FsUO%2F8zhr279CC%2BBchfPQ3CV7yhF8S%2BOU6AvaW1aryG3fO9OLLGPRF3nT21vYO9puOYPe4B0rwzoGI8UdWUPQhrpLwI1SU9U8Q5PWGADj1NSia9x%2BjaPc4iOb1DVGO9MwPSuw47Xry39i%2B9nPIMPRJIVTyeO229OAE1vL0zbDzvoEC903thPN1mQT2Fc%2F88l0QKvKlz8z0pNPq9kGVVve85pbybf8q%2BBLOgvZL2TL3dAPc7AfVtPgpd6rwlva28%2Fe%2BuvVGK8z3jJEc8%2FdWOPCbF8j2R%2BCK9JYq2vcvm8j221cg8m00API8Laj1j3049p3QkvPYU6Dypd989Dt1ivWpCcD2mDIM9%2FHEPPOAArL0vzcs8q9nNO5s2872VvP495HumPPk3Hj30SqU83xKHPUMIEDxW%2Fm%2B8n79YvUgFKj0KR6s8K4IJvavWO71Qgh%2B9PnpnvUwWwb2iomo9PRZ5PWNC%2Bz36NzC9kjkPclx673bjus9TarcPT64vD2WDzm8SmXGPODWkD2EkPO9mqrnvXsxuzy2T8a8akhfPTr697zhYjW9G%2FDKvUE9cL4YYbA9zltfPQ4sU70YLyy9rVsHu72%2BYL2GZ1o9RdtXPa7Y5z4LiDS%2BIL7uvCXmK73IfKo8ue3KPZCQAT2Cni88ToYTPV27BL3XN7C9HNhNvZS%2FLz0sLo48Jeg9vbNPUD3IXRg9ePstPBz2RL1ls9U8qxNRPXNyit2Sowy9AcIXvH1GLLzMb8k9yNg5PbOgPL38m%2F%2B9b3eUvbqpajxadsQ9u%2F5kPe%2F2Fz3cZ0s8hVHJPaLQWzuoTGu9XdGavVK</pre>			

s5z0bcwM7LqqAOxPayrqJsPe9RudTPVakiT0wooa8iVyGvI4xFD1FAkQ8svnhPLUT9r3k6ia7ca
O4PLEGAD1p6wU9KWIBPY0agjsLWjE9ZTHLPF6XbT1HCfC9hS5IPYqZjBwKp7%2B9BMKjPSiu8
Dysy3A9pspivWJ9Zb2A%2FmC8AQ2PPQLMwL2ZOoe7x7thPMjQSjwqG8S8kitMu66aBT2u9
ws8LaPOPUNyszWuUf882hs3PUcUQj3Fw069mKVou9Yi6708TXq9G22ZvaTPozsEHY88mfWm
POHIBz1wqBK9dKaFPQhWkr2pN2y64NoyPSy4zj2QcFm8lWhqPRwBU7xobc28%2BZWYPX6
B5Tz0FEI9B26jPC57aTyXkeS86MqPvVFGt72MQ%2Bs9QsPDu7b7Qb0T46c85hb4vLAu4D1gb
lI9gEOvPNMfsT0q%2F%2FO9MgCPvJDle7v%2F2kA8QbQbPHPX2TxB5Zq8td1zvYoKAg%3D%
3D",

"detect": [

[

27,

59

],

[

457,

477

]

],

"attributes": {

"age_lower_limit": {

"age_lower_limit": 15

},

"age_up_limit": {

"age_up_limit": 25

},

"respirator_color": {

"color_type_other": 0.00027386393048800528,

"color_type_none": 0.99972611665725708

},

"st_expression": {

"st_angry": 0.00069544912548735738,

"st_sorrow": 0.00010877022577915341,

"st_calm": 0.99858838319778442,

"st_scared": 0.00010238654795102775,

"st_yawn": 0.0001024908633553423,

"st_surprised": 0.00017854649922810495,

"st_happy": 0.00012150364636909217,

"st_disgust": 0.0001024030934786424

```
},
"ethic_code": {
  "uygur": 0.952105700969696,
  "ethic_code_type_other": 0.04789423942565918
},
"black_type": {
  "st_brown": 0.0079759117215871811,
  "st_black": 0.99202406406402588
},
"skin_color": {
  "white": 0.074003152549266815,
  "yellow": 0.89875364303588867,
  "black": 0.027243111282587051
},
"glass_style": {
  "sunglasses": 0.017305755987763405,
  "glasses_style_type_none": 0.90273618698120117,
  "transparent_glasses": 0.079958021640777588
},
"st_age_value": {
  "st_age_value": 20
},
"gender_code": {
  "female": 0.9937012791633606,
  "male": 0.0062987585552036762
},
"st_age": {
  "st_adult": 1,
  "st_child": 0,
  "st_old": 0
},
"cap_style": {
  "cap": 0.001062926952727139,
  "hat_style_type_none": 0.99893707036972046
},
"st_helmet_style": {
  "st_helmet_style_type_none": 0.99924498796463013,
  "st_helmet": 0.00075507804285734892
}
```

```

    },
    "mustache_style": {
      "whiskers": 0.017389640212059021,
      "mustache_style_type_none": 0.98261034488677979
    }
  },
  "id": 0
}
]
},
"success": true
}

```

8.1.7 Get Face Angle from Image Base64

API Endpoint	{{studio_base_url}}/COGNITIVESVC/cognitive/face/getHeadposeByBase64			
Method	POST			
Request Format	form-data			
Request Parameters	Name	Required	Type	Description
	figureImageBase64	Yes	String	Image in Base64
Response Parameters	Name	Required	Type	Description
	data		FaceAngle	
FaceAngle	face		Array[Array[Double]]	Face angles of the detected face
Sample Request	--form 'figureImageBase64={{image in Base64}}'			
Sample Response	<pre> { "code": "0000", "msg": "成功", "data": { "face": [[14.468512, 1.490012, -2.1727037]] } }, </pre>			

	<pre>"success": true }</pre>
--	------------------------------

8.1.8 Get Image Quality of Image Base64

API Endpoint	{{studio_base_url}}/COGNITIVESVC/cognitive/face/getImageQualityByImageBase64			
Method	POST			
Request Format	form-data			
Request Parameters	Name	Required	Type	Description
	figureImageBase64	Yes	String	Image Base64
Response Parameters (data)	Name	Required	Type	Description
	face		Array[Face]	
Face	score		Double	Image quality score
	detect		Array[Array[Double]]	Face coordinates
	id		String	Face ID
Sample Request	--form 'figureImageBase64={{image in Base64}}'			
Sample Response	<pre>{ "code": "0000", "msg": "success", "data": { "face": [{ "score": 0.14015938, "detect": [[45.0, 36.0], [105.0, 98.0]], "id": 0 }] } },</pre>			

```
"success": true
```

```
}
```

8.1.9 Detect Face from Image Base64

API Endpoint	{{studio_base_url}}/COGNITIVESVC/cognitive/face/getDetectByBase64			
Method	POST			
Request Format	form-data			
Request Parameters	Name	Required	Type	Description
	figureImageBase64	Yes	String	Image in Base64
Response Parameters	Name	Required	Type	Description
	data		Detect	
Detect	face_X		Array[Array[Integer]]	Face coordinates of the detected face
Sample Request	--form 'figureImageBase64={{image in Base64}}'			
Sample Response	<pre>{ "code": "0000", "msg": "成功", "data": { "face_3": [[56, 94], [98, 136]], "face_2": [[196, 99], [238,</pre>			

```
        141
      ]
    ],
    "face_1": [
      [
        110,
        121
      ],
      [
        148,
        161
      ]
    ],
    "face_0": [
      [
        268,
        95
      ],
      [
        309,
        135
      ]
    ]
  },
  "success": true
}
```

8.1.10 Detect Face from Image Path

API Endpoint	{{studio_base_url}}/COGNITIVESVC/cognitive/face/getDetectByImagePath			
Method	POST			
Request Format	form-data			
Request Parameters	Name	Required	Type	Description
	figureImageBase64	Yes	String	Image in Base64
Response Parameters	Name	Required	Type	Description
	data		Detect	

Detect	face_X		Array[Array[Integer]]	Face coordinates of the detected face
Sample Request	--form 'figureImageUrl=/images/temp/20200628/93/97e11cdcca271f7973d523b587674af7.jpg'			
Sample Response	{ "code": "0000", "msg": "成功", "data": { "face_3": [[56, 94], [98, 136]], "face_2": [[196, 99], [238, 141]], "face_1": [[110, 121], [148, 161]], }, }			

```
        "face_0": [
            [
                268,
                95
            ],
            [
                309,
                135
            ]
        ]
    },
    "success": true
}
```

8.1.11 Detect Face from Image Base64 with Confidence Score

API Endpoint	{{studio_base_url}}/COGNITIVESVC/cognitive/image/cutFaceByImageBase64			
Method	POST			
Request Format	form-data			
Request Parameters	Name	Required	Type	Description
	figureImageBase64	Yes	String	Image Base64
Response Parameters	Name	Required	Type	Description
	data		Result	
Result	results		Array[Result]	Face coordinates and confidence score of the detected face
Result	id		Integer	Face id
	x		Integer	Face X coordinates in the upper left corner
	y		Integer	Face Y coordinates in the upper left corner
	width		Integer	Face width
	height		Integer	Face height
	confidence		Double	Confidence score
Sample Request	--form 'figureImageBase64={{image in Base64}}'			
Sample Response	{ "code": "0000", "msg": "成功",			

```
"data": {
  "results": [
    {
      "id": 0,
      "x": 108,
      "y": 119,
      "width": 40,
      "height": 45,
      "confidence": 0.920668
    },
    {
      "id": 1,
      "x": 272,
      "y": 93,
      "width": 40,
      "height": 44,
      "confidence": 0.90789336
    },
    {
      "id": 2,
      "x": 199,
      "y": 97,
      "width": 42,
      "height": 45,
      "confidence": 0.89403945
    },
    {
      "id": 3,
      "x": 57,
      "y": 90,
      "width": 43,
      "height": 45,
      "confidence": 0.87219876
    }
  ]
},
"success": true
}
```

8.1.12 Detect Face from Image Path with Confidence Score

API Endpoint	{{studio_base_url}}/COGNITIVESVC/cognitive/image/cutFaceByImagePath			
Method	GET			
Request Format	form-data			
Request Parameters	Name	Required	Type	Description
	figureImageUrl	Yes	String	Image path
Response Parameters	Name	Required	Type	Description
	data		Result	
Result	results		Array[Result]	Face coordinates and confidence score of the detected face
Result	id		Integer	Face id
	x		Integer	Face X coordinates in the upper left corner
	y		Integer	Face Y coordinates in the upper left corner
	width		Integer	Face width
	height		Integer	Face height
	confidence		Double	Confidence score
Sample Request	--form 'figureImageUrl=/images/temp/20200628/93/97e11cdcca271f7973d523b587674af7.jpg'			
Sample Response	{ "code": "0000", "msg": "成功", "data": { "results": [{ "id": 0, "x": 108, "y": 119, "width": 40, "height": 45, "confidence": 0.920668 }, { "id": 1, 			

```
        "x": 272,  
        "y": 93,  
        "width": 40,  
        "height": 44,  
        "confidence": 0.90789336  
    },  
    {  
        "id": 2,  
        "x": 199,  
        "y": 97,  
        "width": 42,  
        "height": 45,  
        "confidence": 0.89403945  
    },  
    {  
        "id": 3,  
        "x": 57,  
        "y": 90,  
        "width": 43,  
        "height": 45,  
        "confidence": 0.87219876  
    }  
]  
},  
"success": true  
}
```

8.2 1:1 Compare

8.2.1 Compare Feature

API Endpoint	{{studio_base_url}}/COGNITIVESVC/cognitive/face/compareFeatures			
Method	POST			
Request Format	form-data			
Request Parameters	Name	Required	Type	Description
	feature1	Yes	String	Feature 1
	feature2	Yes	String	Feature 2
Response Parameters	Name	Required	Type	Description
	data		Double	Similarity

Sample Request	--form 'feature1={{feature1}}' \ --form 'feature2={{feature2}}'
Sample Response	{ "code": "0000", "msg": "成功", "data": 0.9909673, "success": true }

8.2.2 Compare Face in Image Base64

API Endpoint	{{studio_base_url}}/COGNITIVESVC/cognitive/face/compareImageBase64s			
Method	POST			
Request Format	form-data			
Request Parameters	Name	Required	Type	Description
	figureImageBase64_1	Yes	String	Image 1 in Base64
	figureImageBase64_2	Yes	String	Image 2 in Base64
Response Parameters	Name	Required	Type	Description
	data		Double	Similarity
Sample Request	--form 'figureImageBase64_1={{image 1 in Base64}}' \ --form 'figureImageBase64_2={{image 2 in Base64}}'			
Sample Response	{ "code": "0000", "msg": "成功", "data": 0.9909673, "success": true }			

8.2.3 Compare Face in Image Path

API Endpoint	{{studio_base_url}}/COGNITIVESVC/cognitive/face/compareImagePaths			
Request Method	POST			
Request Format	form-data			
Request Parameters	Name	Required	Type	Description
	figureImageUrl_1	Yes	String	Image path 1
	figureImageUrl_2	Yes	String	Image path 2

Response Parameters	Name	Required	Type	Description
	data		Double	Similarity
Sample Request	<pre>--form 'figureImageUrl_1=/images/person/20200401/32/091e6a1bb66d720e51a71832cc904875.jpg' \ --form 'figureImageUrl_2=/images/person/20200615/32/091e6a1bb66d720e51a71832cc904875.jpg'</pre>			
Sample Response	<pre>{ "code": "0000", "msg": "成功", "data": 0.9953087, "success": true }</pre>			

8.3 1:N Search

8.3.1 Search Person by Image Base64

API Endpoint	{{studio_base_url}}/COGNITIVESVC/cognitive/face/compareFacelImage			
Method	POST			
Request Format	form-data			
Request Parameters	Name	Required	Type	Description
	figureImageBase64	Yes	String	Image in Base64
	personType	Yes	String	Person type Target: target group Passer: stranger
	personGroup	No	String	Group unique ID
	tag	No	String	Person tag
	count	No	String	Search for top K results
	threshold	No	String	Comparison threshold score
	faceAttr	No	Boolean	Whether to respond the attributes (true/false)
Response Parameters	Name	Required	Type	Description
	data		Array[SearchResult]	
SearchResult	score		Double	Similarity
	personID		String	Matched person unique ID
	targetType		String	Person type Target: target group

				Passer: stranger
	attribute		Attribute	Person attributes
Sample Request	<pre>--form 'figureImageBase64={{image in Base64}}' \ --form 'personType =Target' \ --form 'personGroup=15' \ --form 'count=1' \ --form 'threshold =0.8' \ --form 'faceAttr=true'</pre>			
Sample Response	<pre>{ "code": "0000", "msg": "成功", "data": [{ "score": 0.9936469, "personID": "1893", "targetType": "Target", "attribute": { "age_lower_limit": { "age_lower_limit": 17 }, "age_up_limit": { "age_up_limit": 27 }, "respirator_color": { "color_type_other": 0.000023760961994412355, "color_type_none": 0.99997627735137939 }, "st_expression": { "st_angry": 0.007050276268273592, "st_sorrow": 0.014857389032840729, "st_calm": 0.9416508674621582, "st_scared": 0.0036149132065474987, "st_yawn": 0.003620693227276206, "st_surprised": 0.0053753373213112354, "st_happy": 0.020214738324284554, "st_disgust": 0.0036157472059130669 }, "ethic_code": {</pre>			

```
"uygur": 0.020860686898231506,  
"ethic_code_type_other": 0.97913932800292969  
},  
"black_type": {  
  "st_brown": 0.22751863300800323,  
  "st_black": 0.772481381893158  
},  
"skin_color": {  
  "white": 0.028056677430868149,  
  "yellow": 0.96879482269287109,  
  "black": 0.0031484670471400023  
},  
"glass_style": {  
  "sunglasses": 0.000071419060986954719,  
  "glasses_style_type_none": 0.99962866306304932,  
  "transparent_glasses": 0.00029991695191711187  
},  
"st_age_value": {  
  "st_age_value": 22  
},  
"gender_code": {  
  "female": 0.99997609853744507,  
  "male": 0.000023927606889628805  
},  
"st_age": {  
  "st_adult": 1,  
  "st_child": 0,  
  "st_old": 0  
},  
"cap_style": {  
  "cap": 0.0017324171494692564,  
  "hat_style_type_none": 0.99826759099960327  
},  
"st_helmet_style": {  
  "st_helmet_style_type_none": 0.99993318319320679,  
  "st_helmet": 0.00006682171078864485  
},  
"mustache_style": {
```

	<pre> "whiskers": 0.015038252808153629, "mustache_style_type_none": 0.98496168851852417 } } }, "success": true } </pre>
--	---

8.3.2 Search Person by Image Base64 in Multiple Groups

API Endpoint	{{studio_base_url}}/COGNITIVESVC/cognitive/face/compareFaceByMultiGroupByBase64			
Method	POST			
Request Format	form-data			
Request Parameters	Name	Required	Type	Description
	figureImageBase64	Yes	String	Image in Base64
	groups	Yes	String	Group unique ID, separated by commas
	thresholds	Yes	String	Comparison threshold score, separated by commas
	count	Yes	String	Search for top K results
Response Parameters	Name	Required	Type	Description
	data		GroupResult	
GroupResult	{{groupID}}		Array[Result]	
Result	{{faceID}}		Double	Similarity
Sample Request	<pre> --form 'figureImageBase64={{image in Base64}}' \ --form 'groups=15,12' \ --form 'thresholds=0.5,0.8' \ --form 'count=1' </pre>			
Sample Response	<pre> { "code": "0000", "msg": "成功", "data": { "12": [{ "1894": 0.9936469 }], "15": [</pre>			

```
{
  "1893": 0.9936469
}
],
},
"success": true
}
```

8.3.3 Search Person by Feature

API Endpoint	{{studio_base_url}}/COGNITIVESVC/cognitive/face/compareFaceImage			
Method	POST			
Request Format	JSON			
Request Parameters	Name	Required	Type	Description
	featureBase64	Yes	String	Feature string
	personType	Yes	String	Person type Target: target group Passer: stranger
	personGroup	No	String	Group unique ID
	personTag	No	String	Person tag
	count	No	String	Search for top K results
	threshold	No	String	Comparison threshold score
Response Parameters	Name	Required	Type	Description
	data		Array[SearchResult]	
SearchResult	score		Double	Similarity
	personID		String	Matched person unique ID
	targetType		String	Person type Target: target group Passer: stranger
Sample Request	{ "count": 1, "personGroup": "9", "personTag": "", "personType": "Target", "threshold": 0.9, "featureBase64": "[the feature string]" }			

Sample Response

```
{
  "code": "0000",
  "msg": "成功",
  "data": [
    {
      "score": 0.97576493,
      "personID": "248",
      "targetType": "Target"
    }
  ],
  "success": true
}
```

9 Analysis by API

9.1 Synchronous Analysis

9.1.1 Synchronous Photo Analysis

API Endpoint	{{studio_base_url}}/APIPKG/apipkg/sync/image/analysis			
Method	POST			
Request Format	JSON			
Request Parameters	Name	Required	Type	Description
	imageId	Yes	String	Image ID
	images	No	Array<String>	Image URL
	processor	No	String	Model name
Response Parameters	Name	Required	Type	Description
	data		String	Analysis result
Sample Request	{ "imageId": "smokeman", "images": ["/images/smoke5.jpg"], "processor": "classifierSmoking" }			
Sample Response	{ "code": "0000", "msg": "成功", "data": {} }			

9.2 Asynchronous Analysis

9.2.1 Add Asynchronous Analysis Task

API Endpoint	{{studio_base_url}}/APIPKG/apipkg/async/video/addTask			
Method	POST			
Request Format	JSON			
Request Parameters	Name	Required	Type	Description
	taskId	Yes	String	Task ID

	processor	Yes	String	Model name
	type	Yes	String	Task type, default is 2
Response Parameters	Name	Required	Type	Description
	data		String	Null
Sample Request	{ "taskId": "", "processor": "", "type": 2 }			
Sample Response	{ "code": "0000", "msg": "success", "data": }			

9.2.2 Set Asynchronous Analysis Callback URL

API Endpoint	{{studio_base_url}}/APIPKG/apipkg/setCallbackUrl			
Method	GET			
Request Format	JSON			
Request Parameters	Name	Required	Type	Description
	callback	Yes	String	Callback URL
Response Parameters	Name	Required	Type	Description
	data		String	Null
Sample Response	{ "code": "0000", "msg": "成功", "data": {} }			

9.2.3 Asynchronous Analysis

API Endpoint	{{studio_base_url}}/APIPKG/apipkg/async/video/analysis			
Method	POST			
Request Format	JSON			
	Name	Required	Type	Description

Request Parameters	taskId	Yes	String	Task ID
	processor	Yes	String	Model name
	video	Yes	String	The path of the target video
Response Parameters	Name	Required	Type	Description
	data		String	
Sample Request	<pre>{ "taskId": "fight", "processor": "falconLiteFight", "video": "/images/fight1.mp4" }</pre>			
Sample Response	<pre>{ "code": "0000", "msg": "成功", "data": { "object_": { "bitField0_": 0, "type_": 0, "portraitImageLocation_": { "panoramicImageSize_": { "width_": 0, "height_": 0, "memoizedIsInitialized": -1, "unknownFields": { "fields": {}, "fieldsDescending": {} }, "memoizedSize": -1, "memoizedHashCode": 0 }, "memoizedIsInitialized": -1, "unknownFields": { "fields": {}, "fieldsDescending": {} }, "memoizedSize": -1, "memoizedHashCode": 0 }, "objectId_": "" } } }</pre>			

```

"associations_": [],
"algo_": {
  "appName_": "falconLiteFight",
  "appVersion_": 1,
  "objectType_": "falconLiteFight",
  "objectVersion_": 1,
  "data_": {
    "typeUrl_": "",
    "value_":
      "[{\\"rois\\":[\\\"1534031947172261890\\\"],\\"confidence\\":0.999832391738892,\\"detect\\":
      {\\"top\\":0,\\"left\\":0,\\"width\\":1920,\\"height\\":1080},\\"targetType\\":\\"falconLiteFight\\",
      \\"attributes\\":[{\\"value\\":\\"fight\\",\\"key\\":\\"falconLiteFight\\",\\"confidence\\":0.9998324
      }],\\"deviceId\\":\\"1534031914442731521\\"}]",
    "memoizedIsInitialized": -1,
    "unknownFields": {
      "fields": {},
      "fieldsDescending": {}
    },
    "memoizedSize": -1,
    "memoizedHashCode": 0
  },
  "memoizedIsInitialized": -1,
  "unknownFields": {
    "fields": {},
    "fieldsDescending": {}
  },
  "memoizedSize": -1,
  "memoizedHashCode": 0
},
"memoizedIsInitialized": -1,
"unknownFields": {
  "fields": {},
  "fieldsDescending": {}
},
"memoizedSize": -1,
"memoizedHashCode": 0
},
"cameraInfo_": {

```

```
"cameraId_": "",
"deviceId_": "1534031914442731521",
"deviceType_": "",
"placeCode_": "",
"placeName_": "",
"tollgateId_": "",
"tollgateName_": "",
"sourceId_": "",
"internalId_": {
  "regionId_": 1,
  "cameraDx_": 1,
  "memoizedIsInitialized": -1,
  "unknownFields": {
    "fields": {},
    "fieldsDescending": {}
  },
  "memoizedSize": -1,
  "memoizedHashCode": 0
},
"zoneId_": "",
"memoizedIsInitialized": -1,
"unknownFields": {
  "fields": {},
  "fieldsDescending": {}
},
"memoizedSize": -1,
"memoizedHashCode": 0
},
"panoramicImage_": {
  "format_": 1,
  "data_": {
    "bytes": [],
    "hash": 0
  },
  "url_":
"/images/cognitivesvc/20220614/ad9dfc3db4044bf5b5c244b98f678efa.jpg",
  "memoizedIsInitialized": -1,
  "unknownFields": {
```

```
        "fields": {},
        "fieldsDescending": {}
    },
    "memoizedSize": -1,
    "memoizedHashCode": 0
},
"capturedTime_": {
    "seconds_": 1655196230616,
    "nanos_": 0,
    "memoizedIsInitialized": -1,
    "unknownFields": {
        "fields": {},
        "fieldsDescending": {}
    },
    "memoizedSize": -1,
    "memoizedHashCode": 0
},
"receivedTime_": {
    "seconds_": 1655196230775,
    "nanos_": 0,
    "memoizedIsInitialized": -1,
    "unknownFields": {
        "fields": {},
        "fieldsDescending": {}
    },
    "memoizedSize": -1,
    "memoizedHashCode": 0
},
"objectIndexInFrame_": 0,
"extraInfo_": "",
"trackEvent_": 1,
"relativeTime_": {
    "seconds_": 0,
    "nanos_": 0,
    "memoizedIsInitialized": -1,
    "unknownFields": {
        "fields": {},
        "fieldsDescending": {}
    }
}
```

```
    },  
    "memoizedSize": -1,  
    "memoizedHashCode": 0  
  },  
  "nsId_": "",  
  "memoizedIsInitialized": 1,  
  "unknownFields": {  
    "fields": {},  
    "fieldsDescending": {}  
  },  
  "memoizedSize": -1,  
  "memoizedHashCode": 0  
}  
}
```

9.2.4 Delete Asynchronous Analysis Task

API Endpoint	{{studio_base_url}}/APIPKG/apipkg/async/task/delete			
Method	POST			
Request Format	JSON			
Request Parameters	Name	Required	Type	Description
	deviceId	Yes	String	Task ID
Response Parameters	Name	Required	Type	Description
	data		String	Null
Sample Request	{ "taskId": "", "processor": "", "type": 2 }			
Sample Response	{ "code": "0000", "msg": "success", "data": }			